

The quest for conserved RNAs in viral genomes

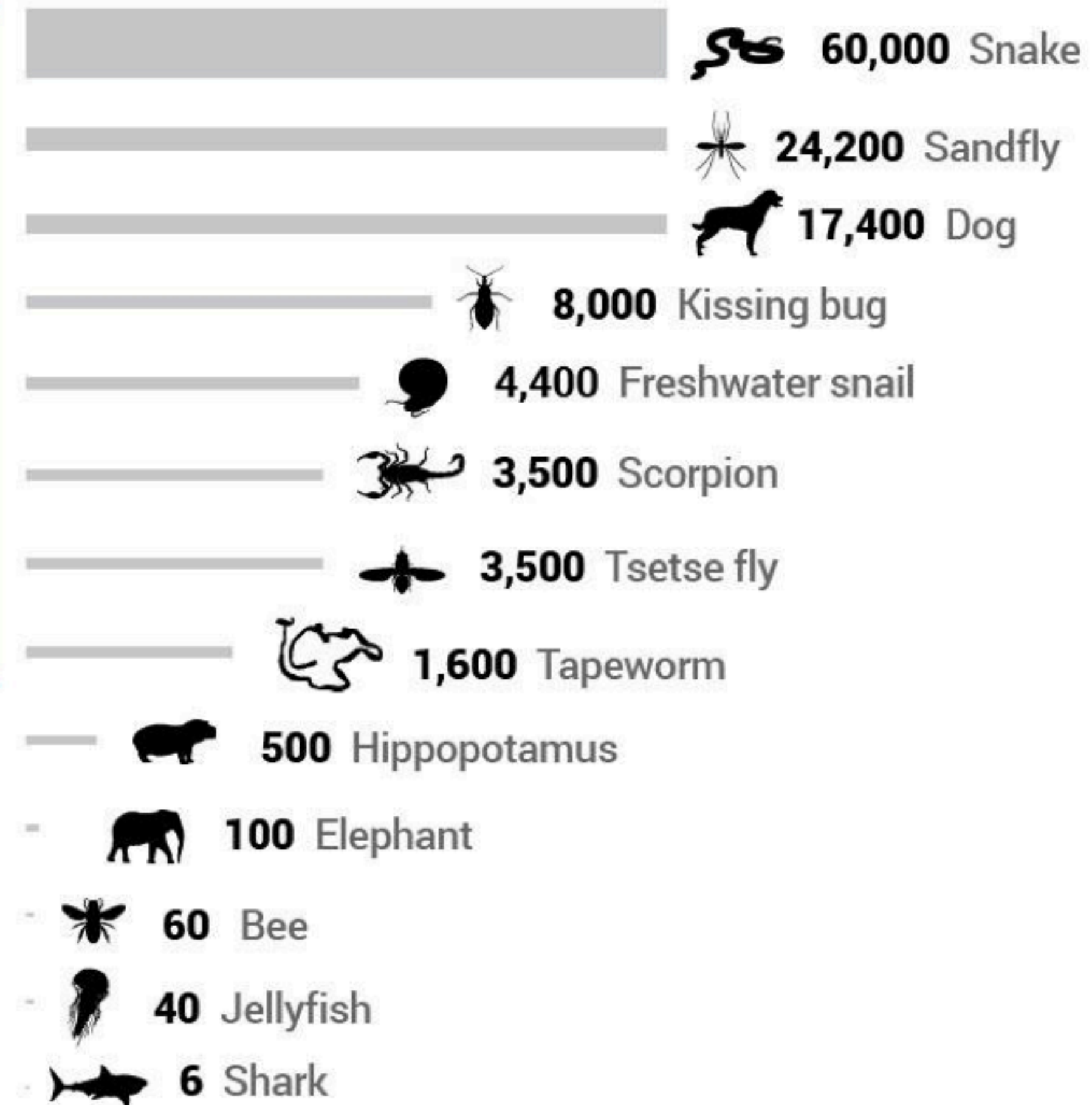
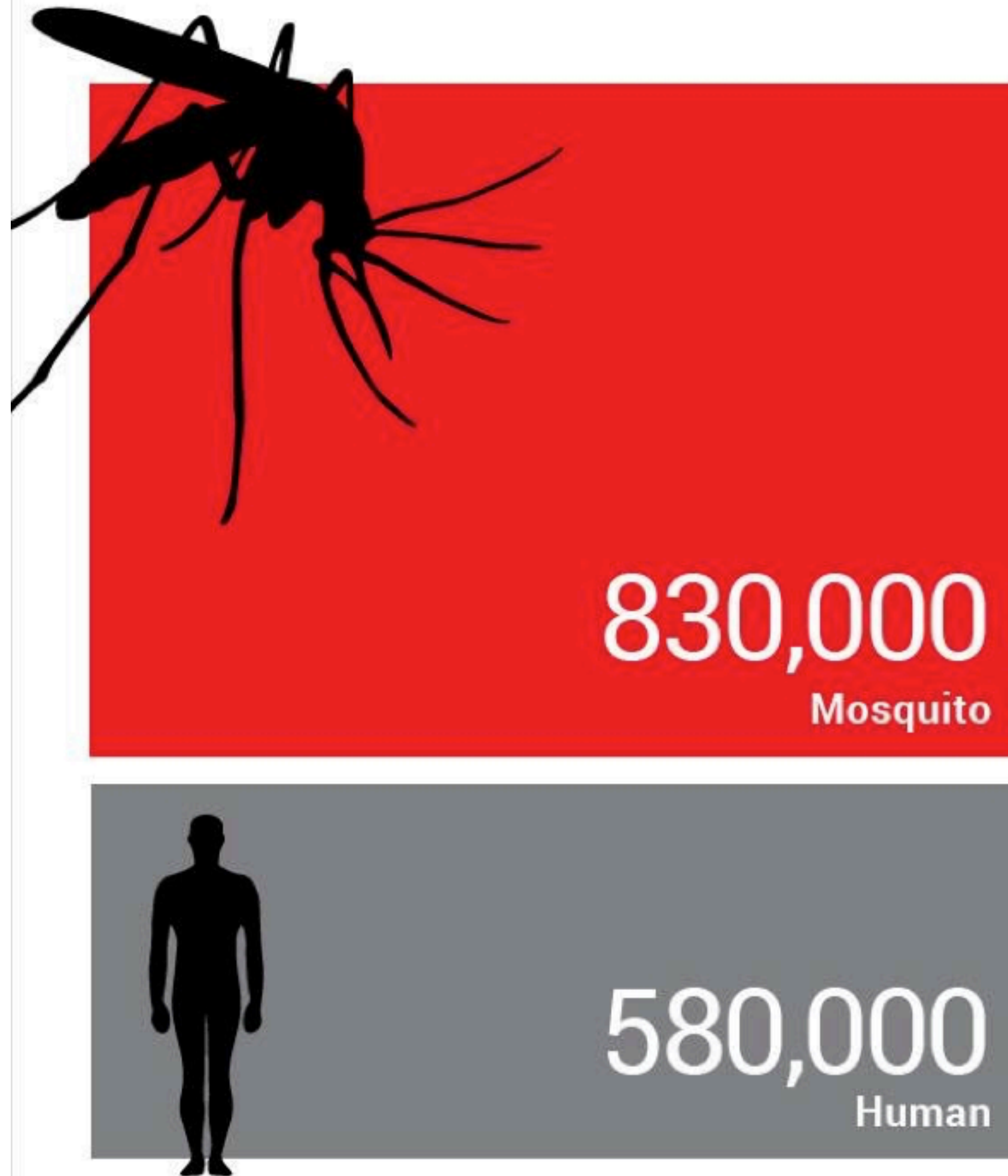
Michael T. Wolfinger

Research Group Bioinformatics and Computational Biology
& Department of Theoretical Chemistry
University of Vienna
Austria

The World's Deadliest Animals

Number of people killed by animals, 2015

gates
notes

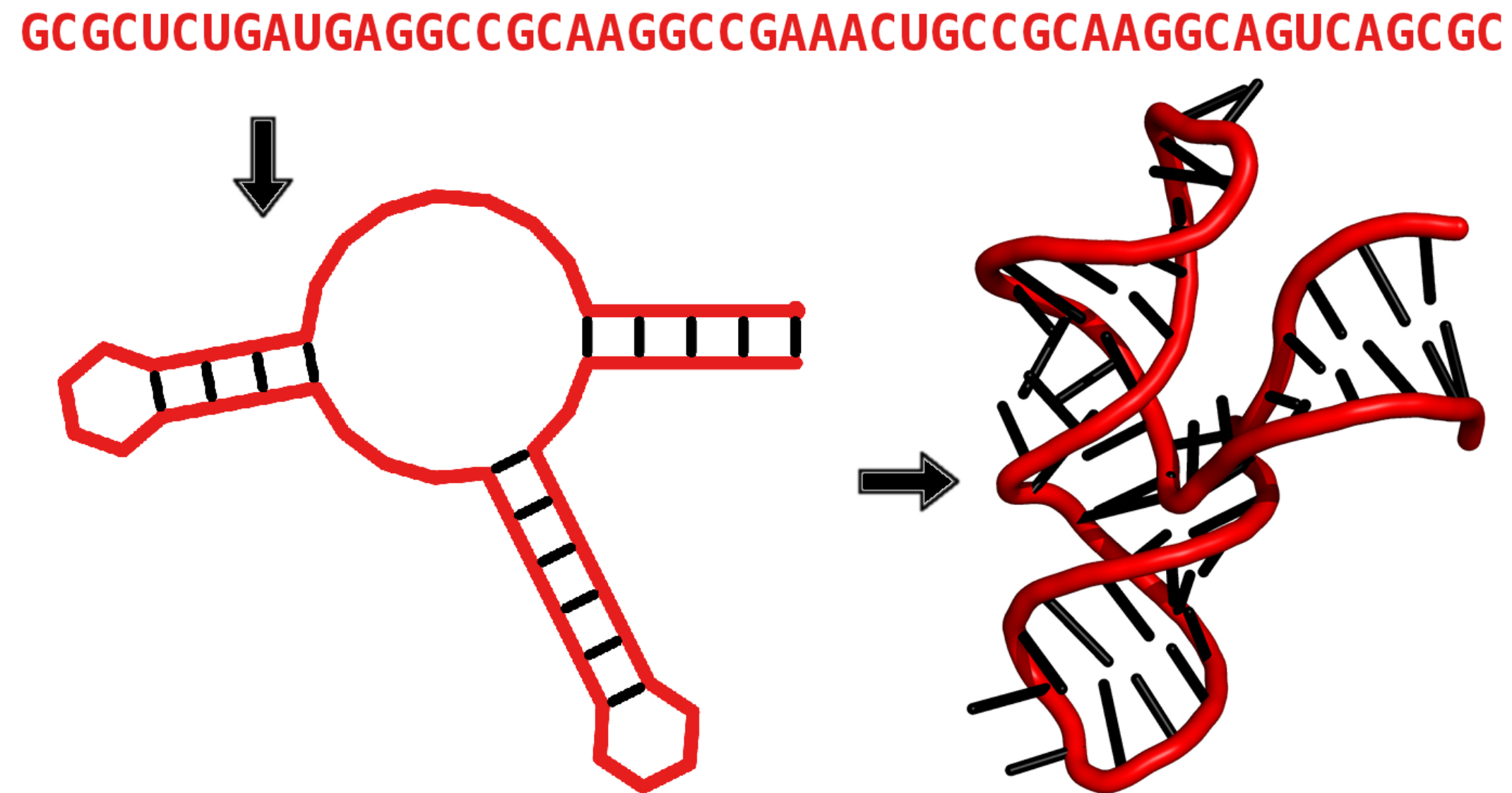


Sources: IHME, WHO, CrocBITE, FAO, Norwegian Institute for Nature Research, International Shark Attack File, National Geographic, PBS, National Science Foundation, CDC, WWF, Wilderness & Environmental Medicine, Nature, French Institute of Research for Development. All calculations have wide error margins.

Part I:

RNA structure prediction

The RNA Folding Problem

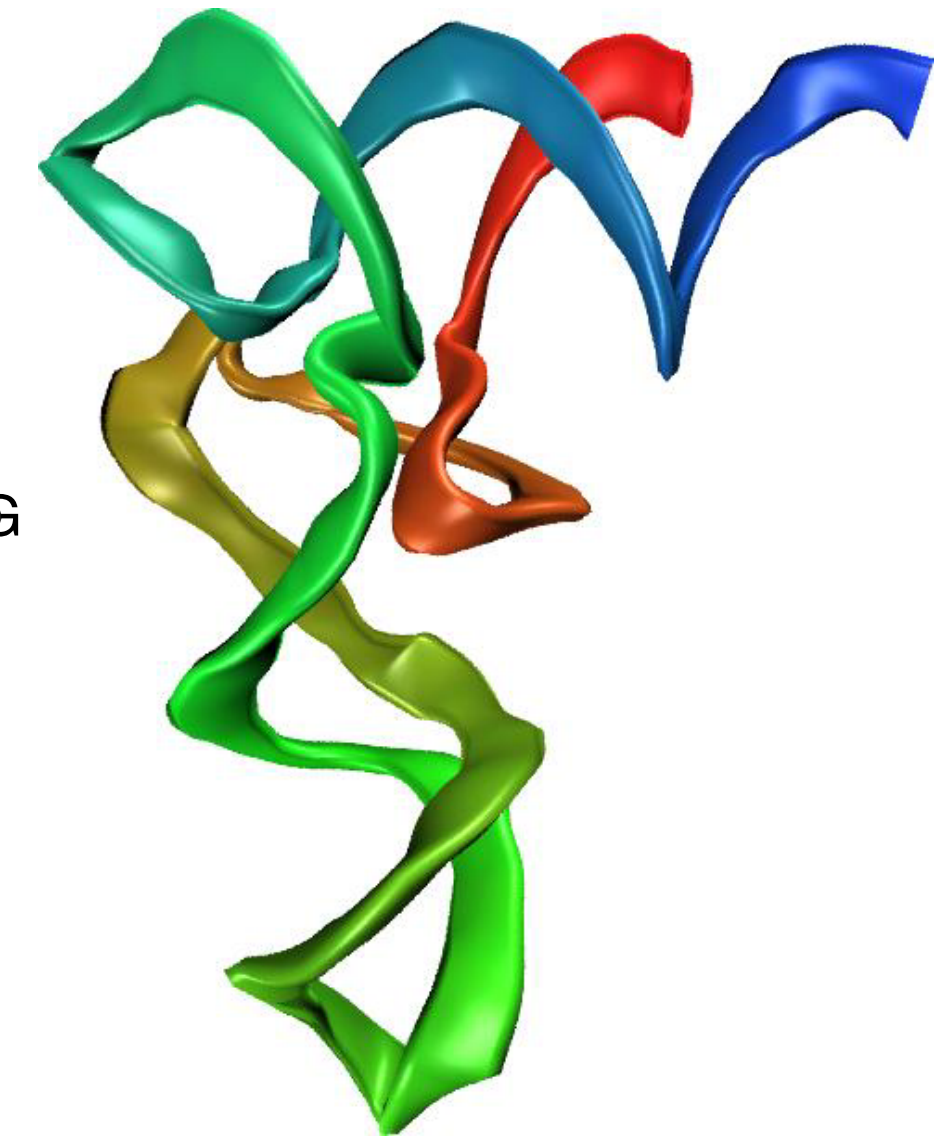
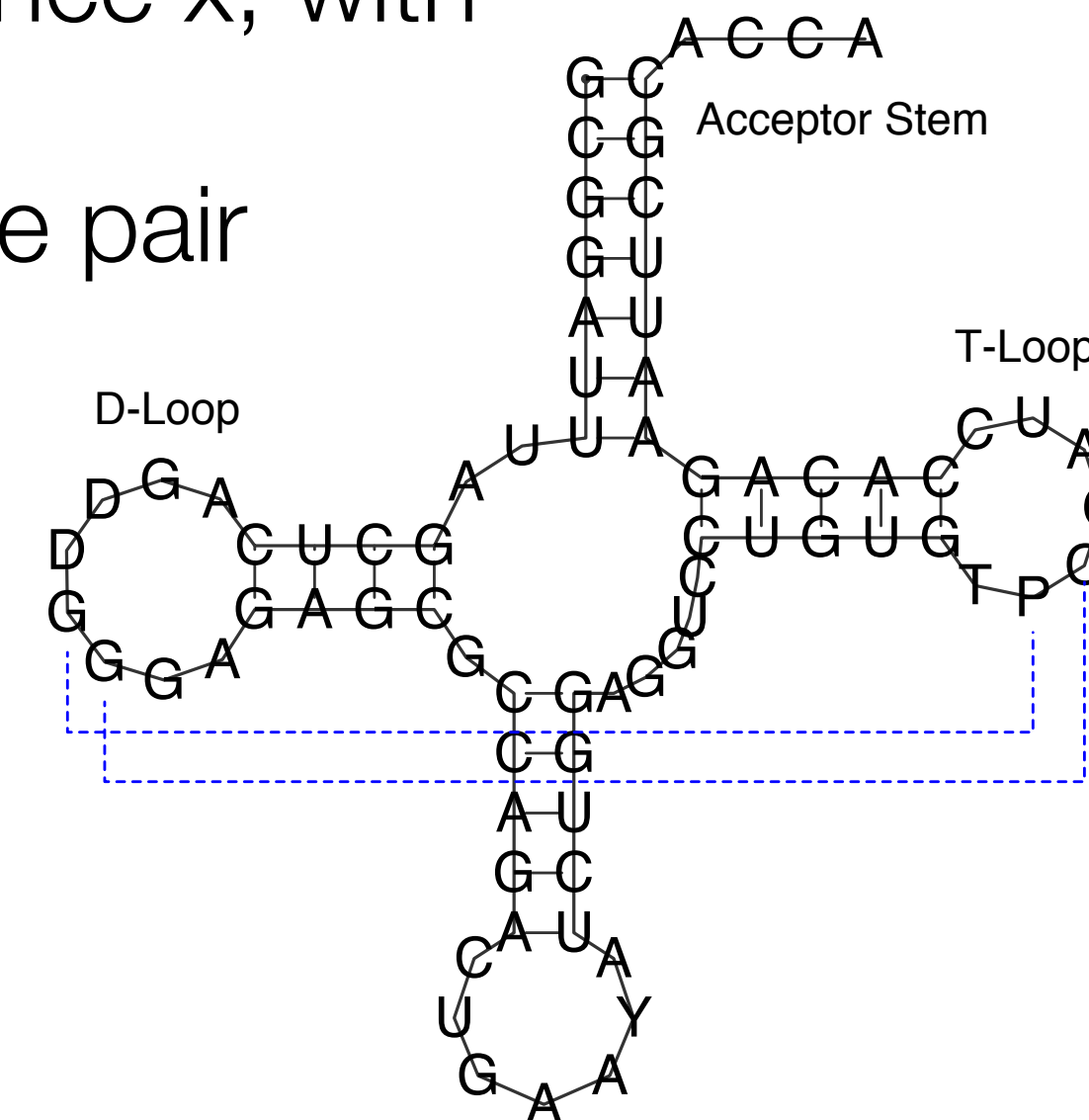


- Hierarchical folding: Secondary structure forms first then helices arrange to form tertiary structure
- Secondary structures cover most most of the folding energy
- Convenient and biologically useful description
- Computationally easy to handle
- Tertiary structure prediction needs knowledge of secondary structure

Secondary Structures

A **secondary structure** is a list of base pairs (i,j) on a sequence x , with

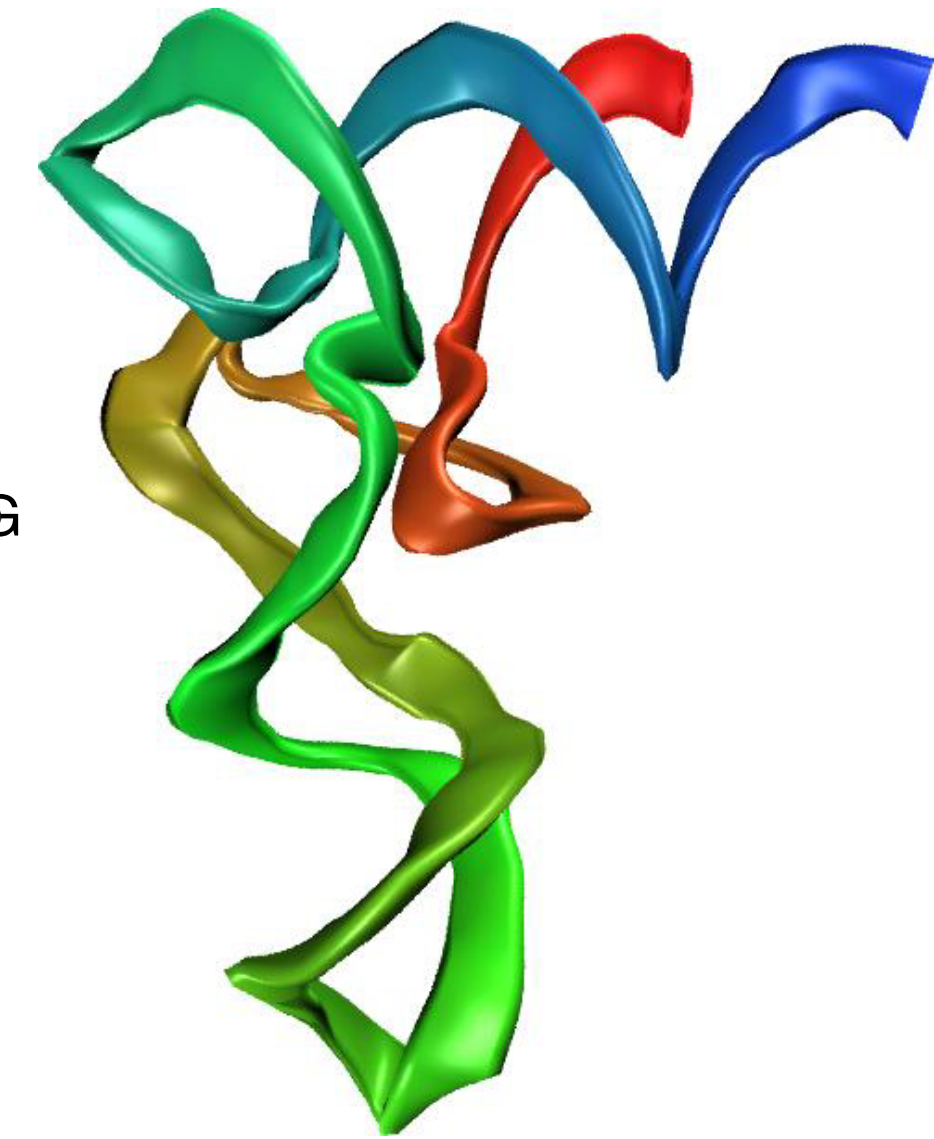
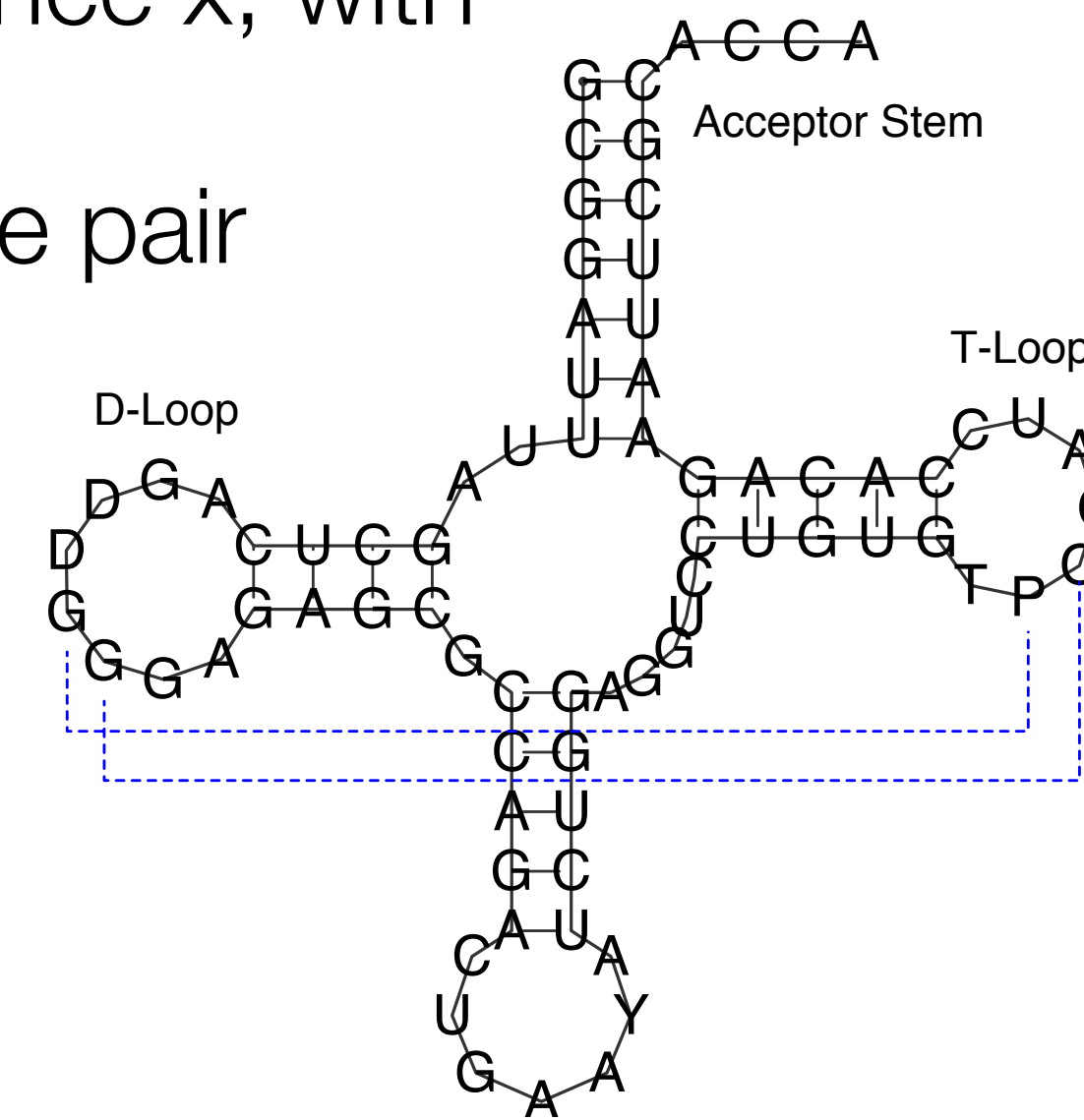
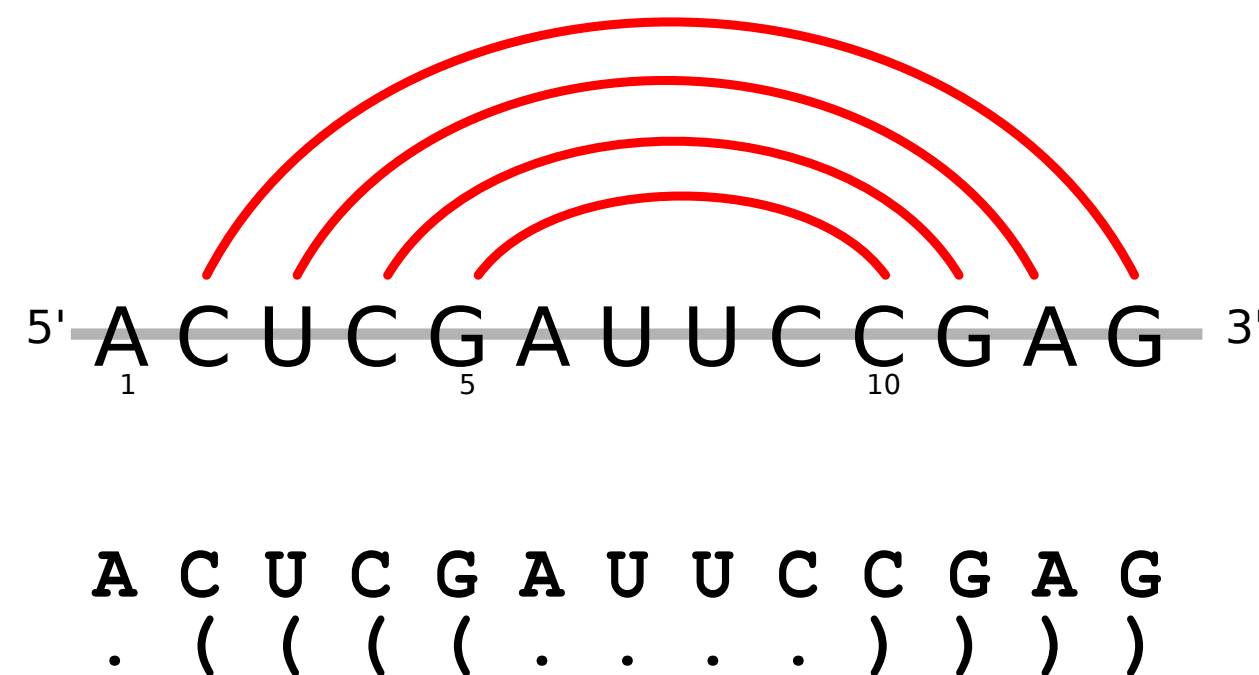
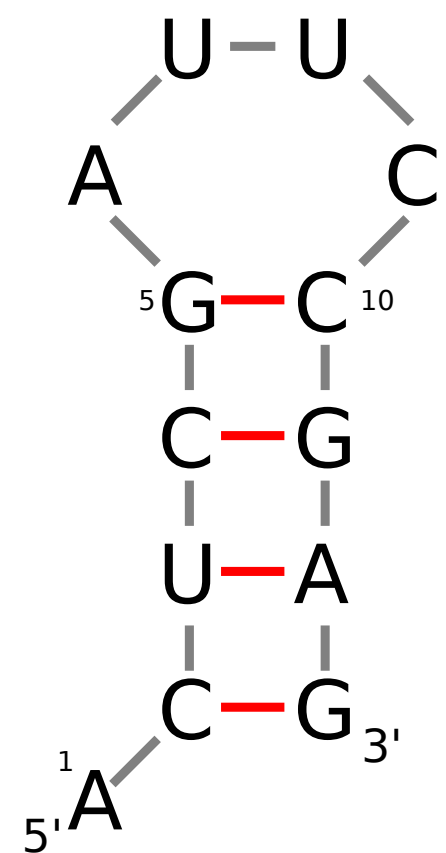
- Any nucleotide (sequence position) can form at most one pair
- No pseudo-knots: No pairs (i,j) and (k,l) with $i < k < j < l$
- If (i,j) is a pair then $x_i x_j \in \{GC, CG, AU, UA, GU, UG\}$
- If (i,j) is a base pair, then $j - i > 3$



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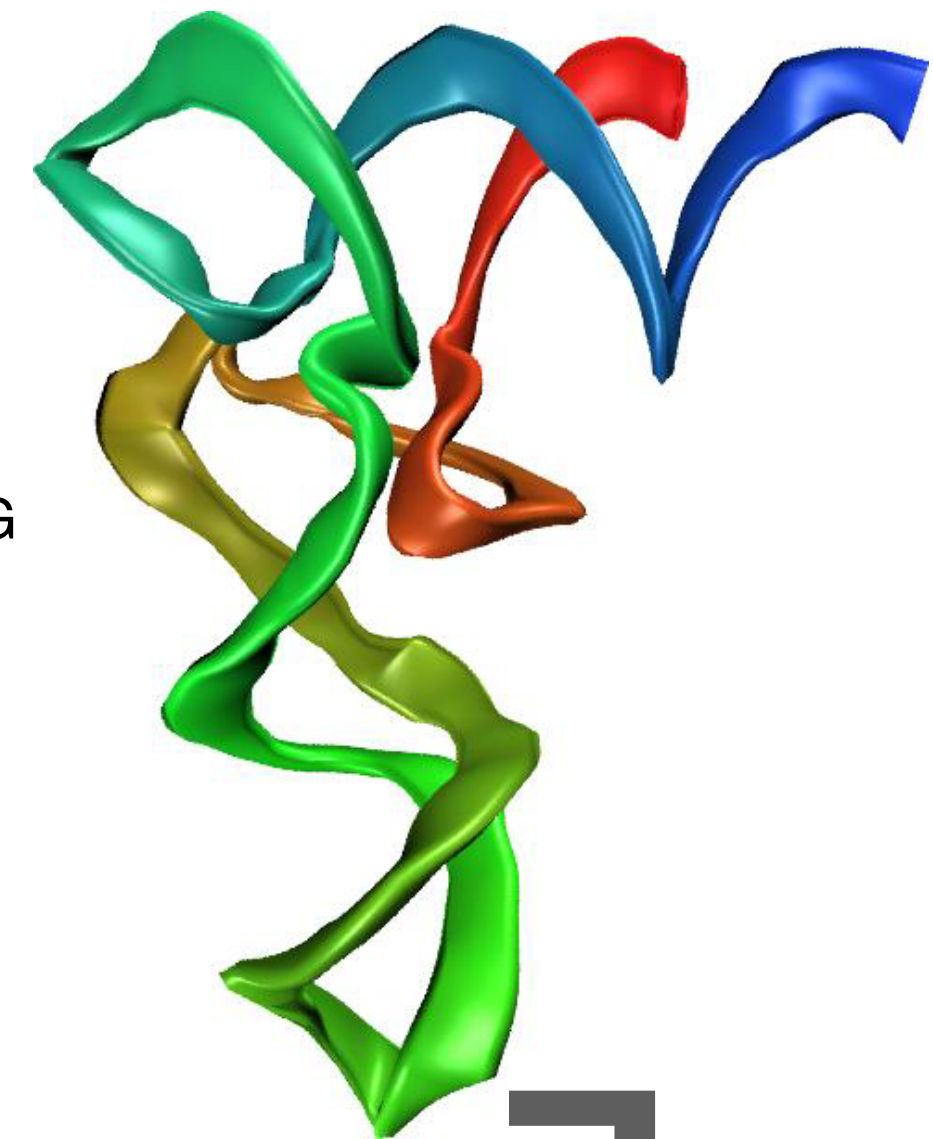
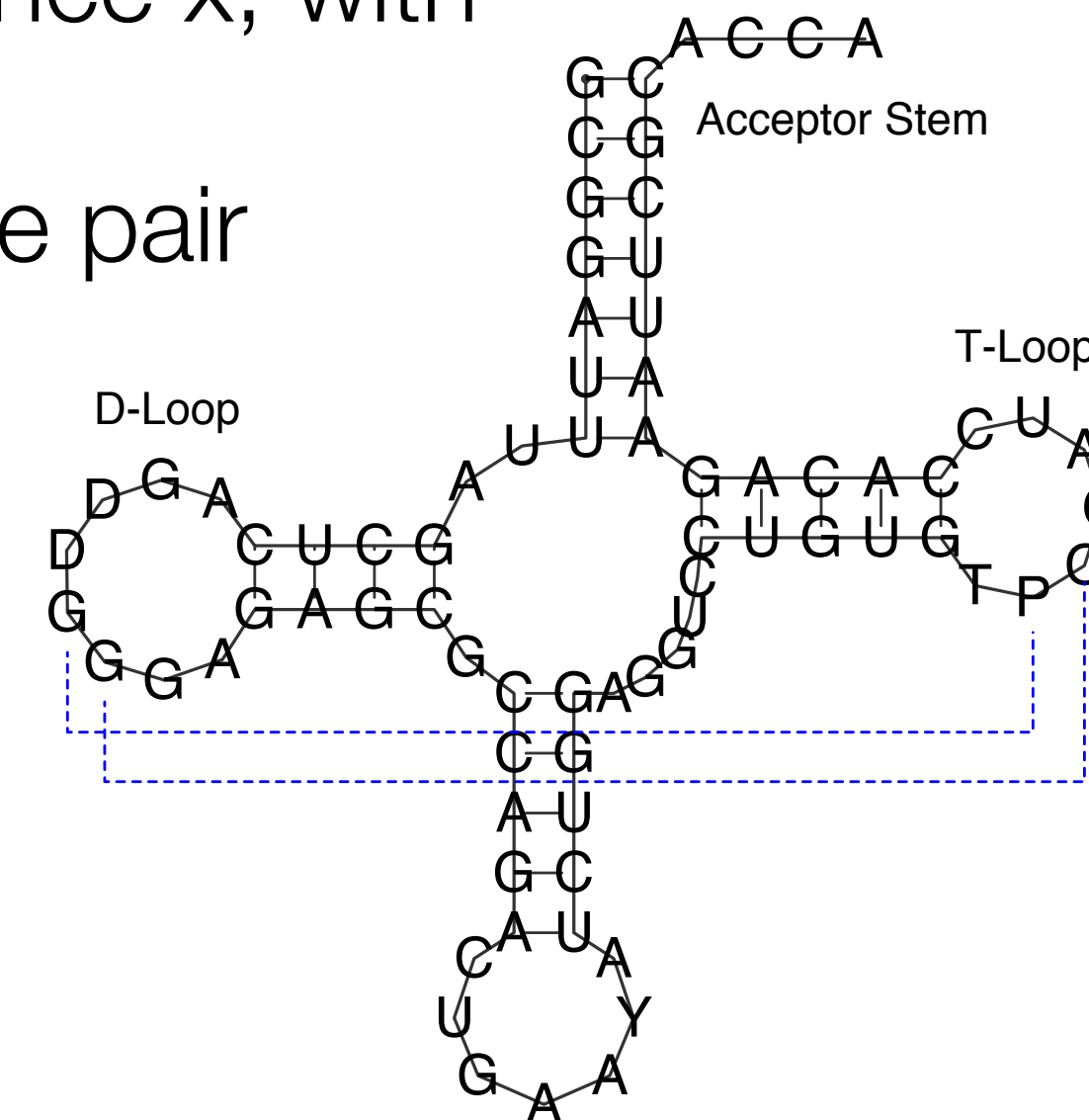
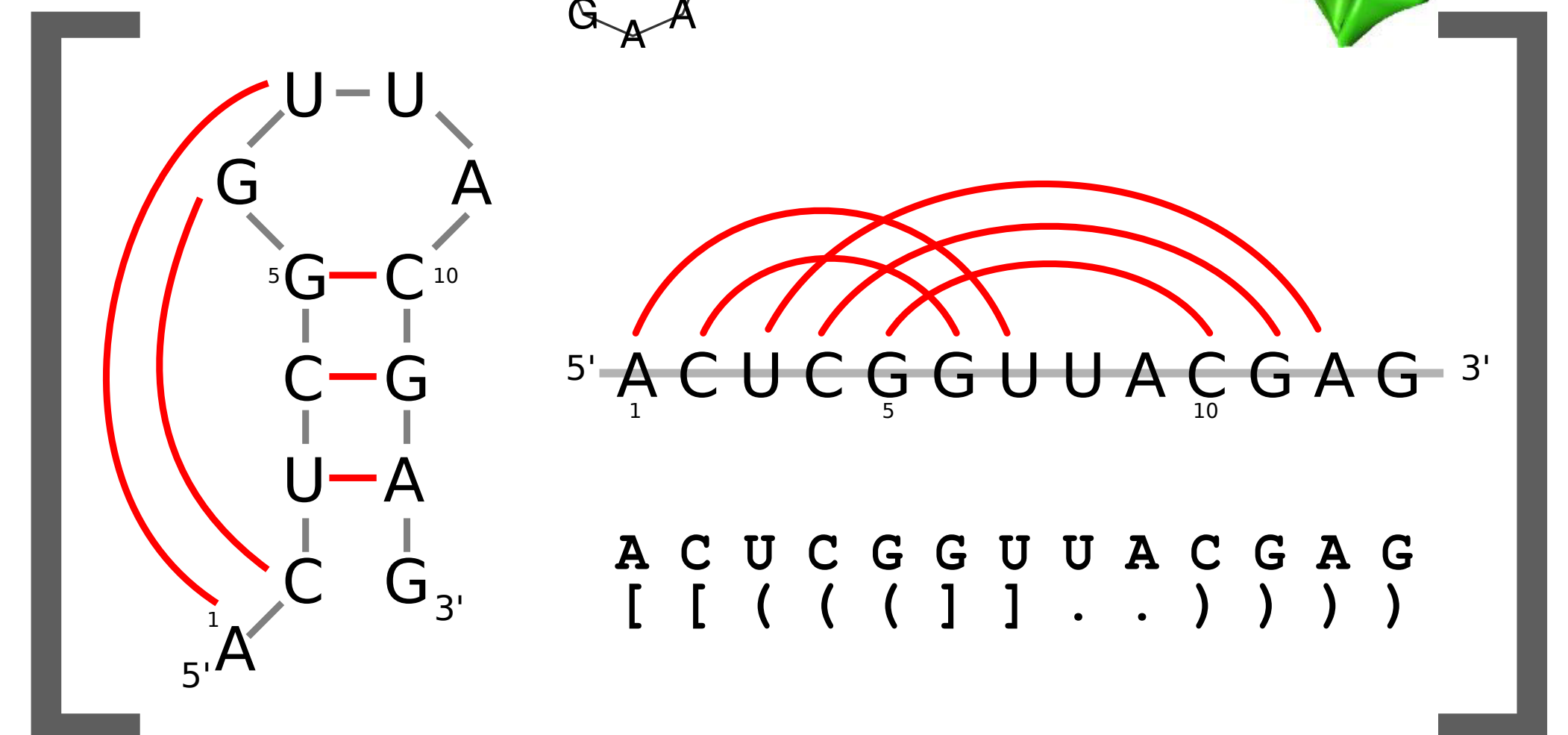
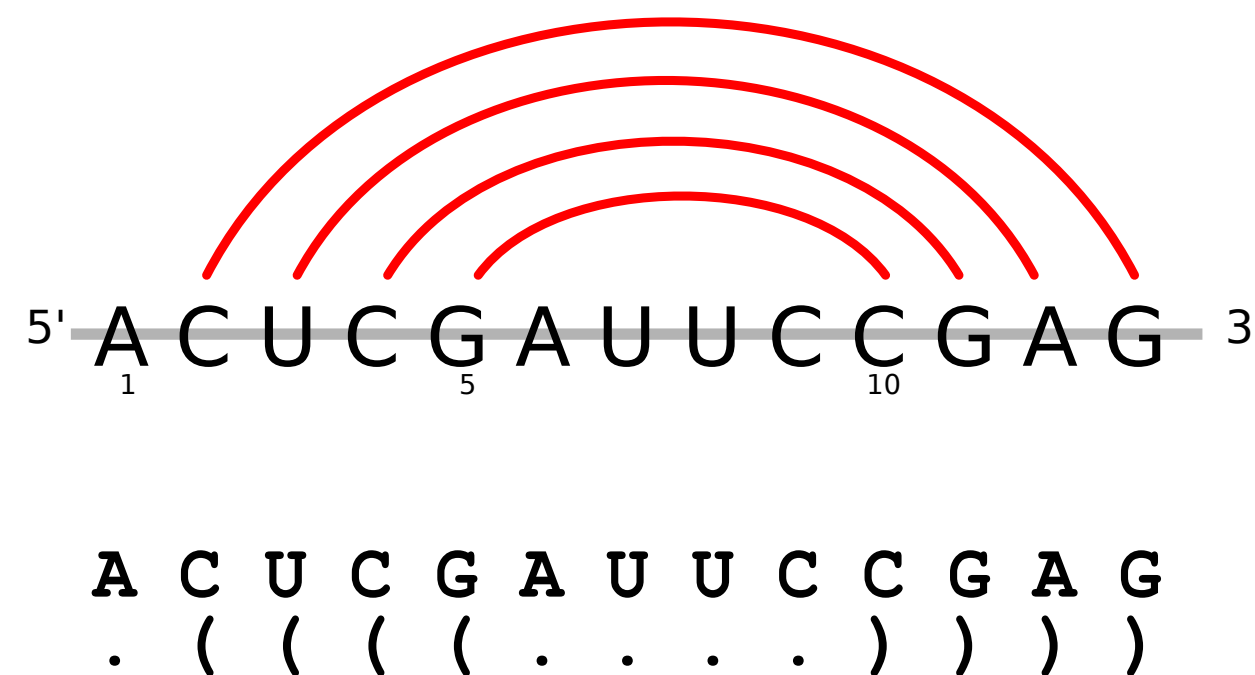
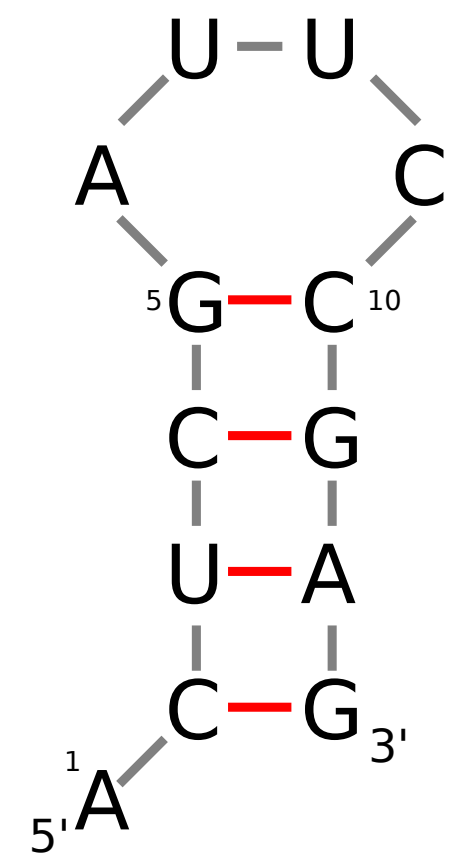
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Conformation Space

The number of secondary structures for a sequence $x = x_1 \dots x_n$ can be computed recursively



$$S_{ij} = S_{i+1,j} + \sum_{k=i+1}^j S_{i+1,k-1} S_{k+1,j} \Pi_{kj}$$

$\Pi_{ik} = 1$ if $x_i x_k \in \{GC, CG, AU, UA, GU, UG\}$, otherwise $\Pi_{ik} = 0$

For sequences with equal $\{A, U, G, C\}$ content, the number of conformations grows asymptotically with sequence length

$$\bar{S}(n) \sim n^{-\frac{3}{2}} 1.85^n$$

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Nature ‘exploits’ this property

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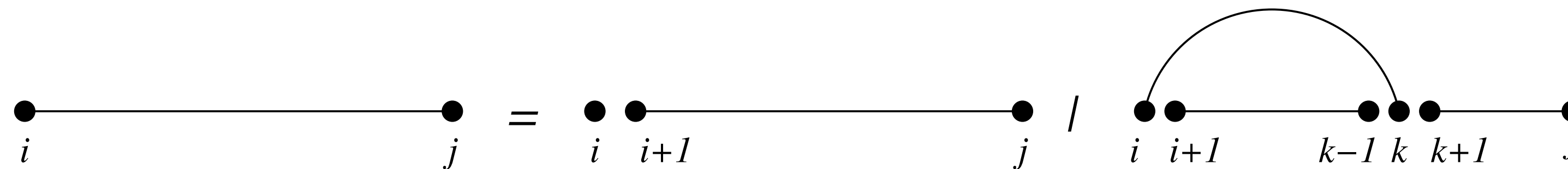
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Solving the RNA Folding Problem

Toy model for RNA folding: assign energies to base pairs $\varepsilon(x, y)$

Easily solved by **Dynamic Programming**: recursive computation with tabulation of intermediate results



$$E_{ij} = \min_{i < k \leq j} \left\{ E_{i+1,j} ; \left(E_{i+1,k-1} + E_{k+1,j} + \varepsilon(x_i, x_k) \right) \right\}$$

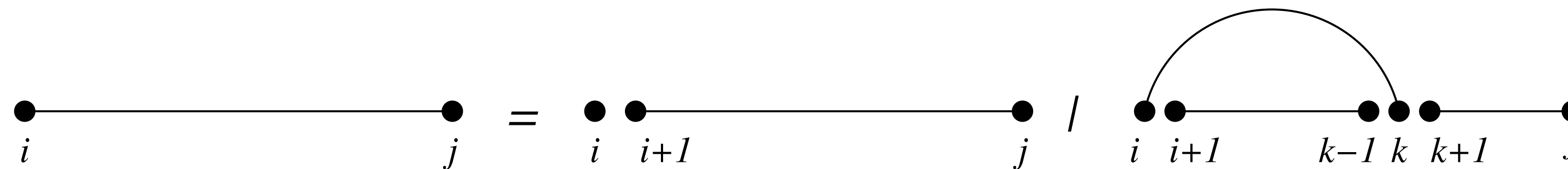
- E_{1n} is the best possible energy for our sequence
- Backtracing through the E table yields the corresponding structure
- The algorithm requires $\mathcal{O}(n^2)$ memory and $\mathcal{O}(n^3)$ CPU time

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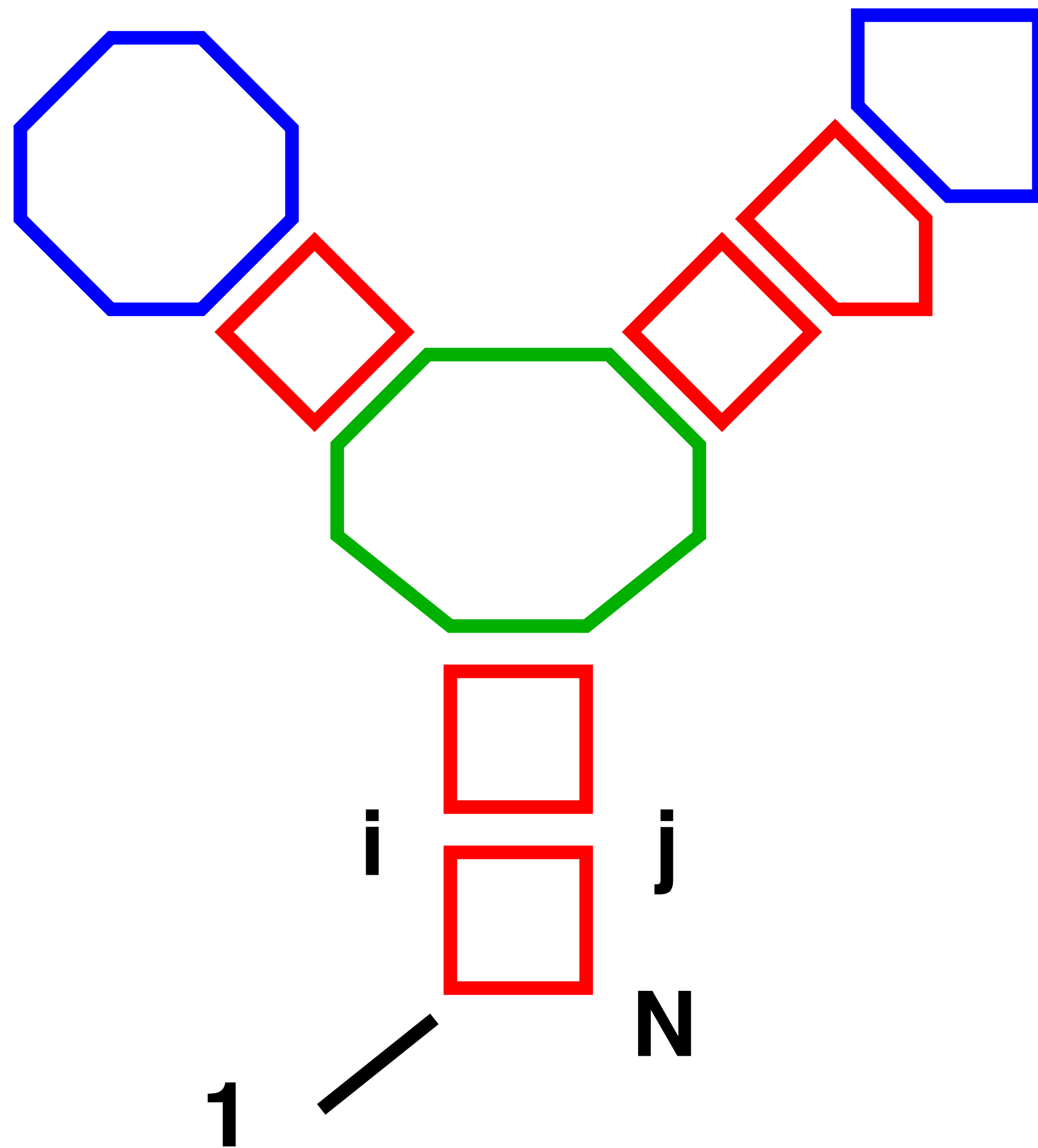


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Loop Decomposition

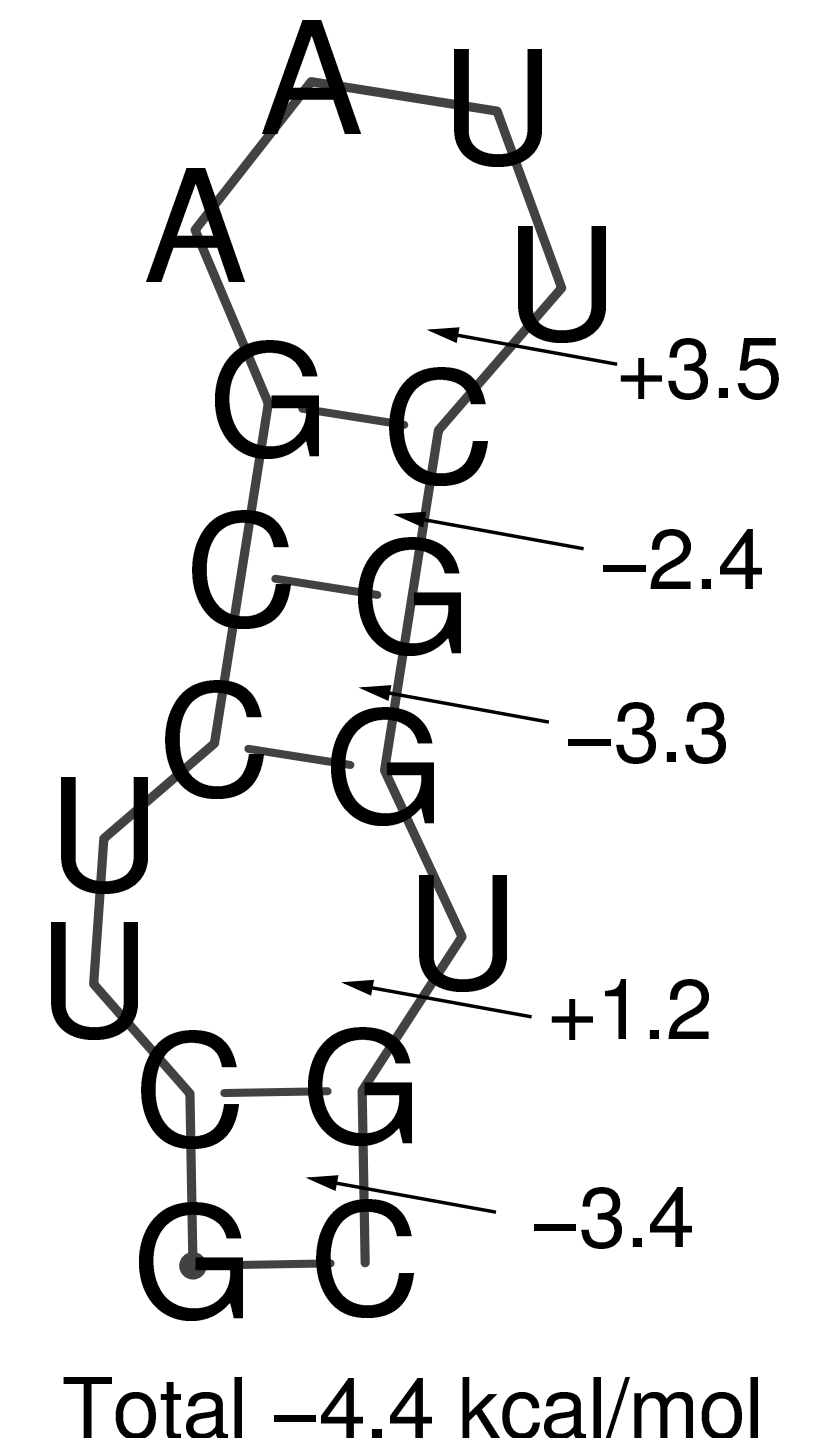


Nearest Neighbour Model

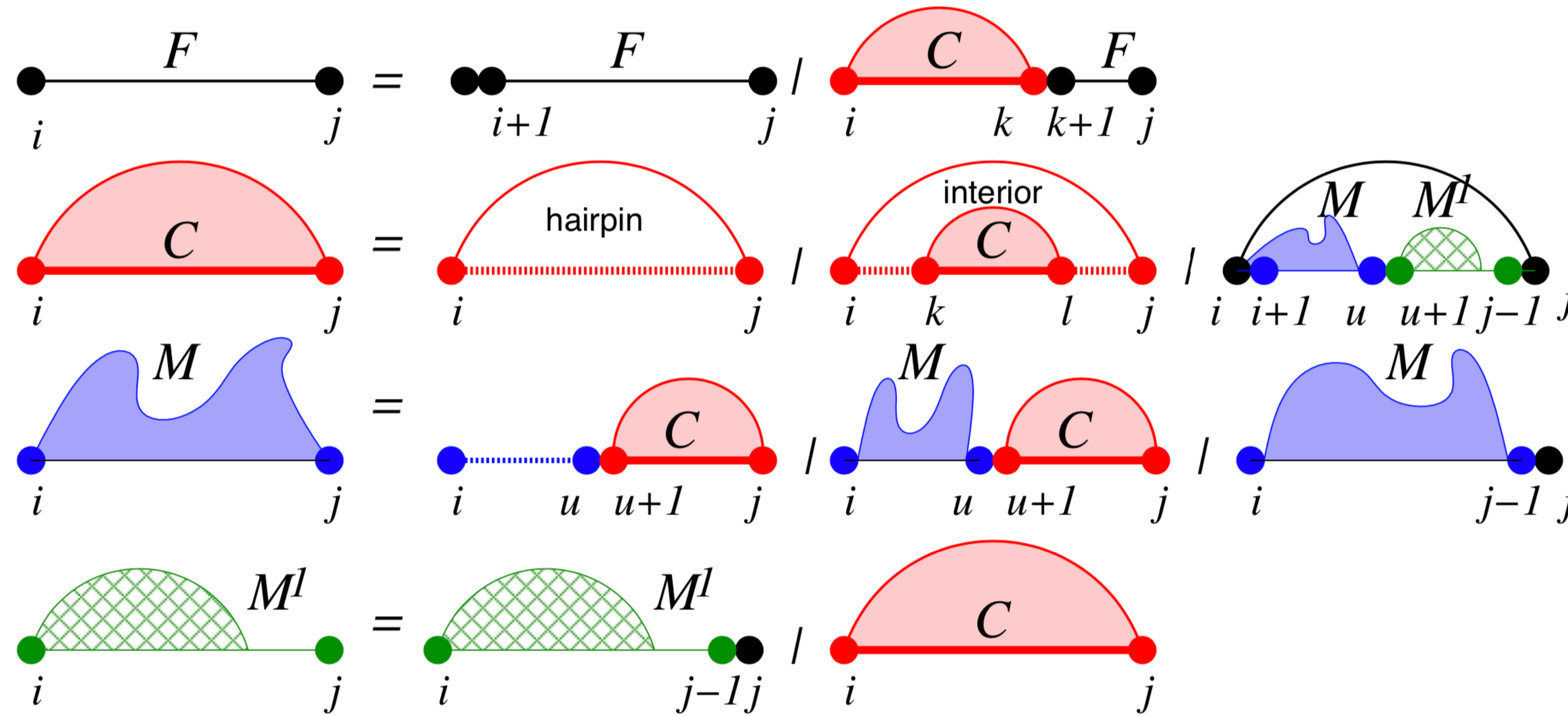
The standard energy model expresses the free energy of a structure as the sum over its loop energies

$$E(\mathcal{S}) = \sum_{l \in \mathcal{S}} E(l)$$

- Good approximation for most oligonucleotides
- Loop energies depend on loop type/size and some sequence dependence
- Most relevant parameters are experimentally measured; some still guesswork
- Secondary structures are macro-states, hence energies are **temperature-dependent free energies**
- Training parameters is becoming a viable alternative to experiment



Folding with Loop Based Energies



F_{ij} free energy of the optimal substructure on the subsequence $x[i..j]$.

C_{ij} optimal free energy on $x[i..j]$, where (i, j) pair.

M_{ij} $x[i..j]$ is part of a multiloop and contains at least one pair.

M_{ij}^1 same as M_{ij} but contains exactly one component closed by (i, h) .

Partition Function

Recall: $\bar{S}(n) \sim n^{-\frac{3}{2}} 1.85^n$

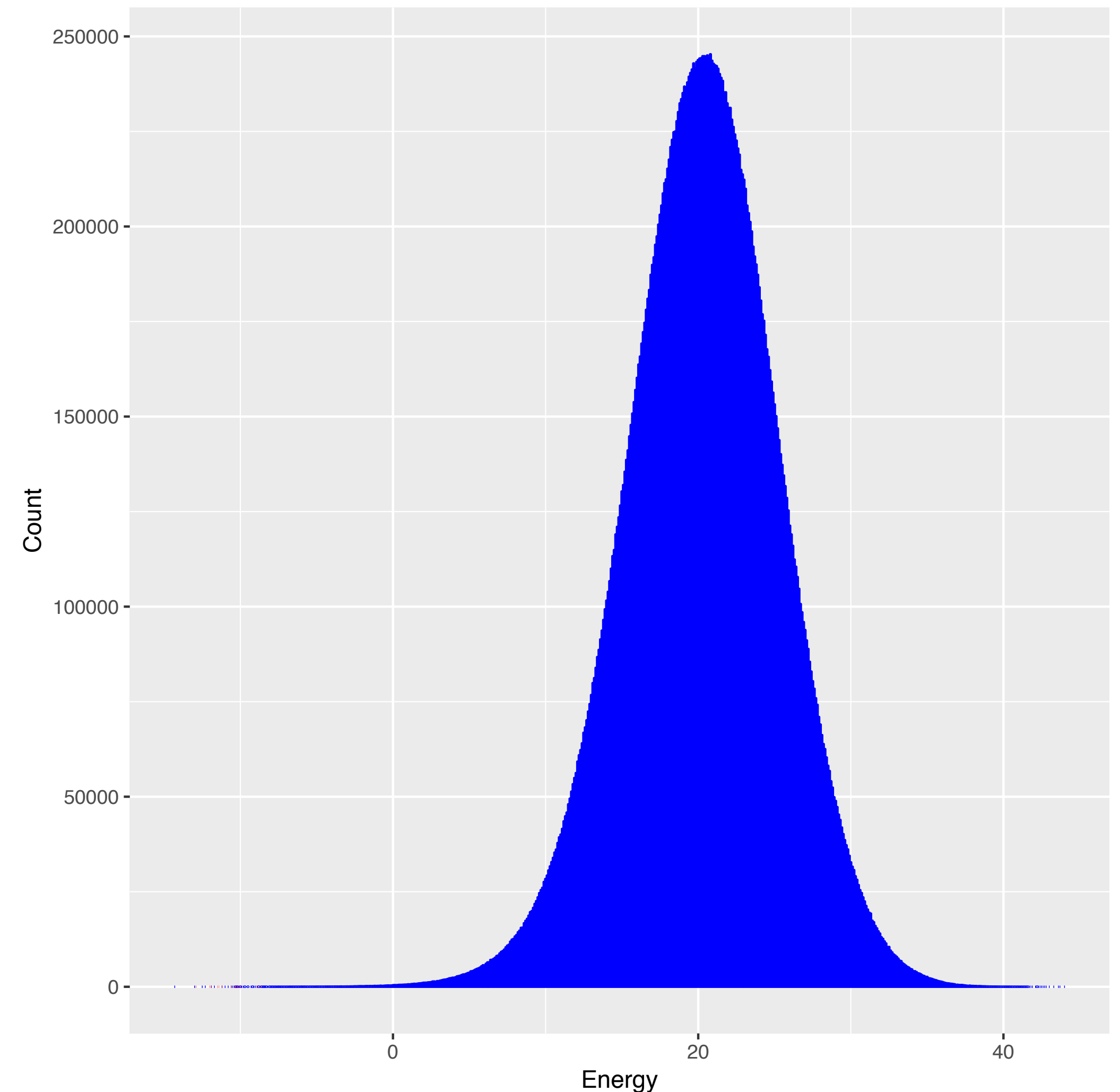
RNA is a biopolymer and ruled by thermodynamics

The **partition function** is the fundamental quantity of statistical mechanics and all thermodynamic properties can be derived from it

$$Z = \sum_{\Psi} \exp\left(-\frac{E(\Psi)}{RT}\right)$$

E.g. the free energy of formation is given by

$$\Delta G = -RT \ln Z$$



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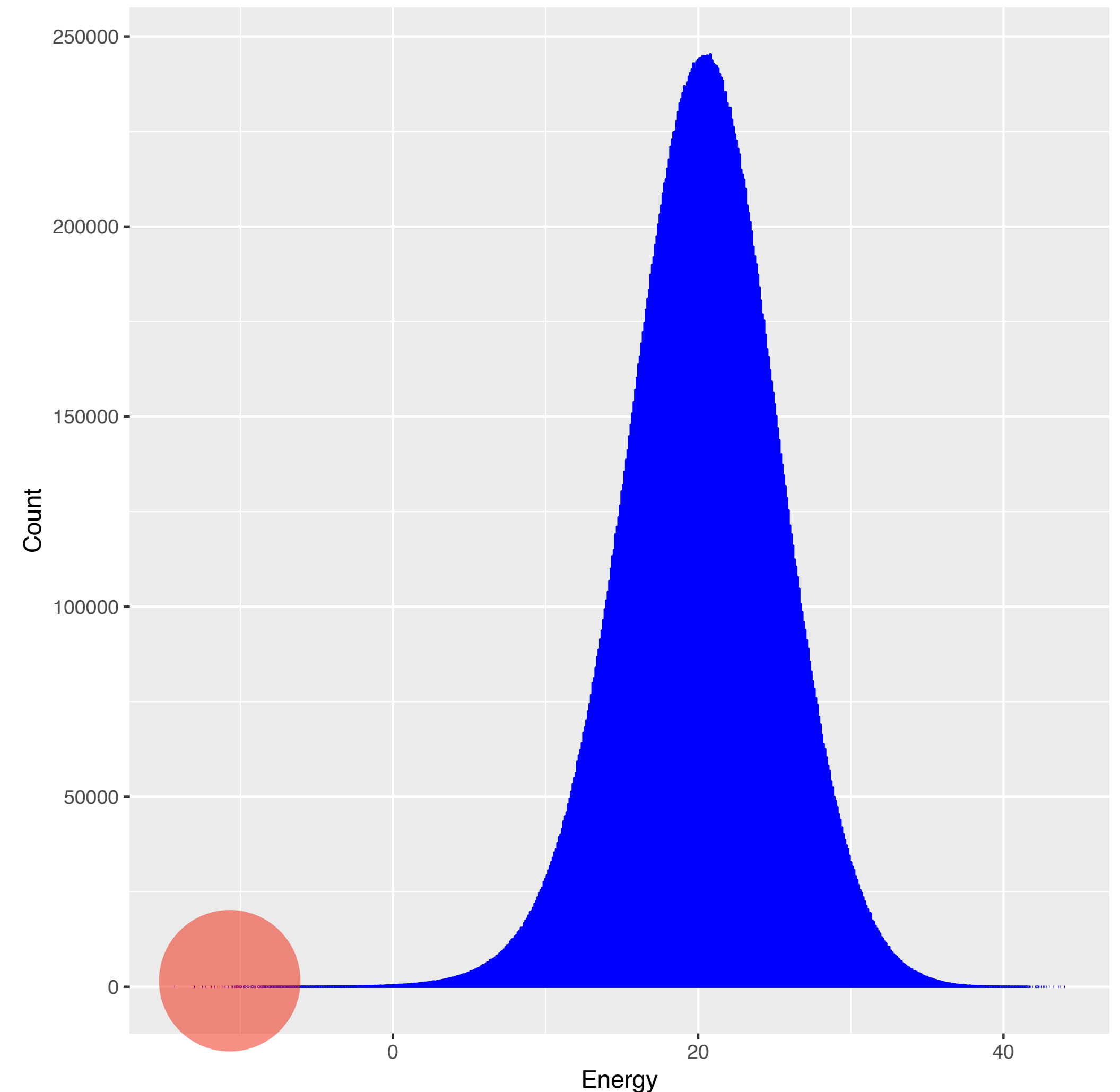
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Computing the Partition Function

The recursion has the same structure as for energy minimisation, with two differences

- replace minimum operation by sums
- addition of energies by products of partition functions

$$E_{ij} = \min_{i < k \leq j} \left\{ E_{i+1,j} ; \left(E_{i+1,k-1} + E_{k+1,j} + \varepsilon(x_i, x_k) \right) \right\}$$

$$Z_{ij} = Z_{i+1,j} + \sum_{k, (i,k) \text{ pairs}} Z_{i+1,k-1} Z_{k+1,j} \exp(-\varepsilon(x_i, x_k)/RT)$$

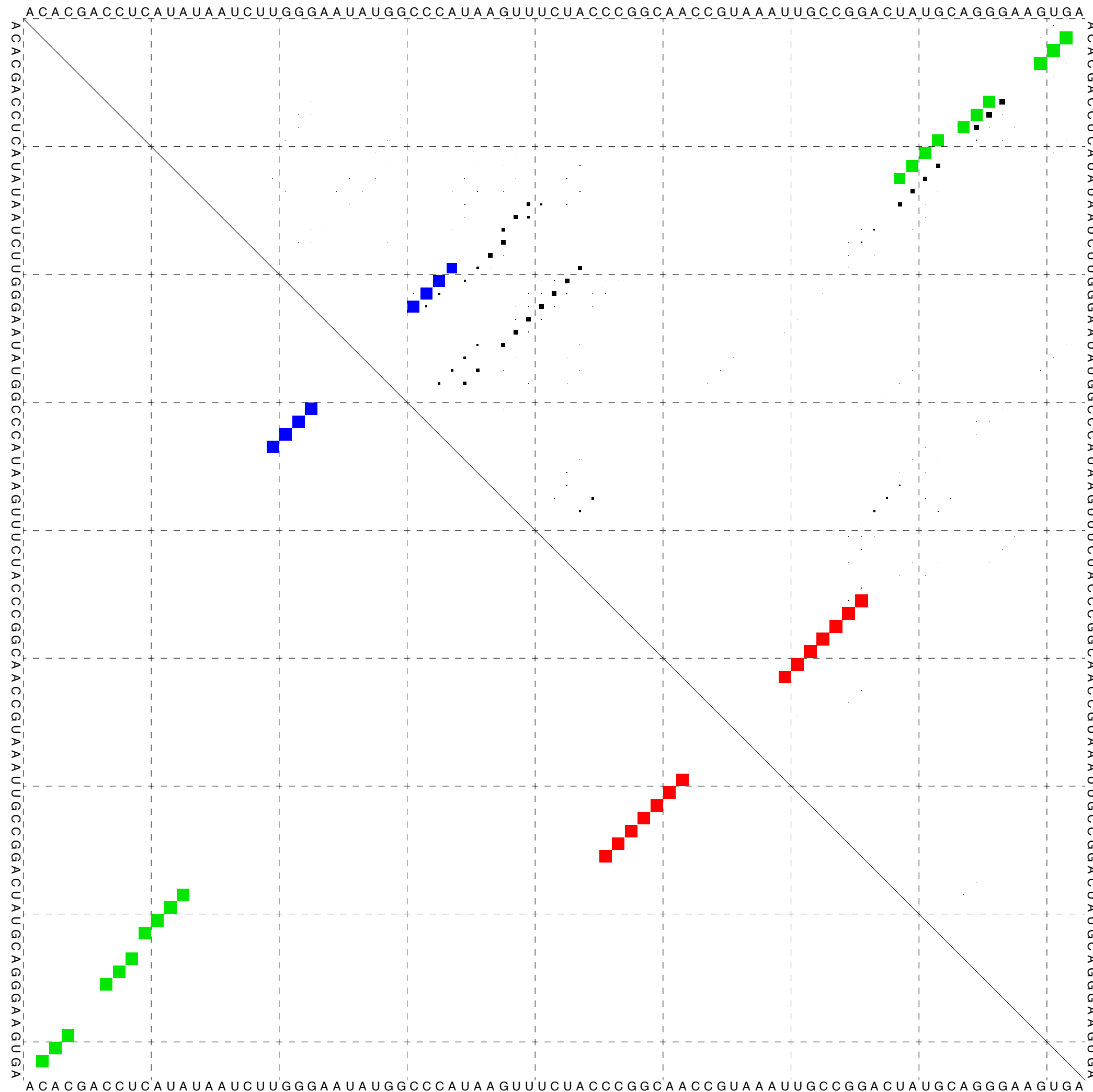
The probability of structure features can be computed from Z, e.g. the probability that a pair is formed

$$p_{ij} = \sum_{\Psi, (i,j) \in \Psi} p(S)$$

For efficient computation define the partition function $\hat{Z}_{ij}$ for structures outside the subsequence $x[i..j]$

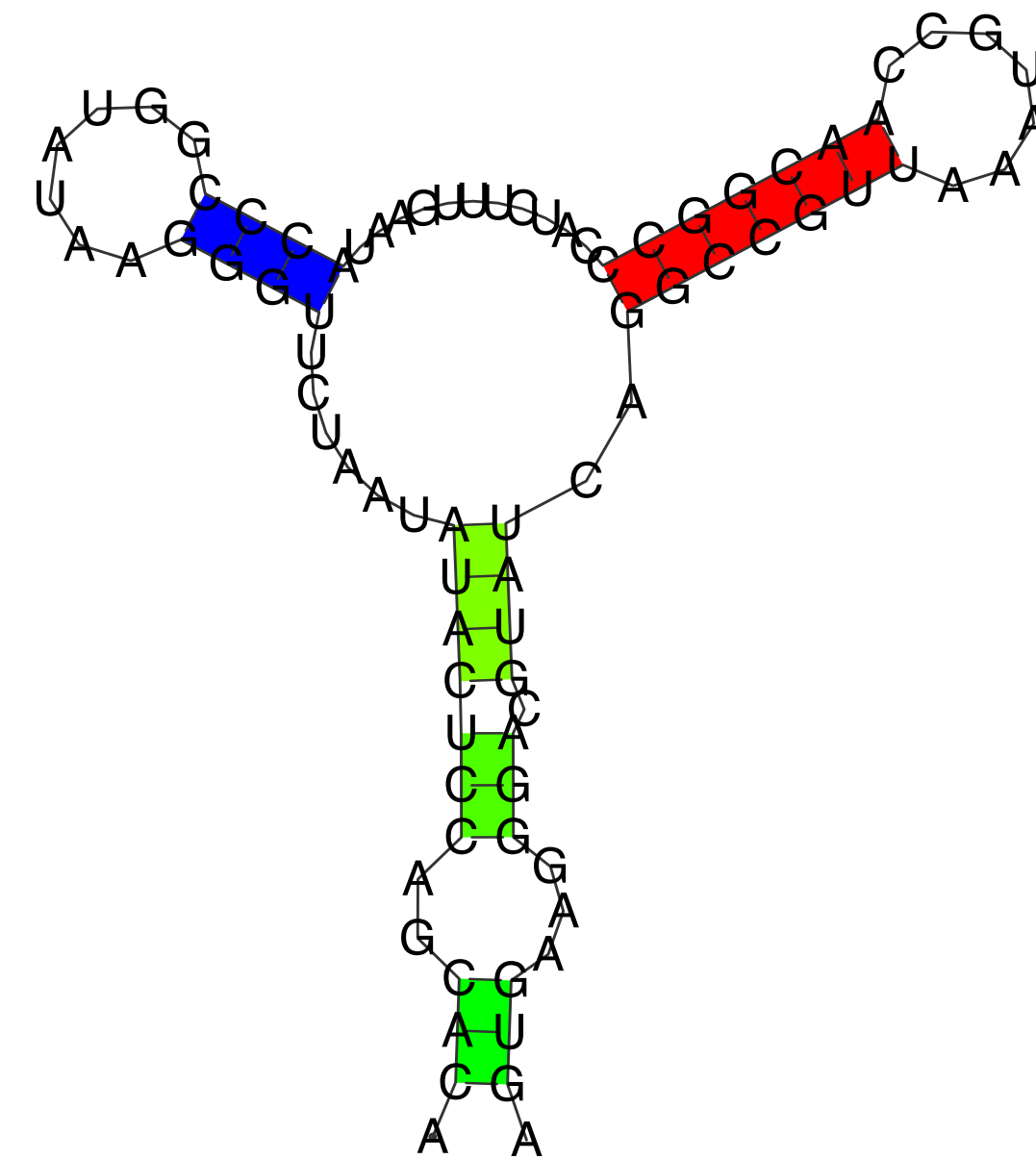
$$p_{ij} = \hat{Z}_{ij} Z_{i+1,j-1} \exp(-\varepsilon_{ij}/RT) / Z$$

Representing Ensembles of RNA Structures



Ensembles of structures (thermodynamic equilibrium) are best represented by base pair probabilities.

A pair (i, j) with probability p is represented by a square in row i and column j with area p .



The Vienna RNA Package

- Minimum free energy and partition function folding
- Complete suboptimal folding
- Inverse folding / RNA design
- Comparison of secondary structures
- Specific heat curves
- Inclusion of structure probing data
- Analysis of folding kinetics / co-transcriptional folding
- Utilities for plotting and annotation structures
- 2.5D prediction: G-quadruplexes and pseudo-knots
- Prediction of consensus structures

For the programmer:

- A C library to link against your programs
- Python/Perl scripting language interface

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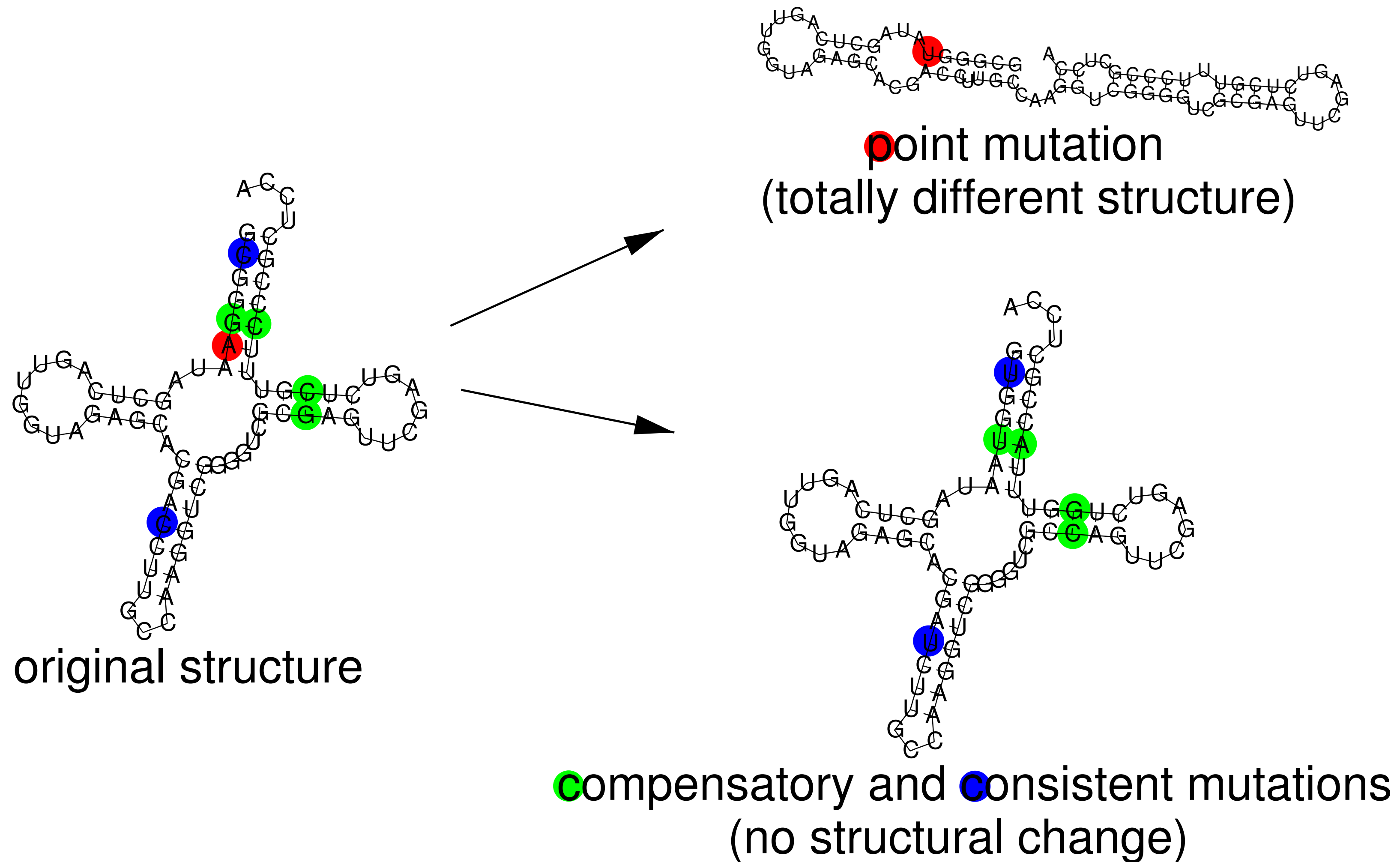
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Free software, C source code and fold server available at

<http://www.tbi.univie.ac.at/RNA/>

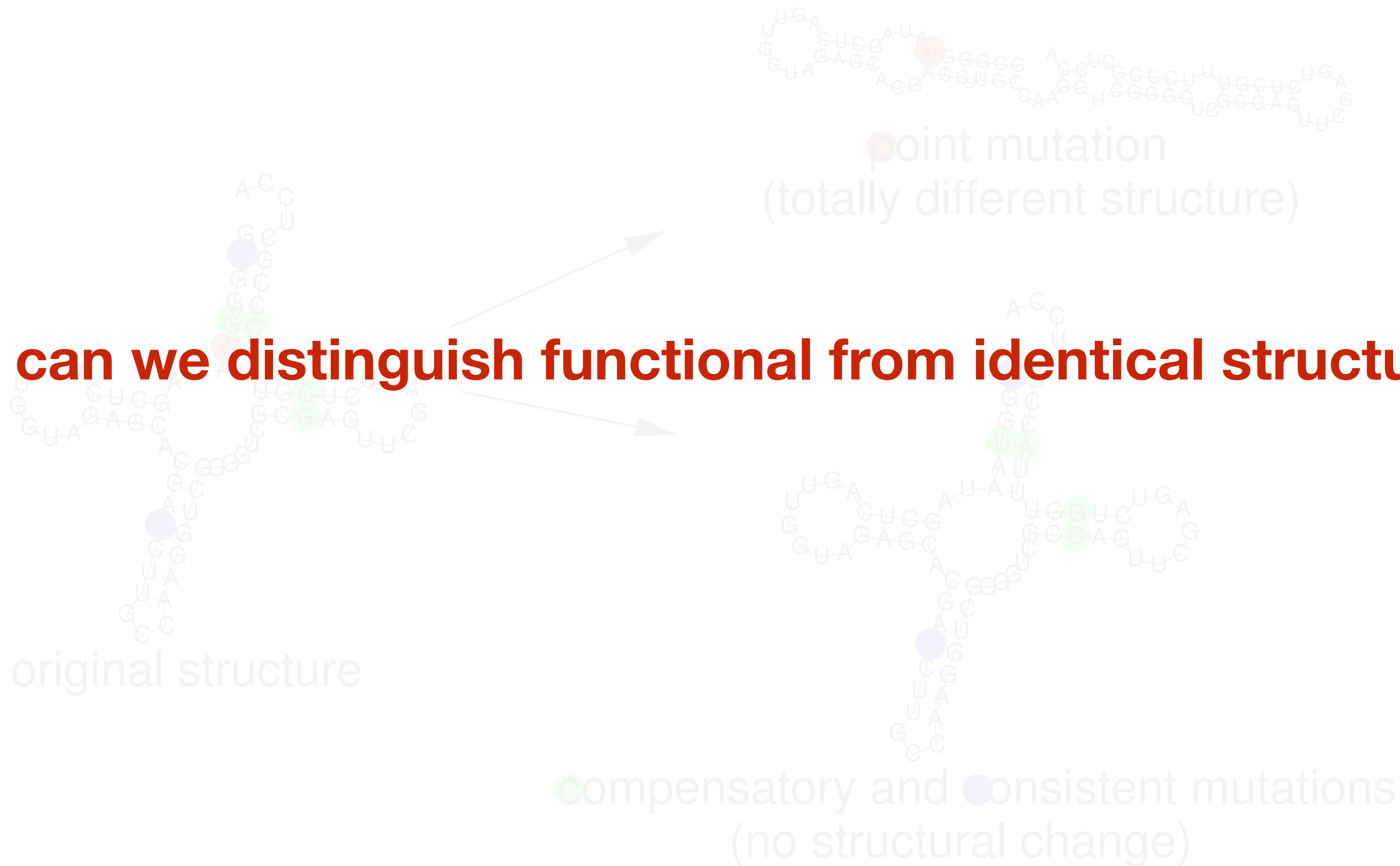
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Functional Structures: Point Mutations



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How can we distinguish functional from identical structures?



Consensus Structures: Alignment Folding

Combine covariance analysis and folding into one DP algorithm

- Apply conventional folding algorithm to alignment
- Use a modified energy function that includes covariance score

$$E_c(A, \Psi) = \sum_k E(A_k, \Psi) + cv \cdot \sum_{(i,j) \in \Psi} B_{ij}$$

- Can be used for all variants: MFE, partition function, ...
- Efficient: $\mathcal{O}(N \cdot n^2 + n^3)$ CPU and $\mathcal{O}(n^2)$ memory for alignment length n and N sequences
- Same results as RNAfold for single sequences

```
GGGCCGGUAGCUCAUUUAGGCAGAGCGUCUGACUCUUAACAGACGGUCGCGUGUUCGAAUCGCGUCCGGCCCA
GCCGAAAUAGCUCAGUU-GGGAGAGCGUUAGACUGAAGACUAAAGGUCCCCGGGUUCAAUCCCGGGUUUCGCCA
GCGGAUUUAGCUCAGUU-GGGAGAGCGCCAGACUGAAGAUUUGGAGGUCCUGUGUUCGAUCCACAGAAUUCGCA
CAGAGUGUAGCUUA---ACACAAAGCACCCAACUUACACUAGGAGAU-UUCAACU-UAACUUGACCGCUCUGA

(((((((..((((.....))))).((((.....))))).(((.....)))))))).((((.....)))))))).
mfe: -29 kcal/mol = (-21.3 + -7.7) kcal/mol
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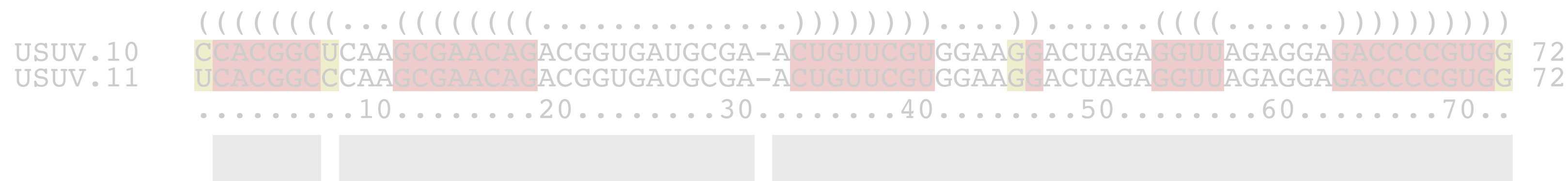
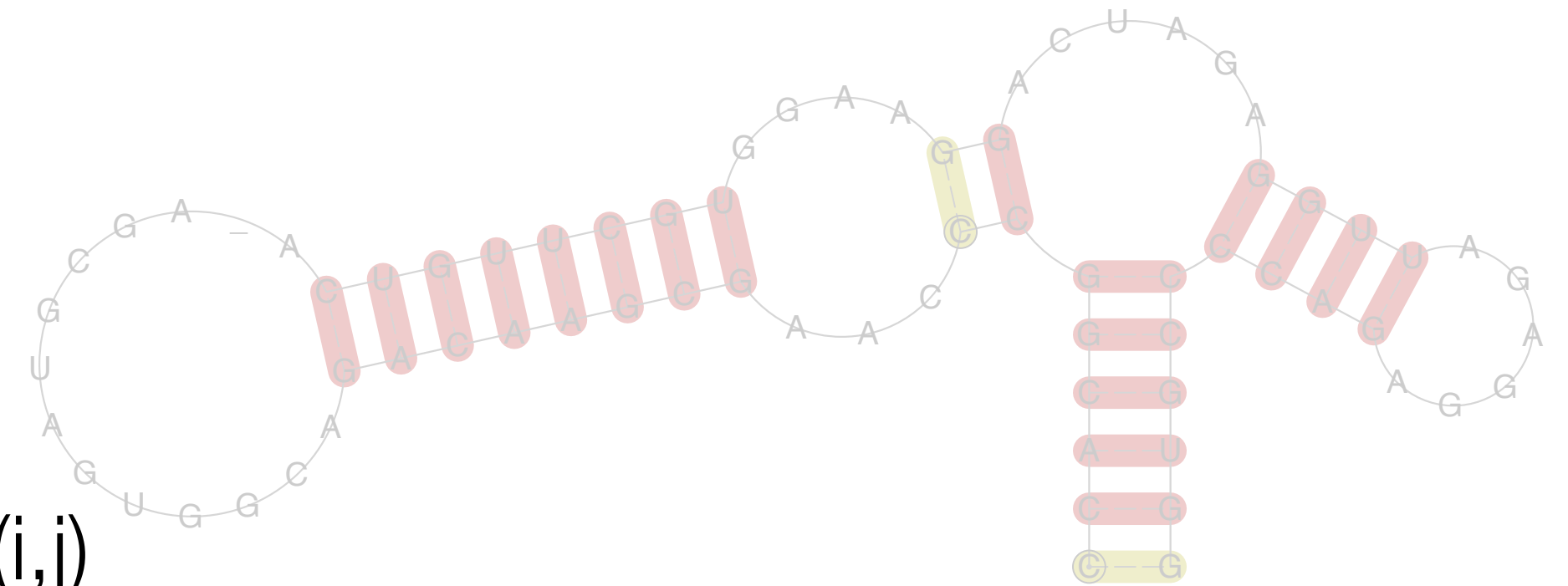
```
GGGCCGGUAAGCUCAUUUAGGCAGAGCGUCUGACUCUUAUUCAGACGGUCGCGUGUUCGAAUCGCGUCCGGCCCA
GCCGAAAUAGCUCAGUU-GGGAGAGCGUUAGACUGAAGAUCUAAAGGUCCCCGGUUCAAUCCCGGGUUUCGGCA
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RNA Covariation as Evolutionary Trait

- High mutation rate in RNA viruses due to error-prone RdRP
- For base pair (i,j): GC/CG/AU/UA/GU/UG
- Consistent mutation: different standard combinations
- Compensatory mutation: both positions are mutated
- Presence of both strongly supports predicted base pair (i,j)



types of pairs:

	1	2	3	4	5	6
0						
1						
2						

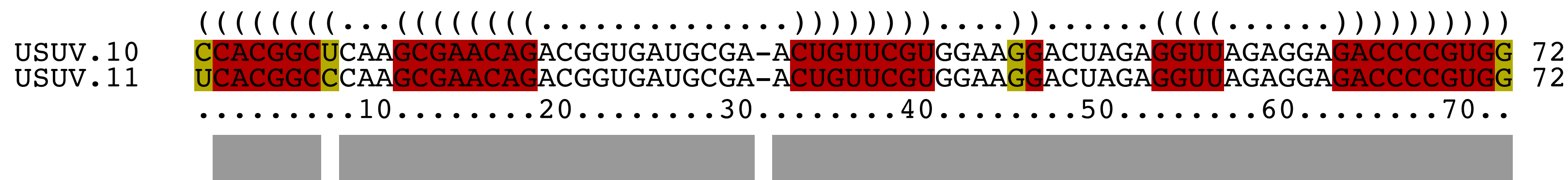
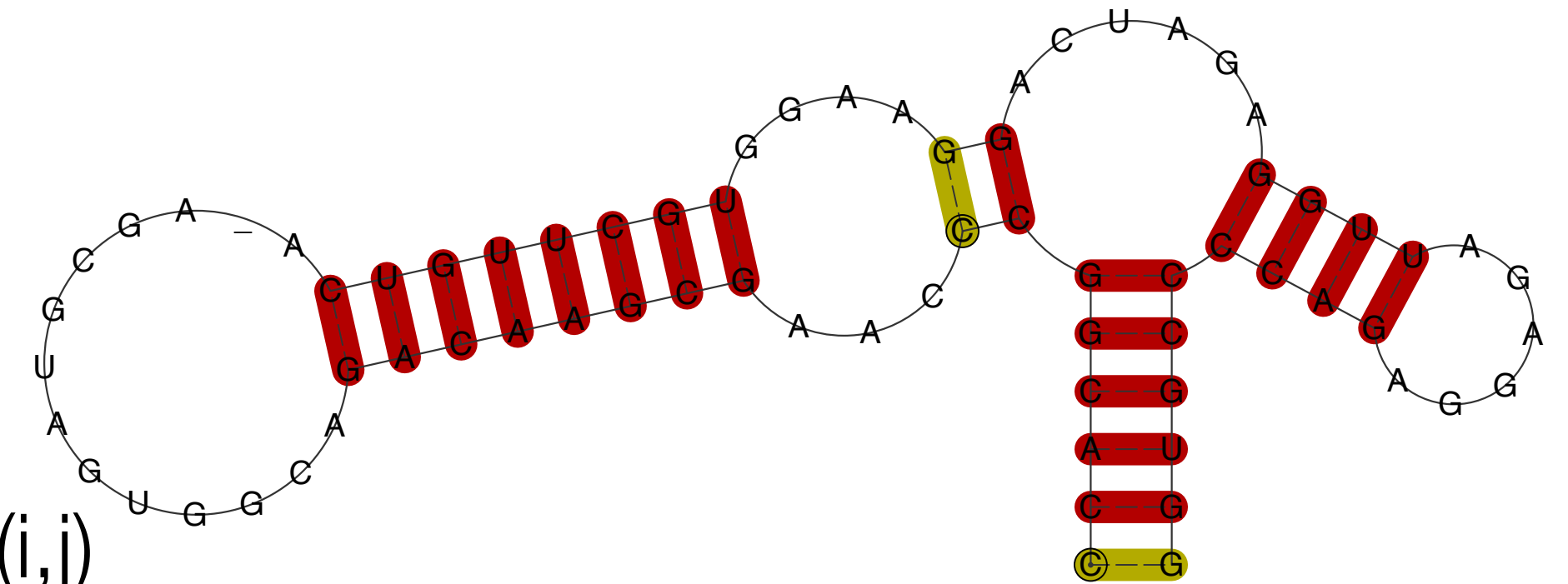
incompatible pairs in column

allowed pairs:

A-U	G-C	G-U
U-A	C-G	U-G

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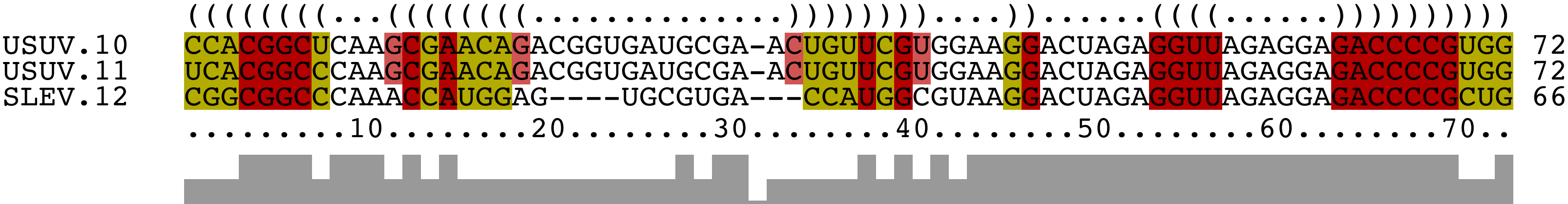
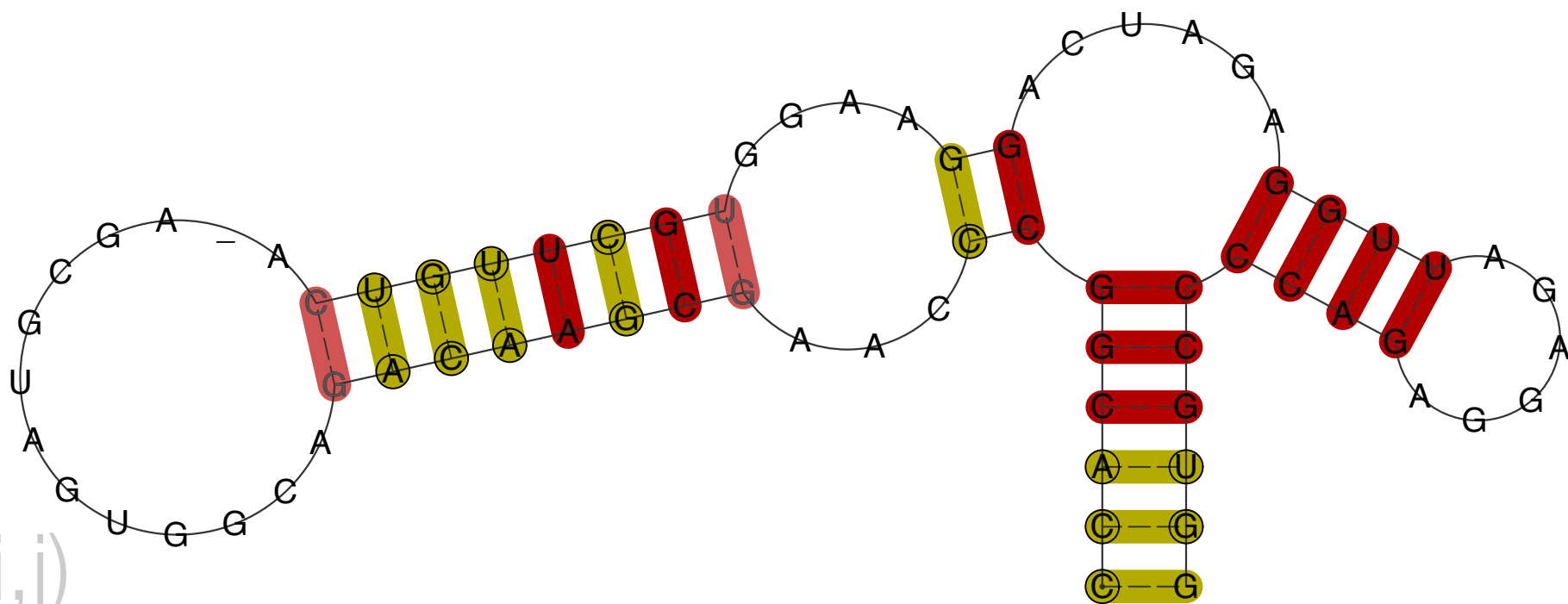
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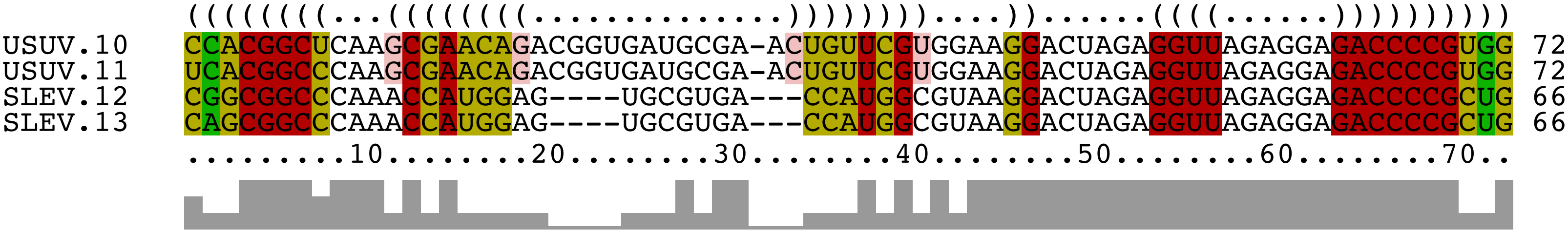
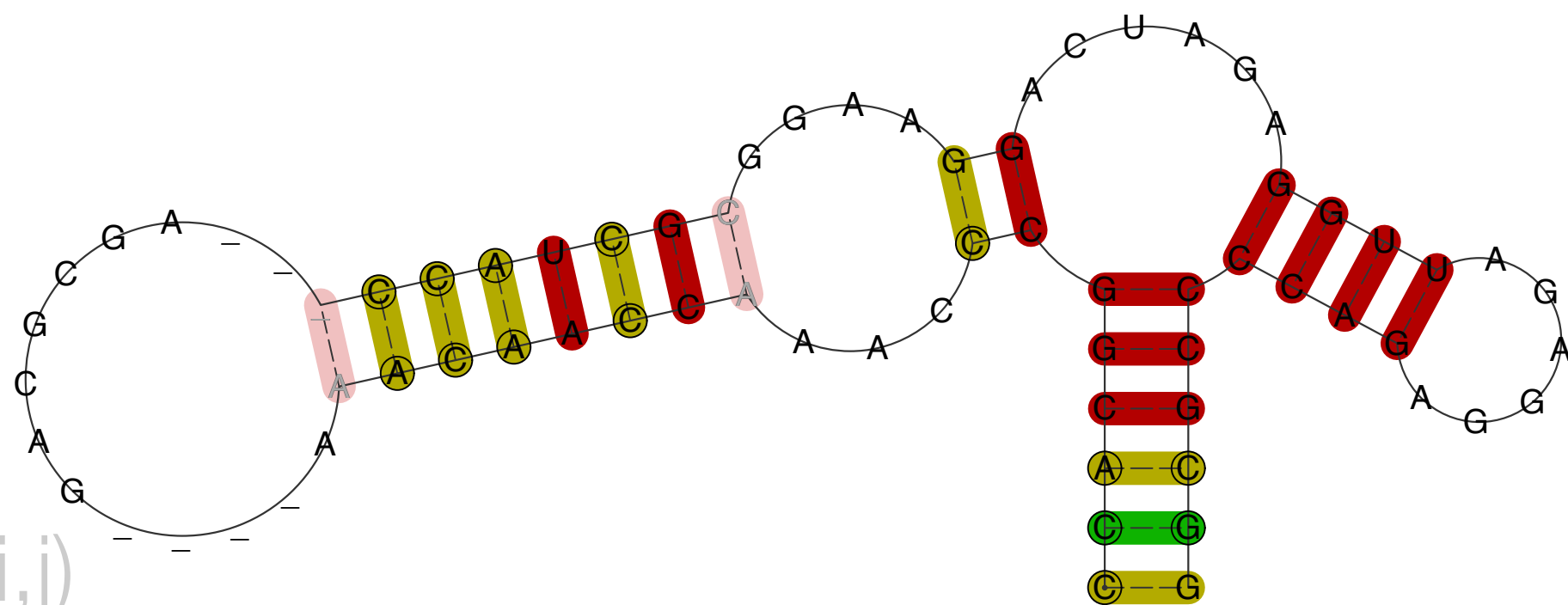
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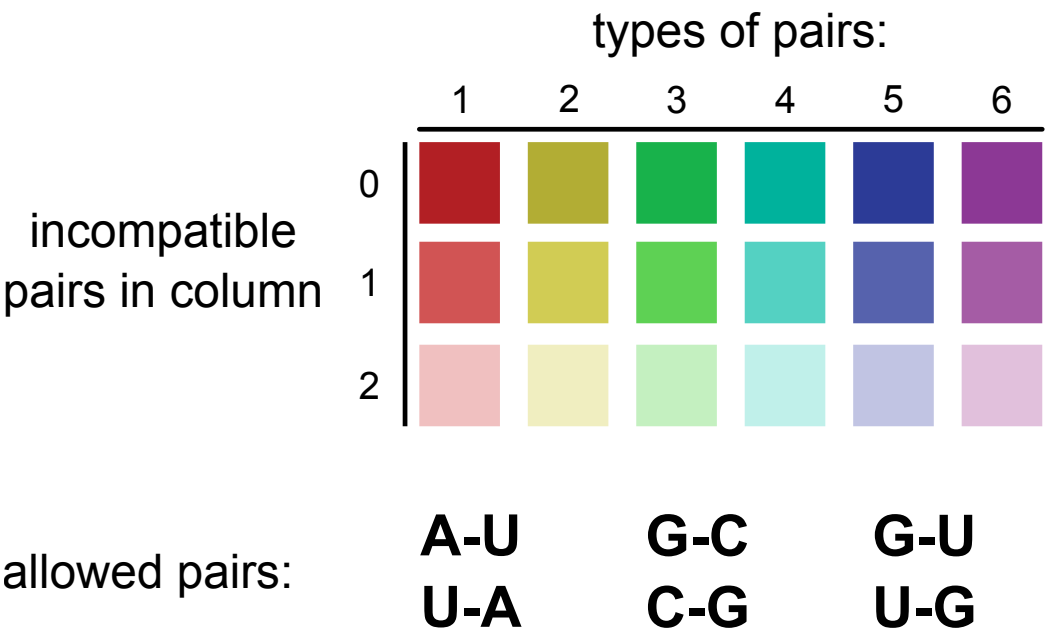
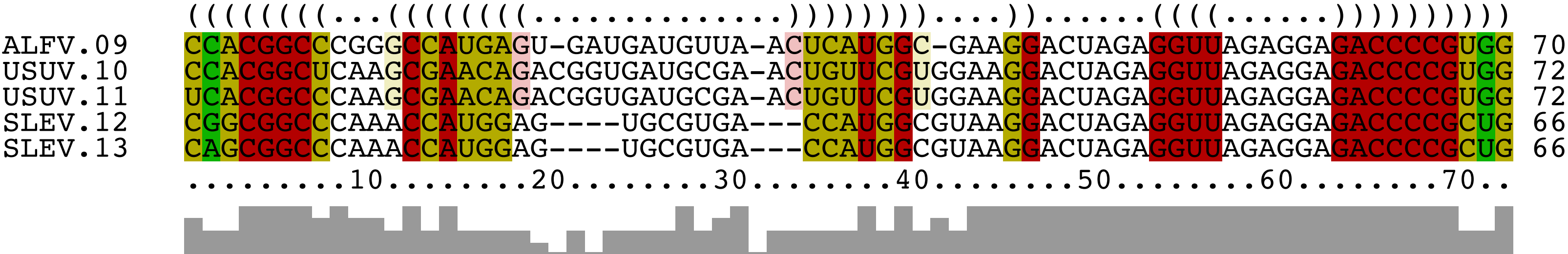
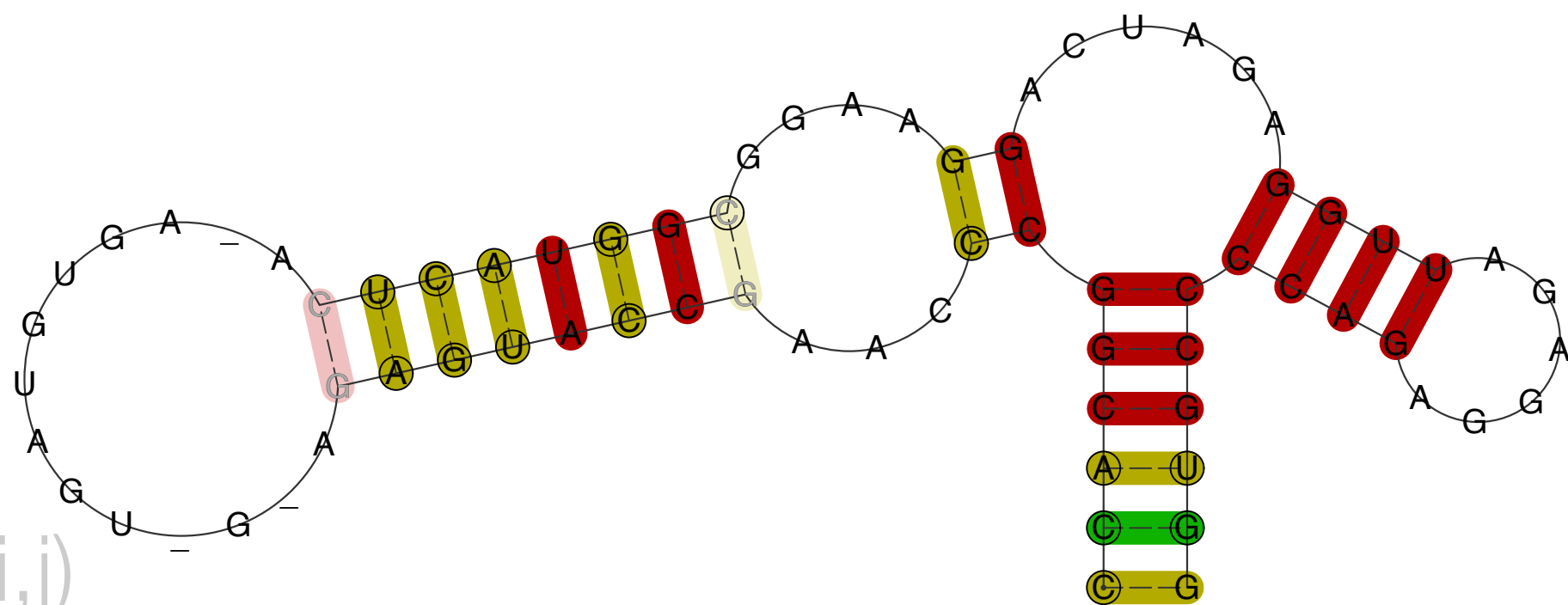
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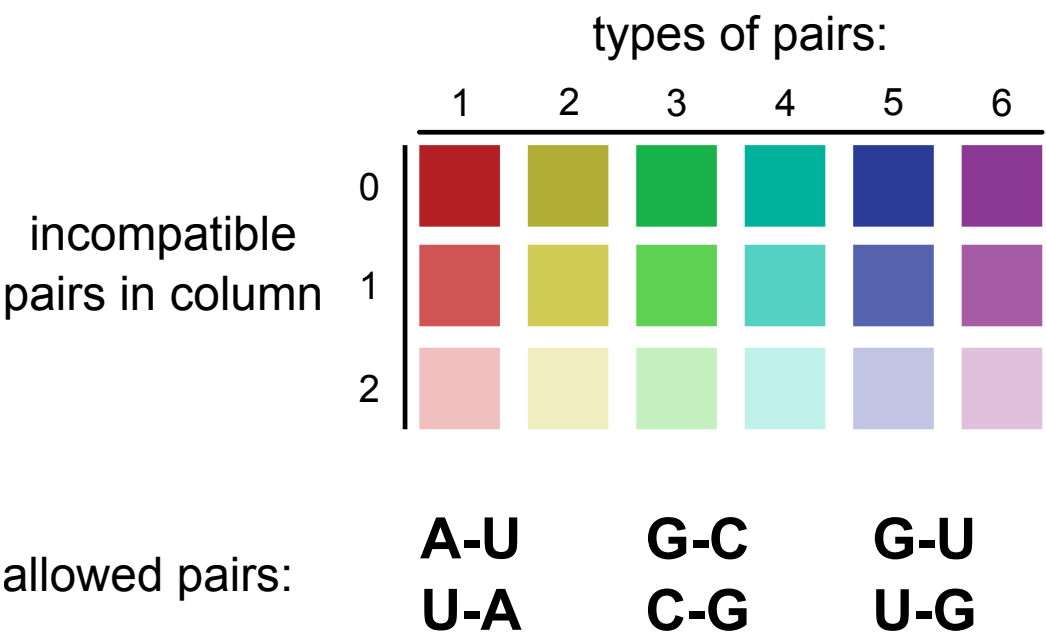
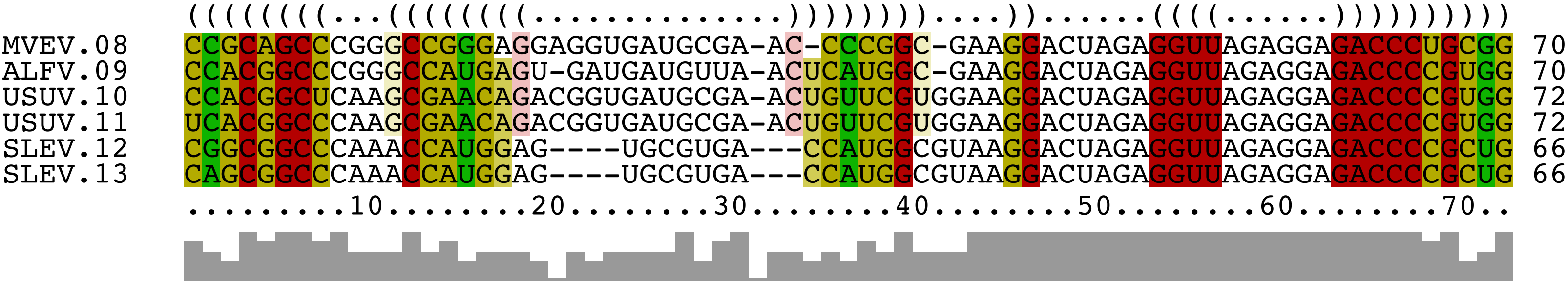
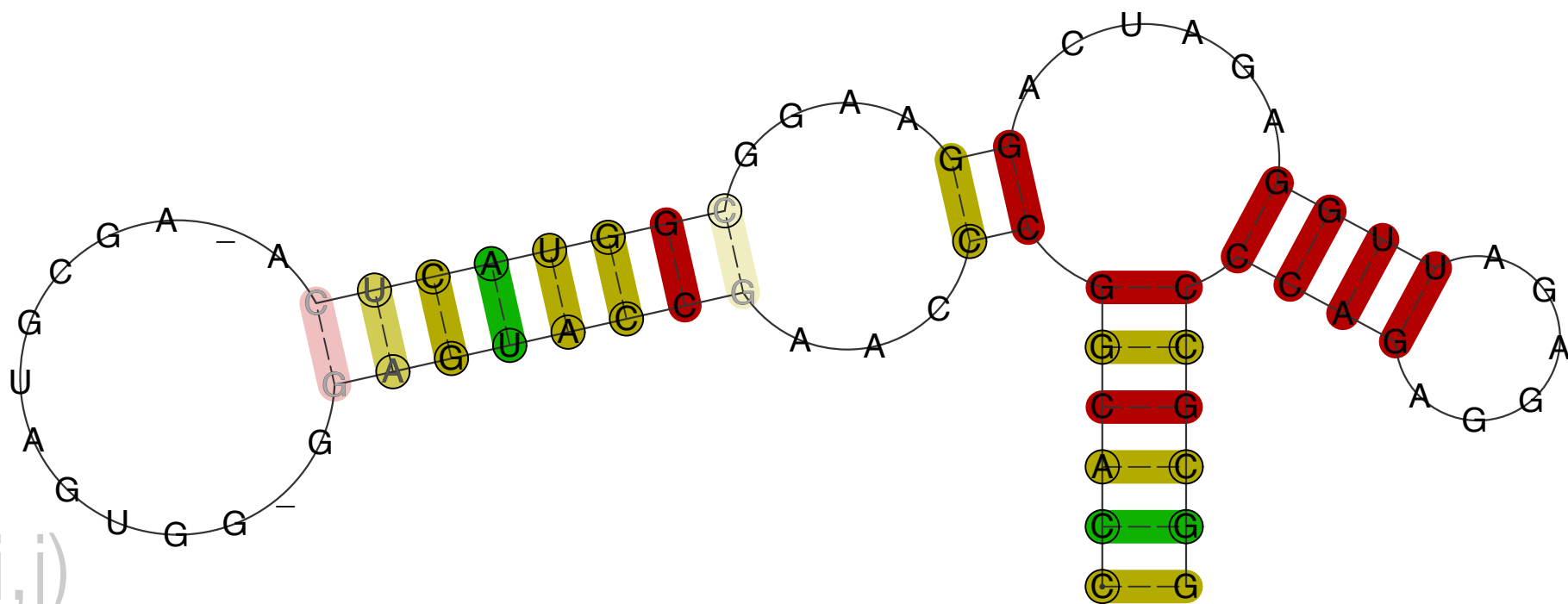
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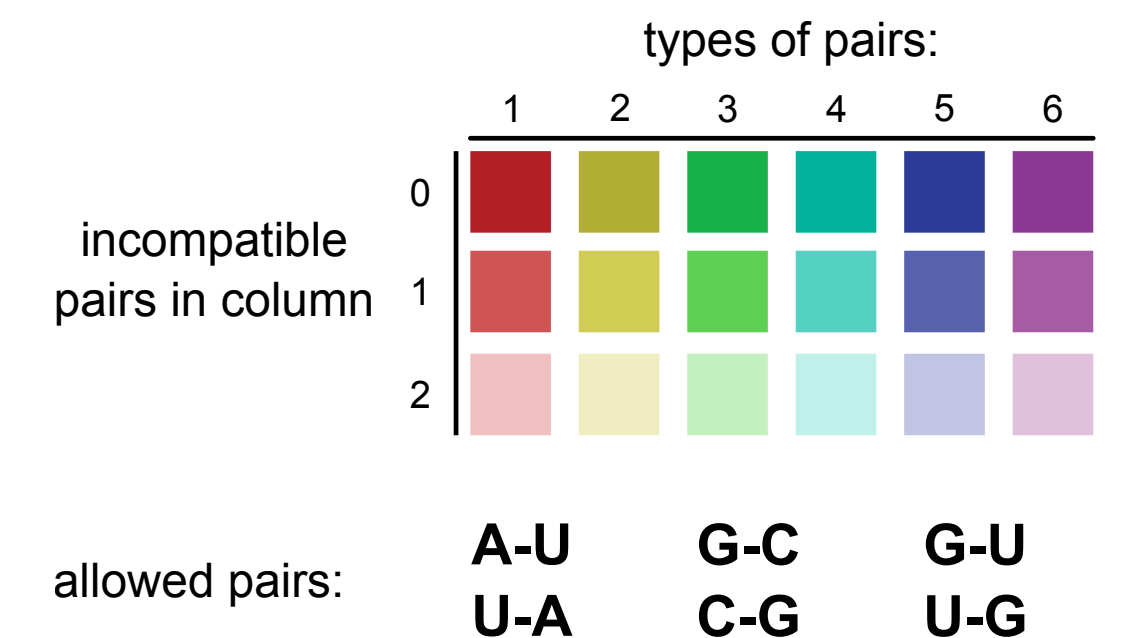
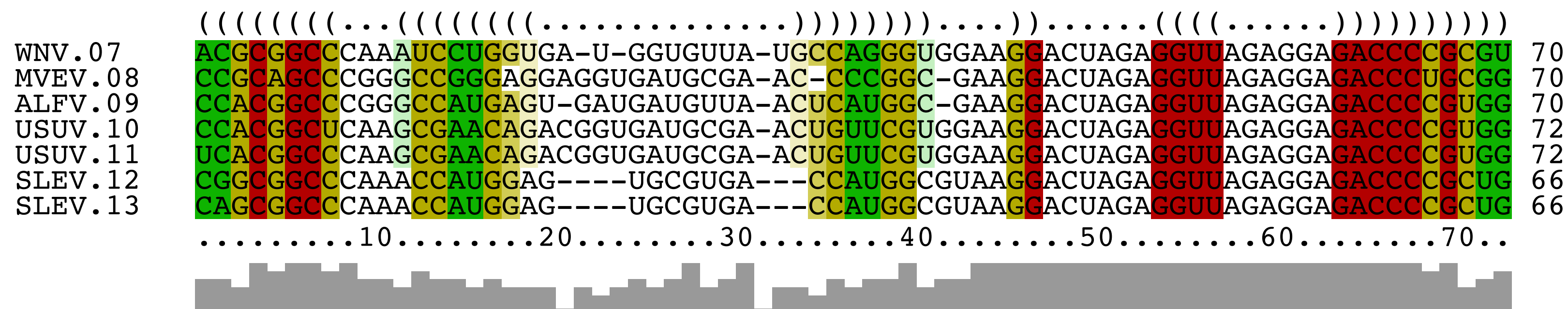
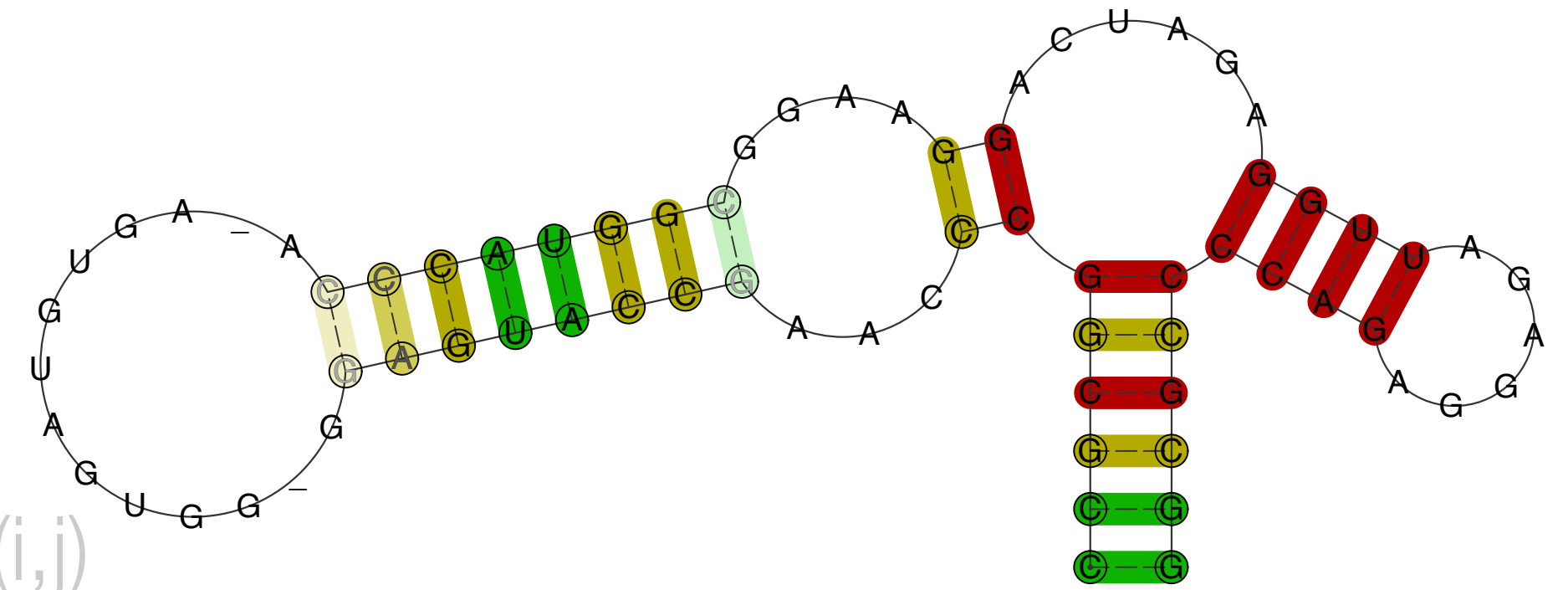


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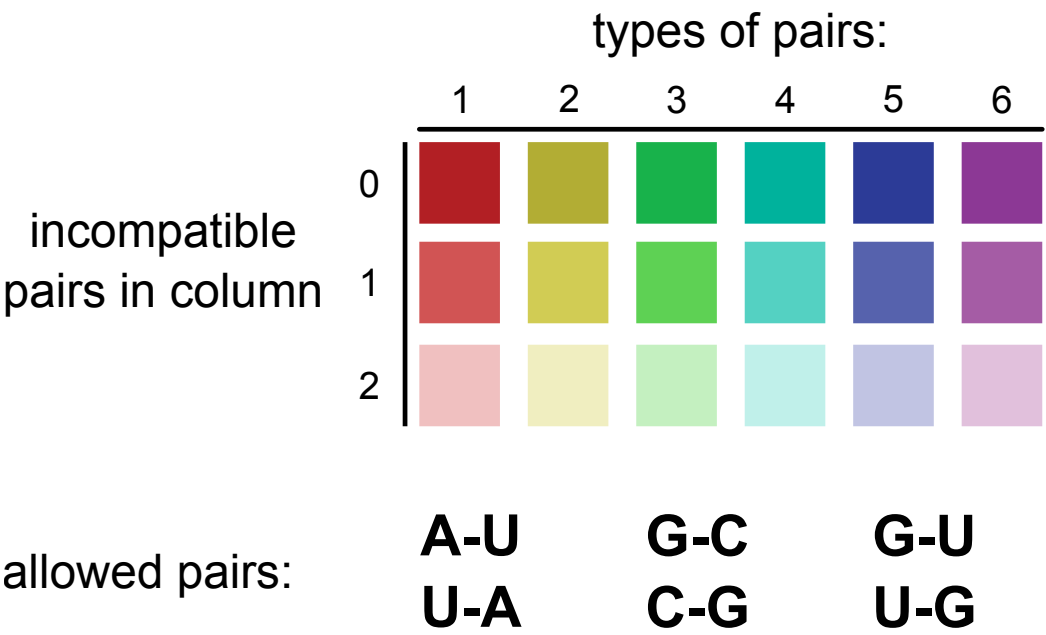
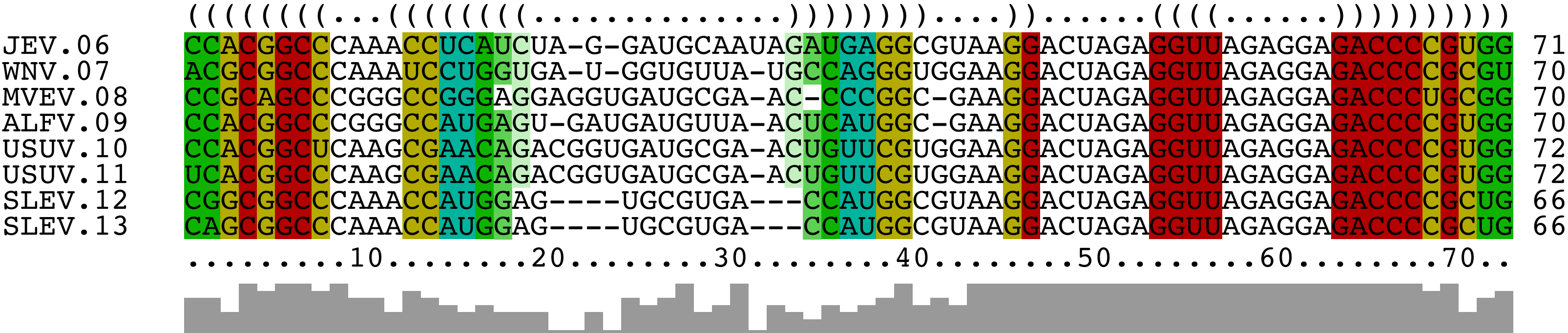
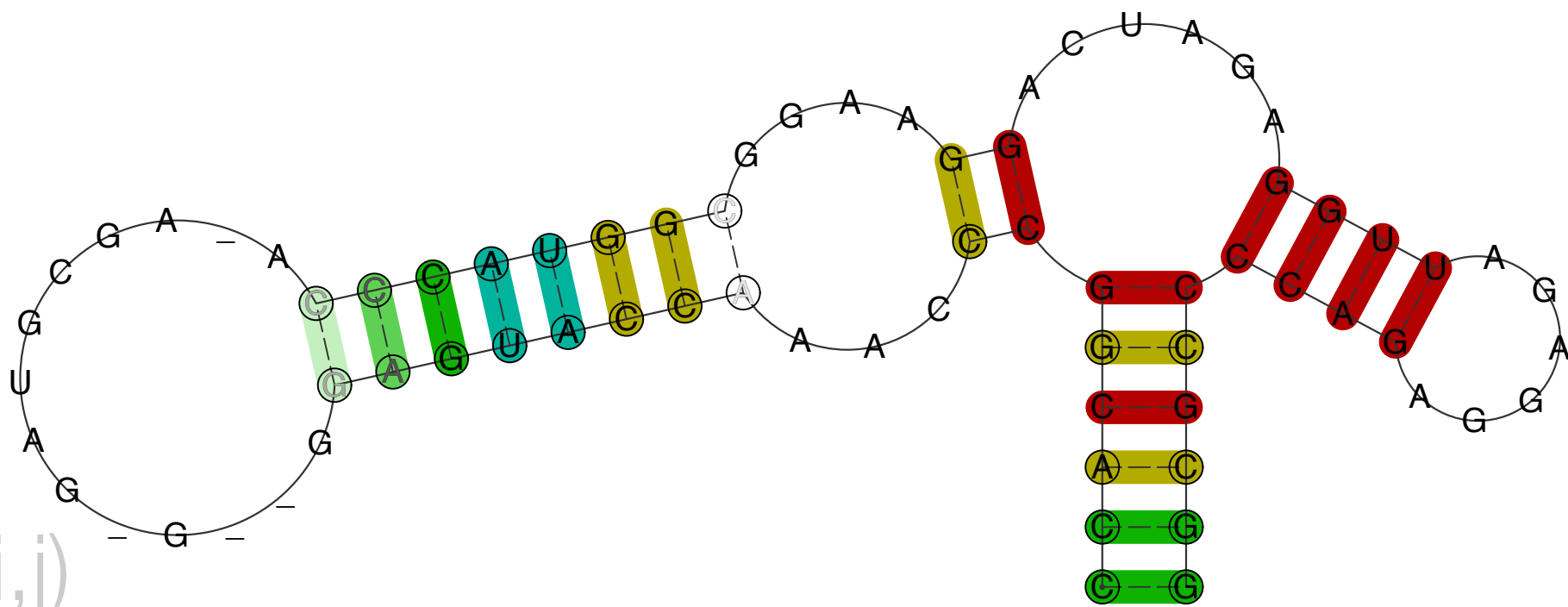


RNA Covariation as Evolutionary Trait

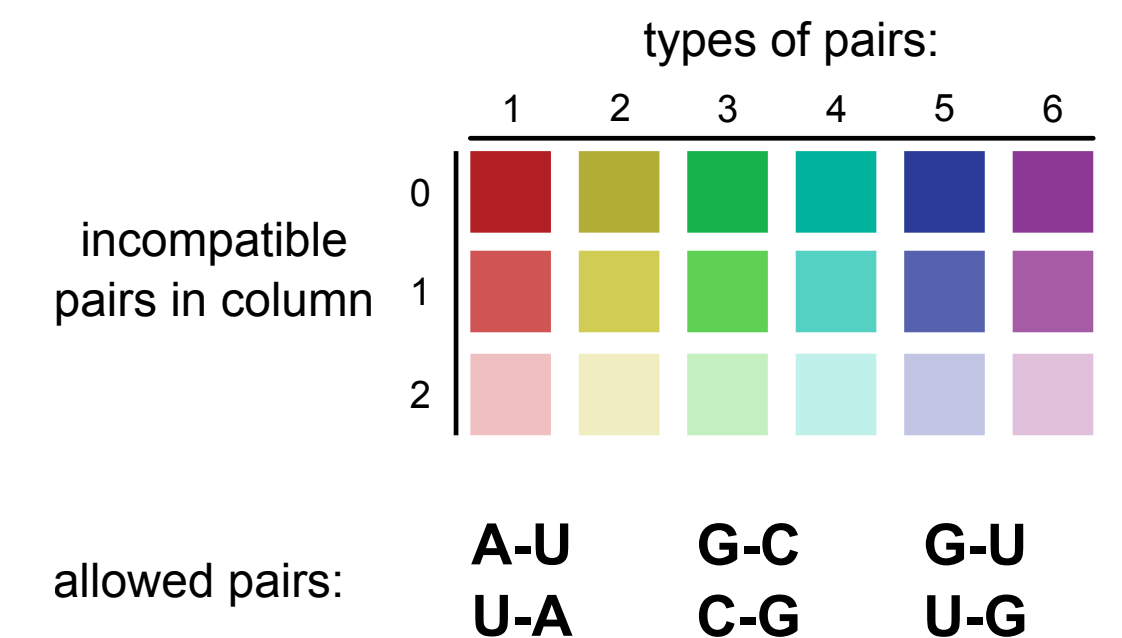
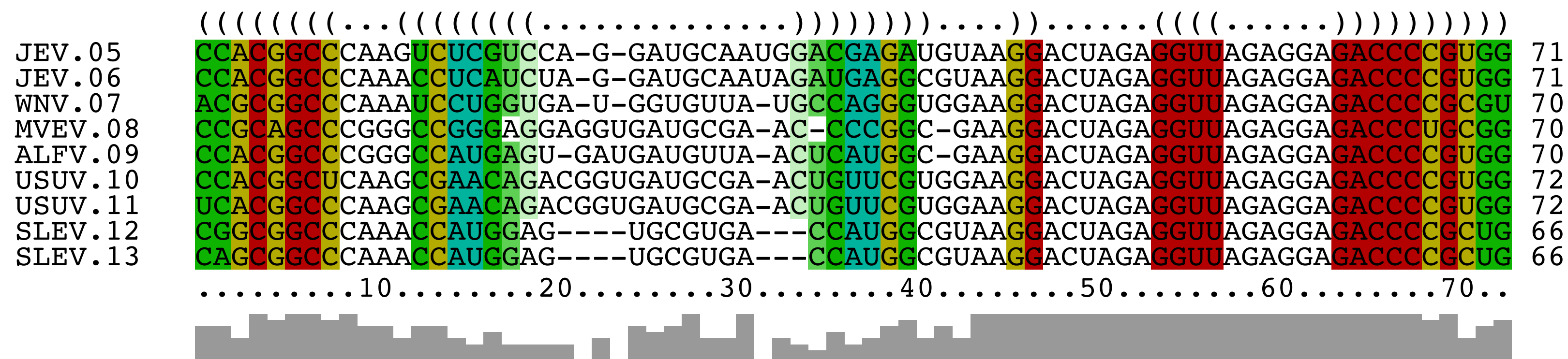
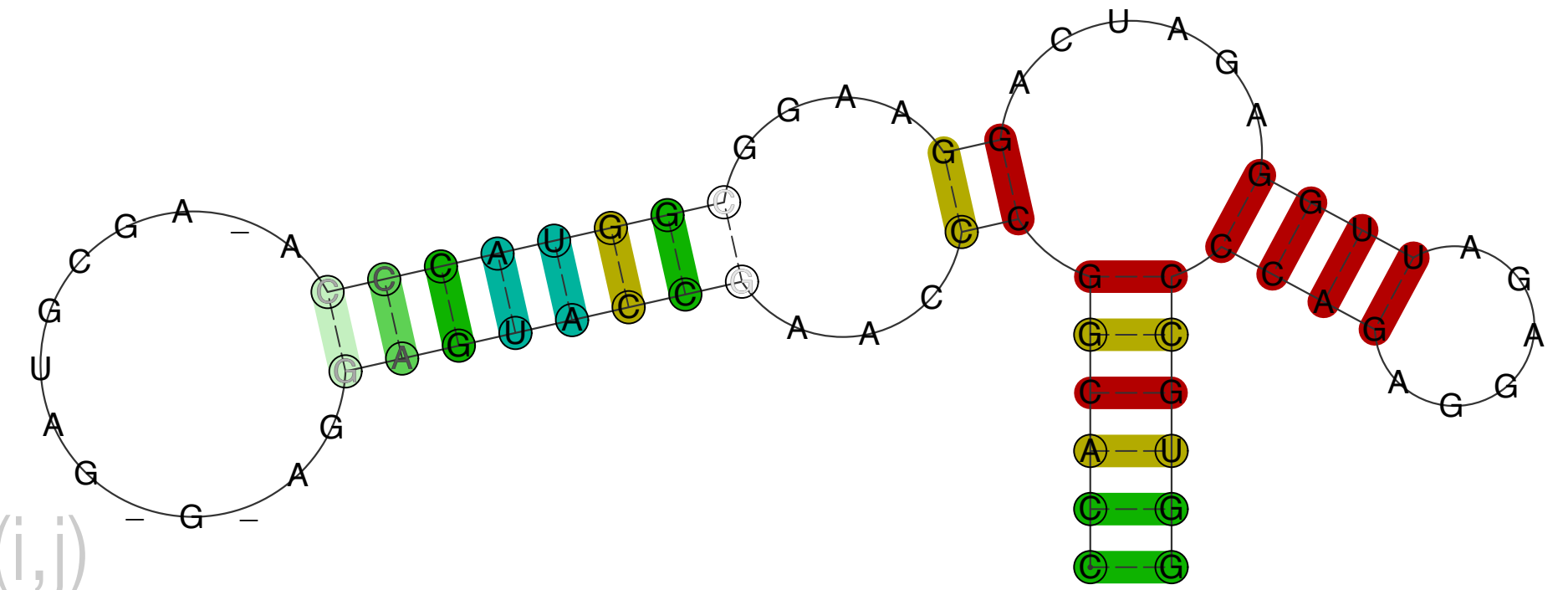


RNA Covariation as Evolutionary Trait

- High mutation rate in RNA viruses due to error-prone RdRP
- For base pair (i,j): GC/CG/AU/UA/GU/UG
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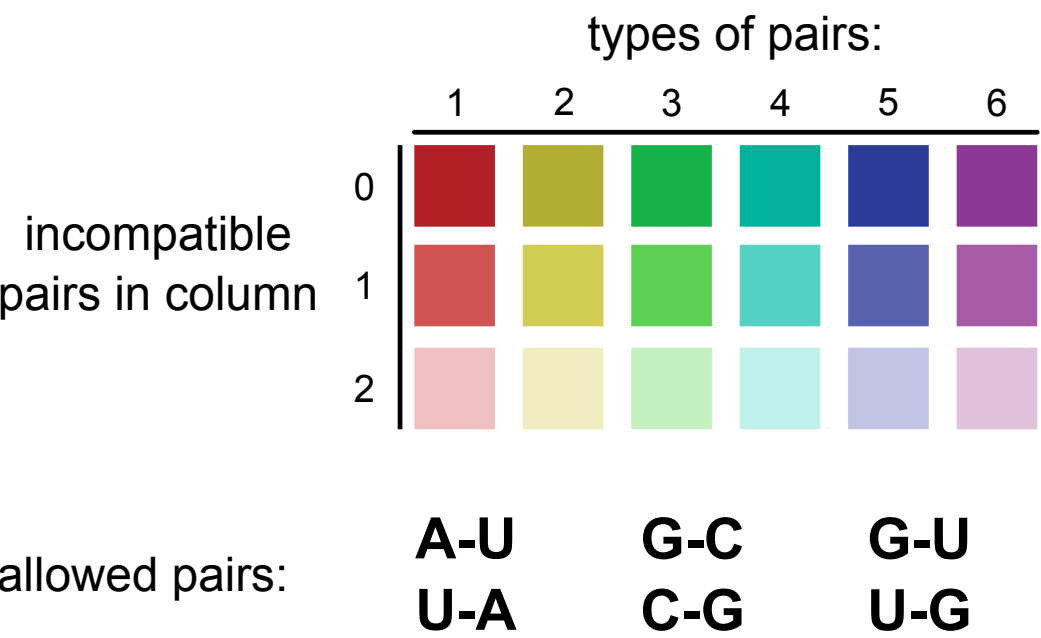
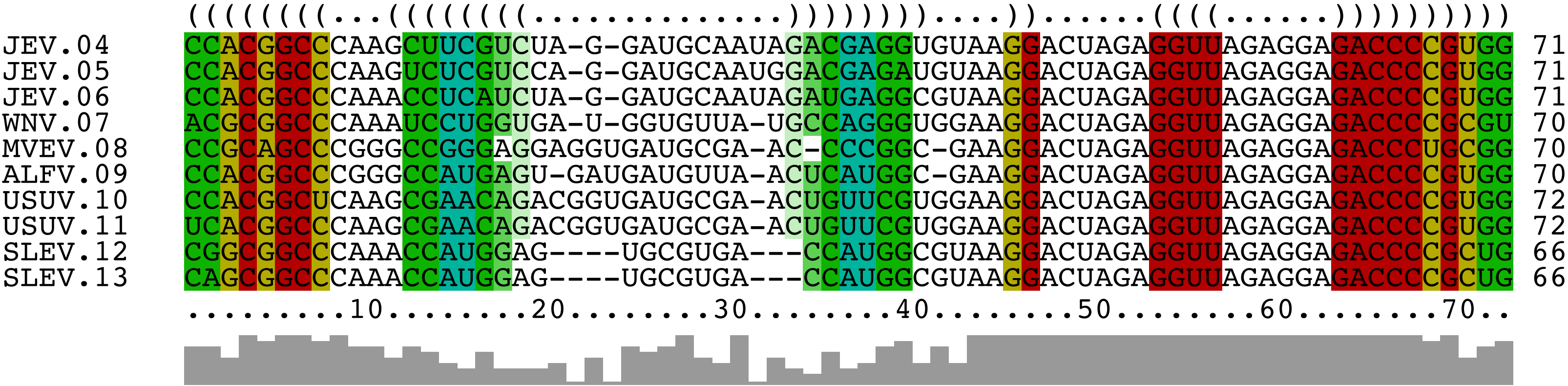
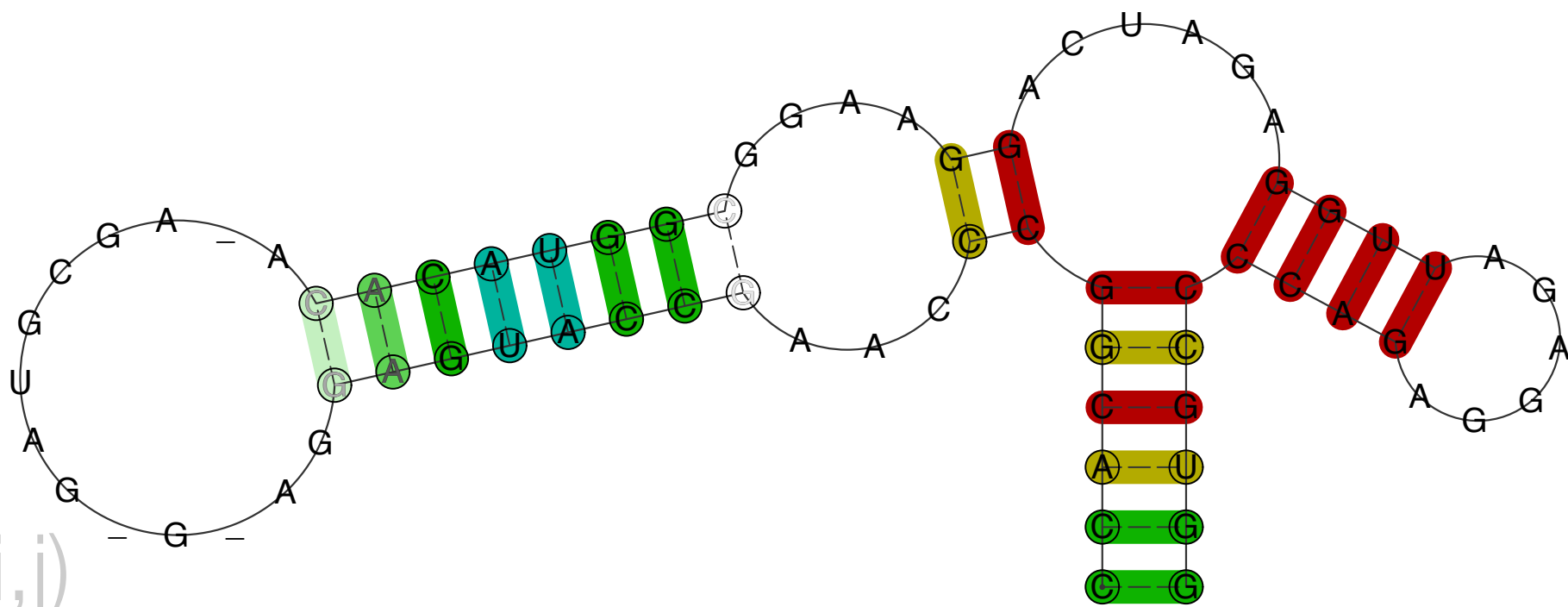


RNA Covariation as Evolutionary Trait

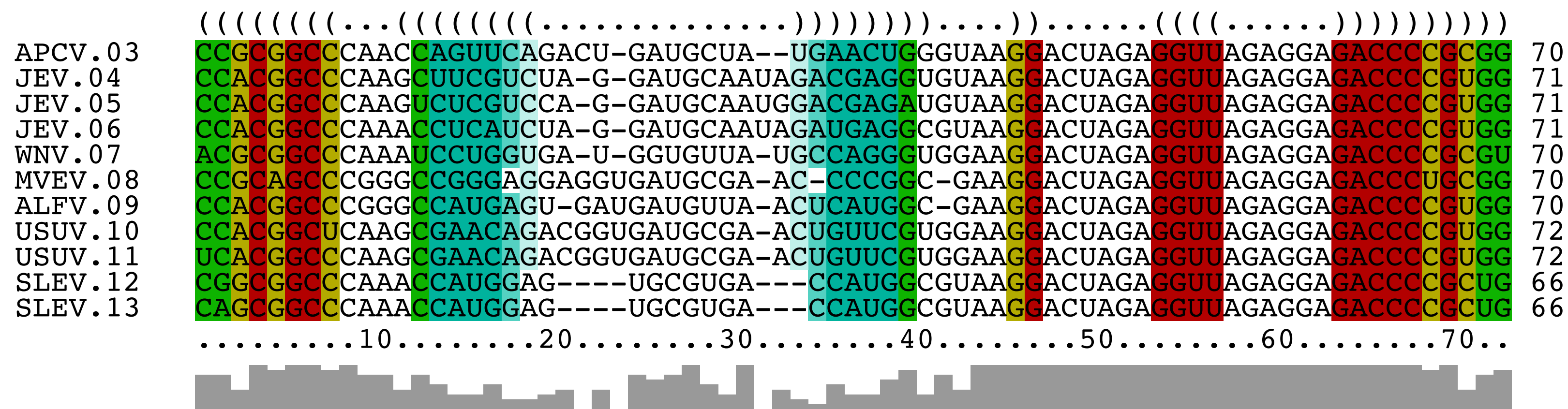
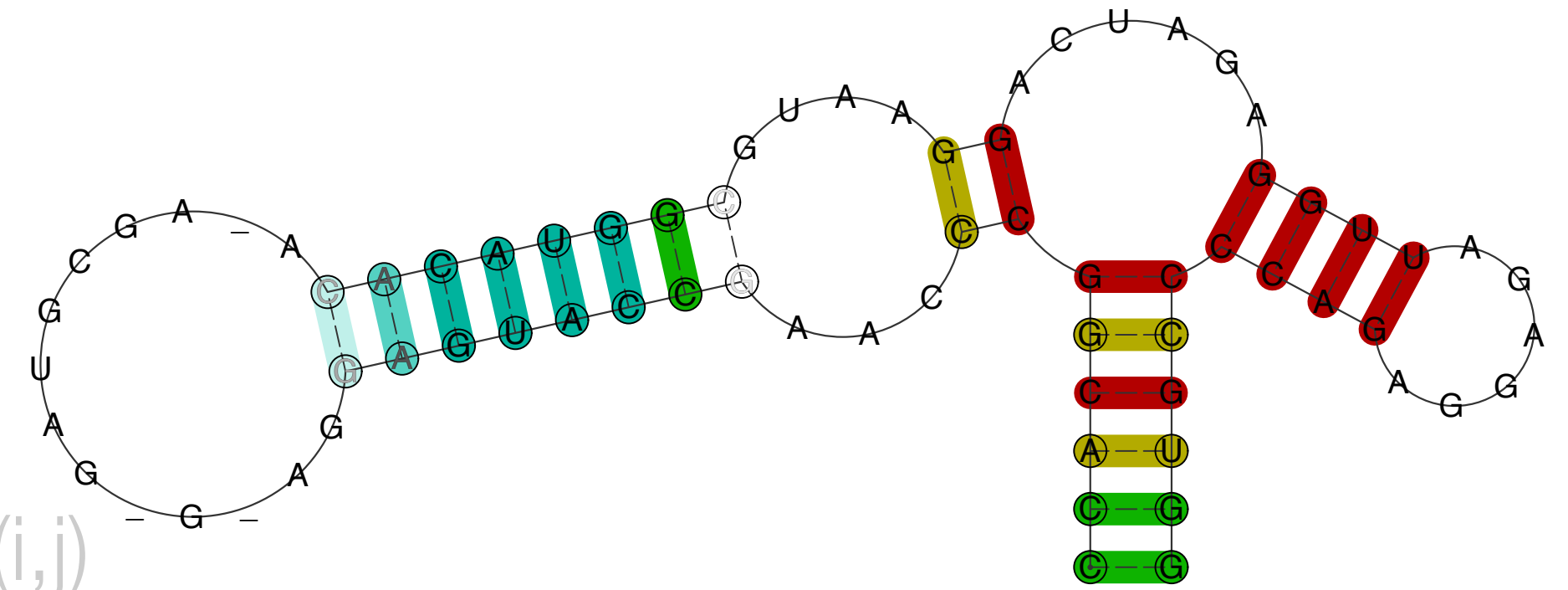


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RNA Covariation as Evolutionary Trait



types of pairs:

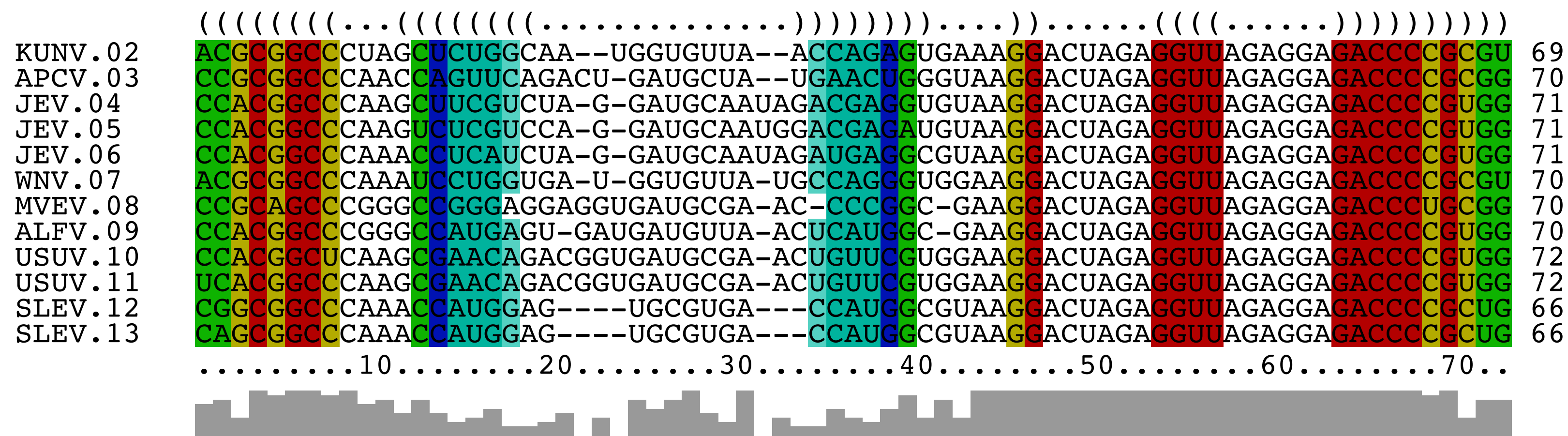
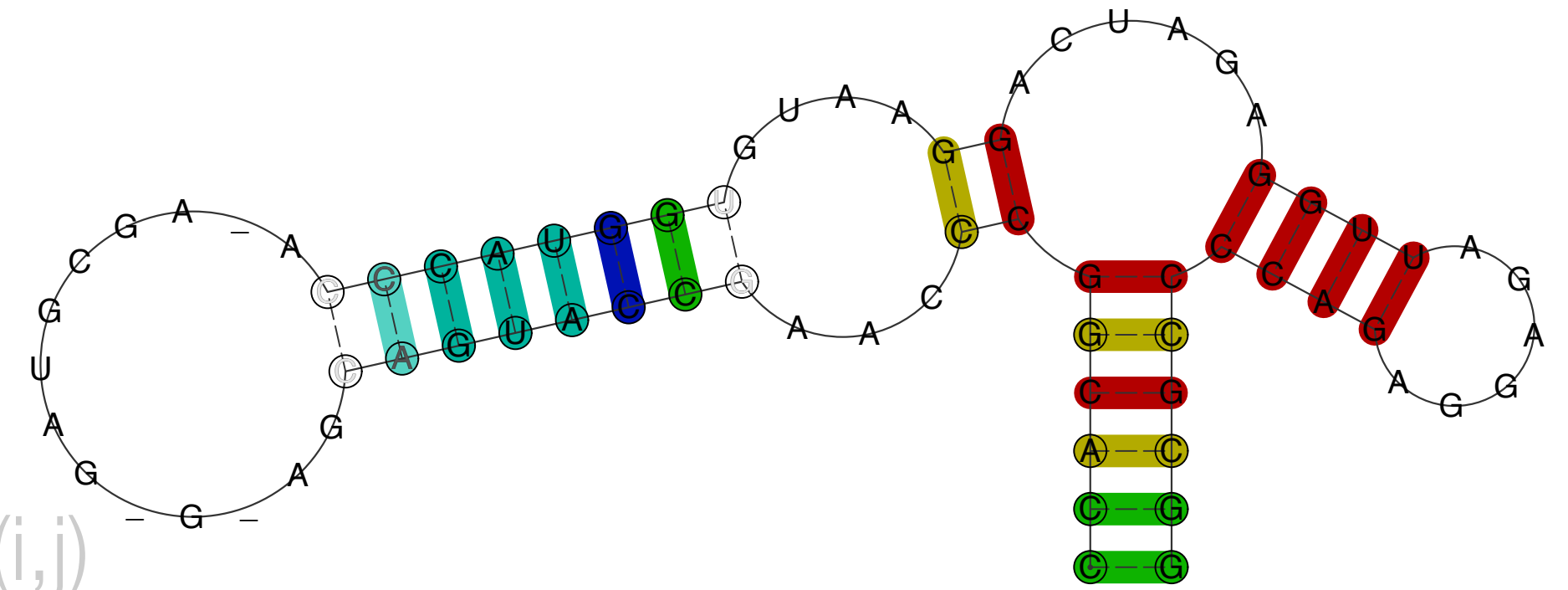
	1	2	3	4	5	6
0						
1						
2						

incompatible pairs in column

allowed pairs:

A-U	G-C	G-U
U-A	C-G	U-G

RNA Covariation as Evolutionary Trait



types of pairs:

	1	2	3	4	5	6
0						
1						
2						

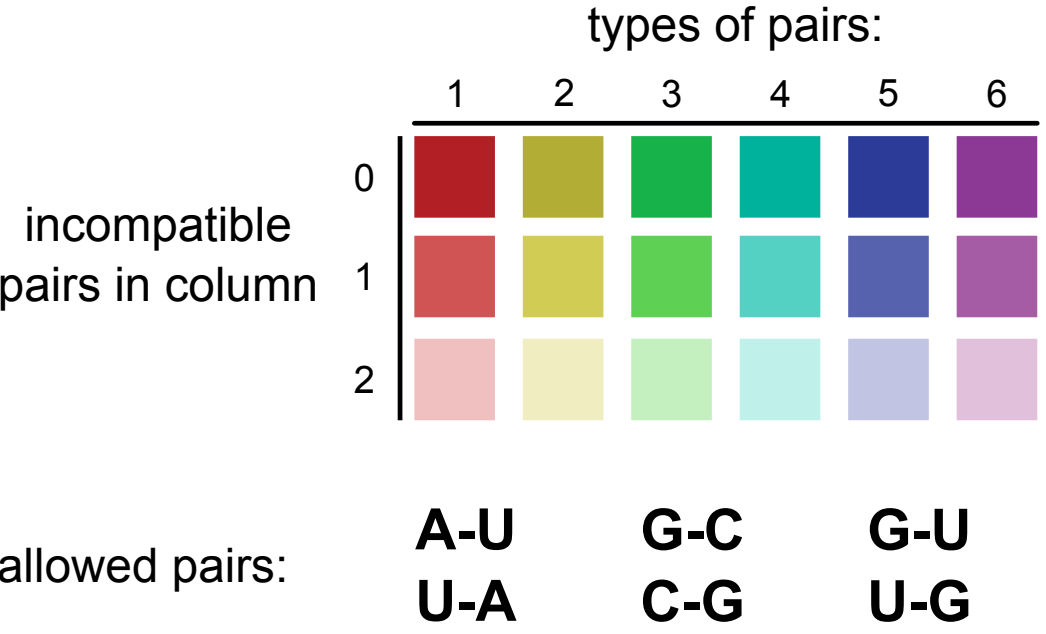
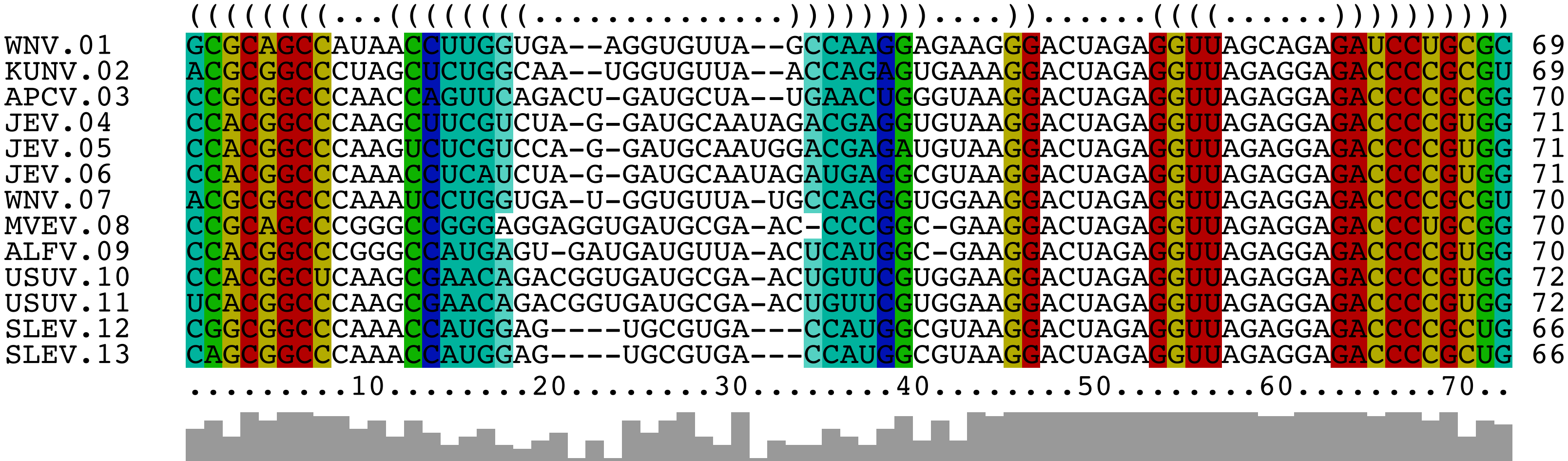
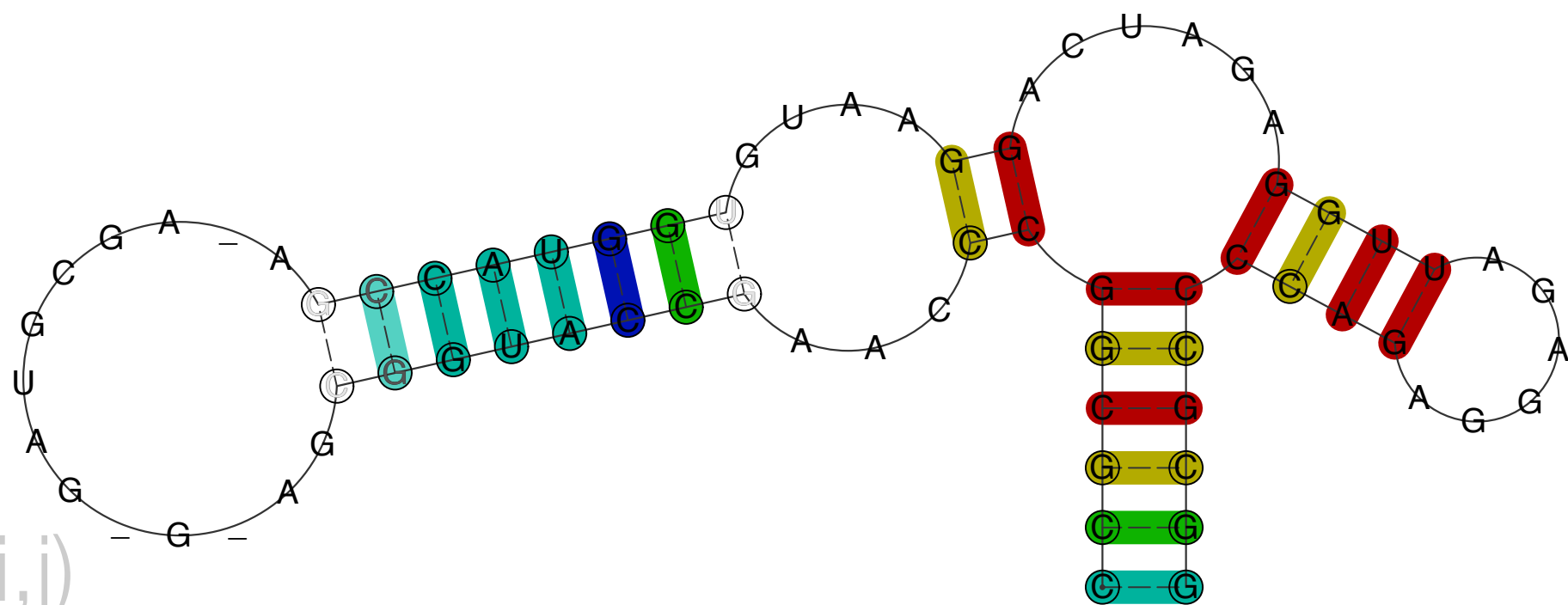
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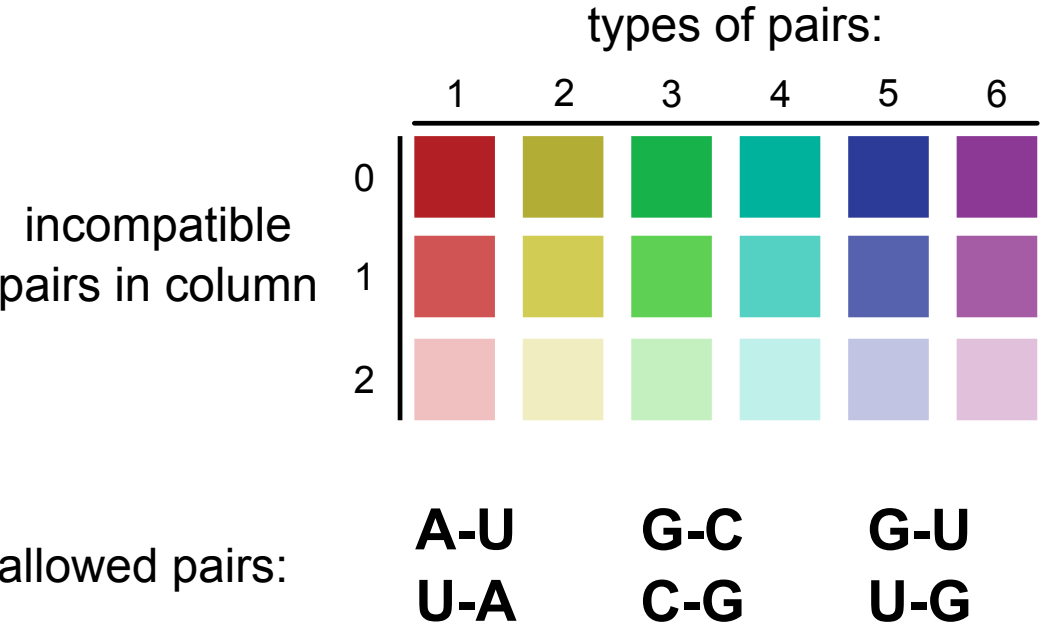
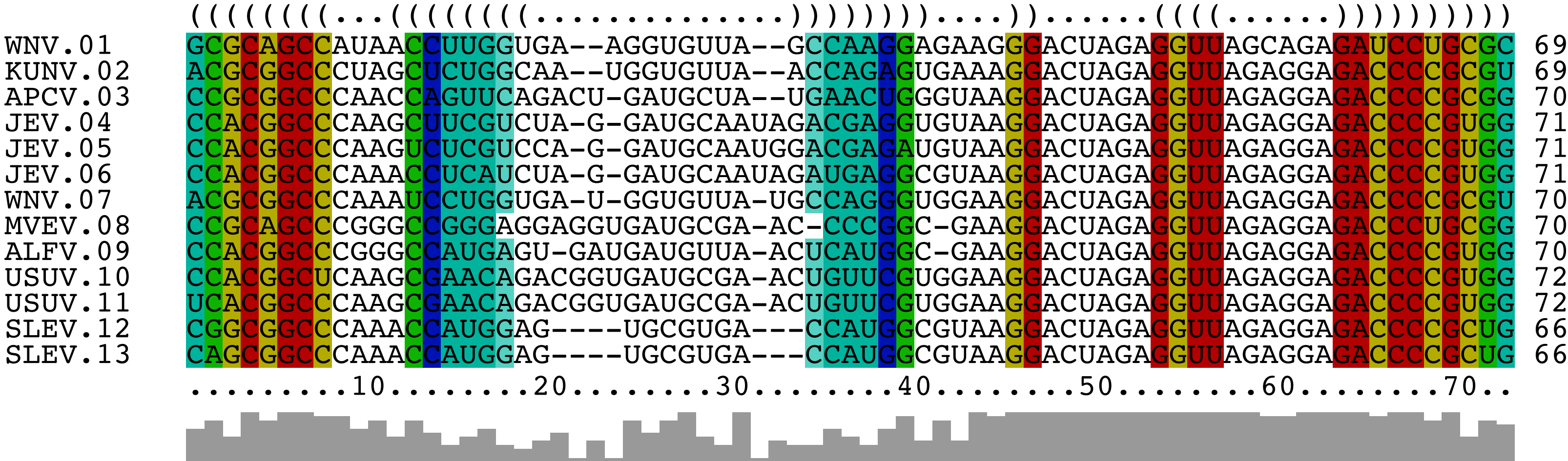
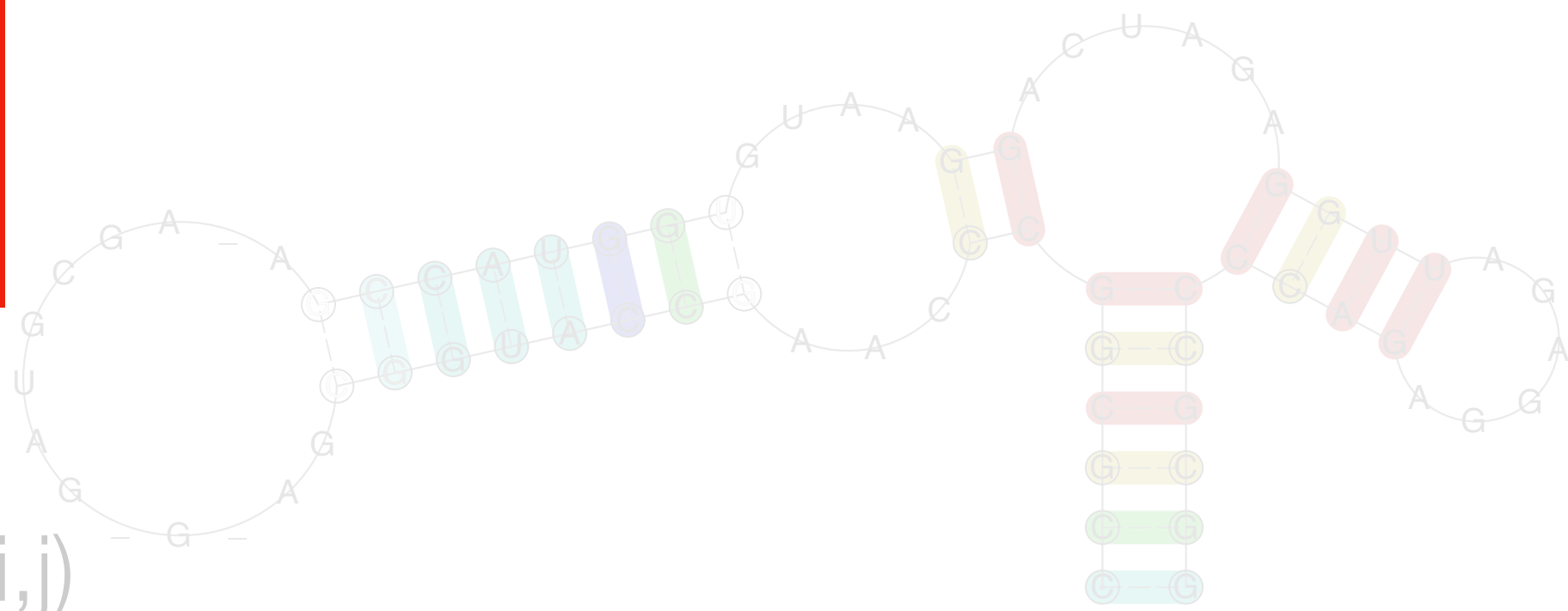
RNA Covariation as Evolutionary Trait

- High mutation rate in RNA viruses due to error-prone RdRP

- **We use structural RNA alignments and Covariance Models to find conserved elements**

- Compensatory mutation: both positions are mutated

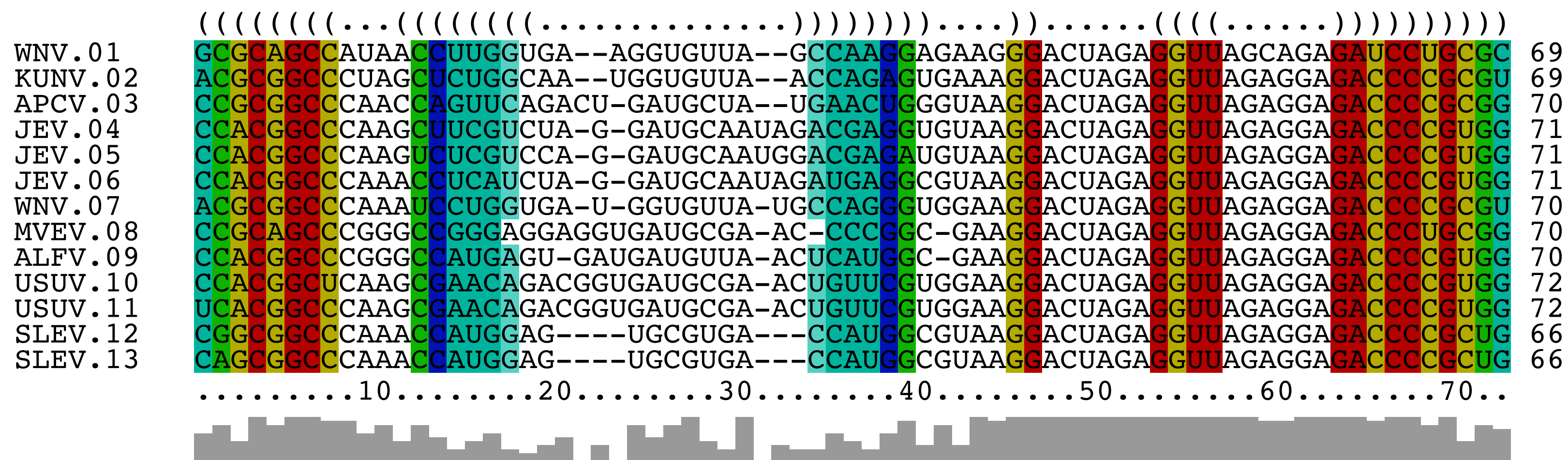
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RNA Covariation as Evolutionary Trait

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	1	2	3	4	5	6
0						
1						
2						

incompatible pairs in column

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A-U	G-C	G-U
U-A	C-G	U-G

Part II:

Conserved RNA structures in flaviviruses

Flaviviruses as Global Health Threat

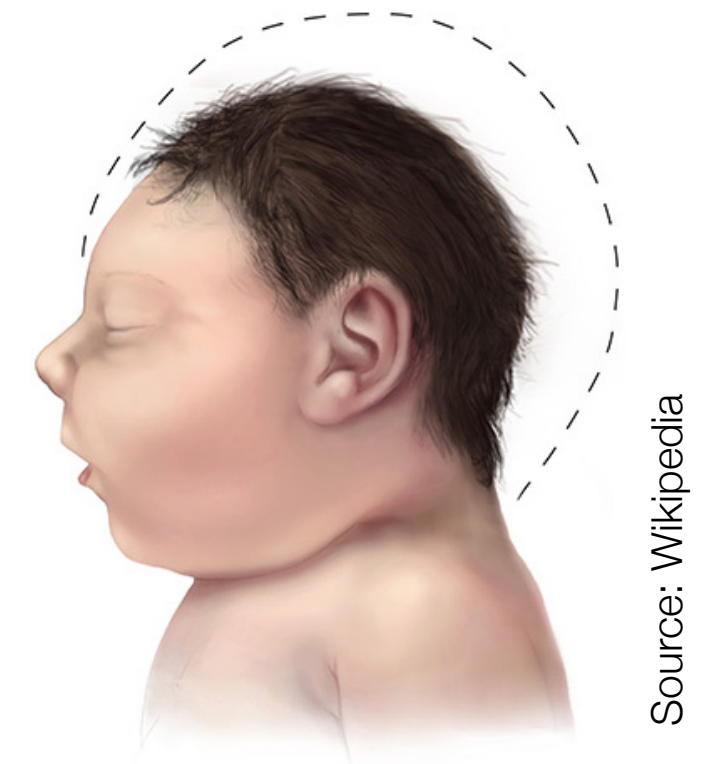
- Family *Flaviviridae* / genus *Flavivirus*; arthropod-borne (mainly mosquitoes & ticks)
- FV include (re-)emerging human pathogens like YFV, DENV, JEV, WNV, TBEV
- Clinical manifestations: headache, rash, (hemorrhagic) fever, meningitis, encephalitis
- FV neurotropism reported for YFV, WNV, TBEV, DENV, ZIKV
- ZIKV outbreak in the Americas 2015-2017
- Congenital neurotropism: high increase in microcephaly cases in newborns



Source: Wikipedia

Flaviviruses as Global Health Threat

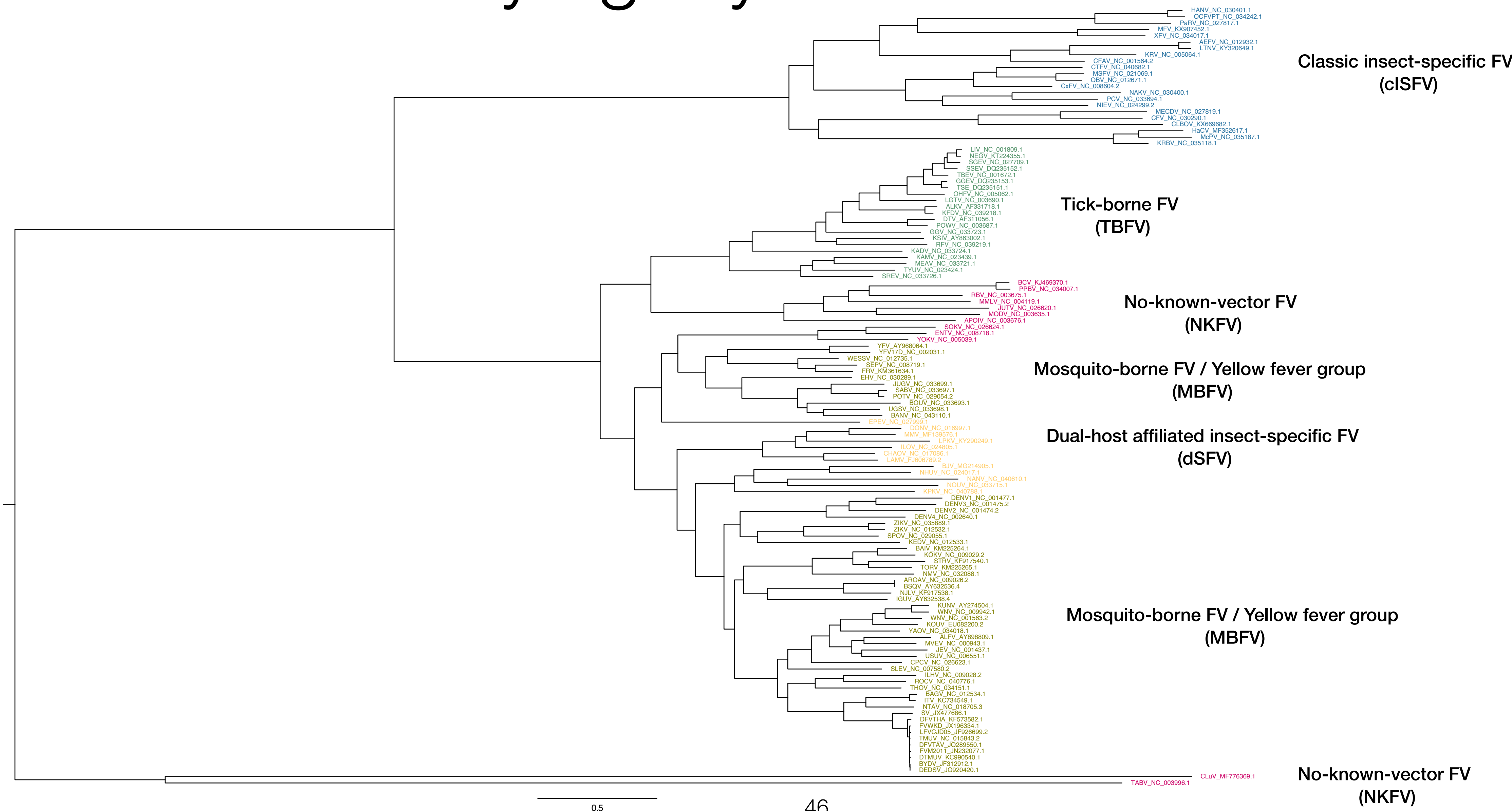
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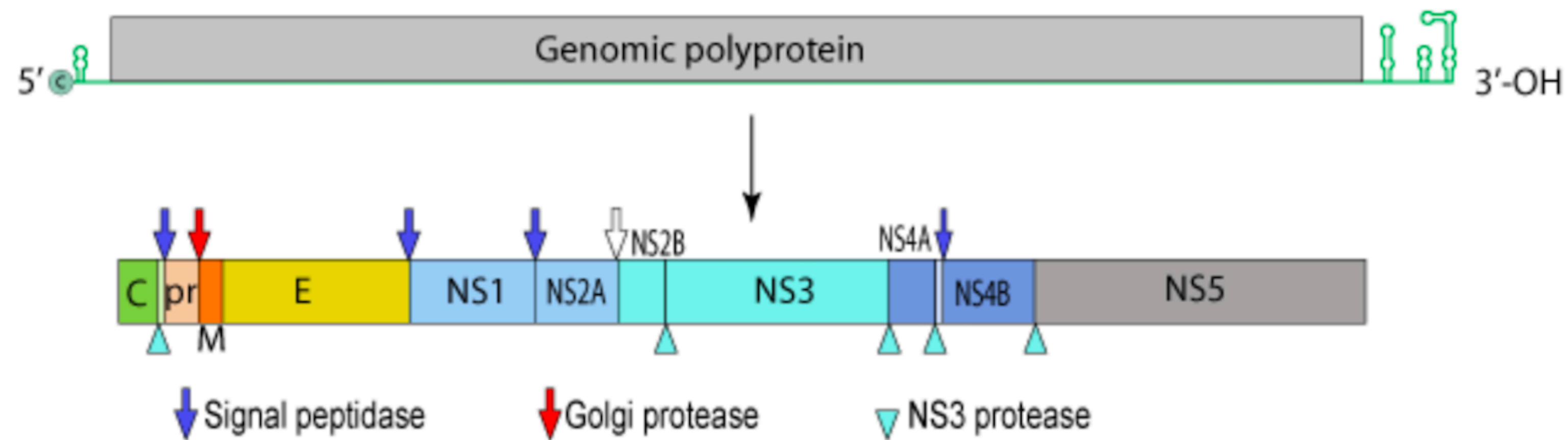
Source: Wikipedia

Vaccine only available for YFV

Flavivirus Phylogeny



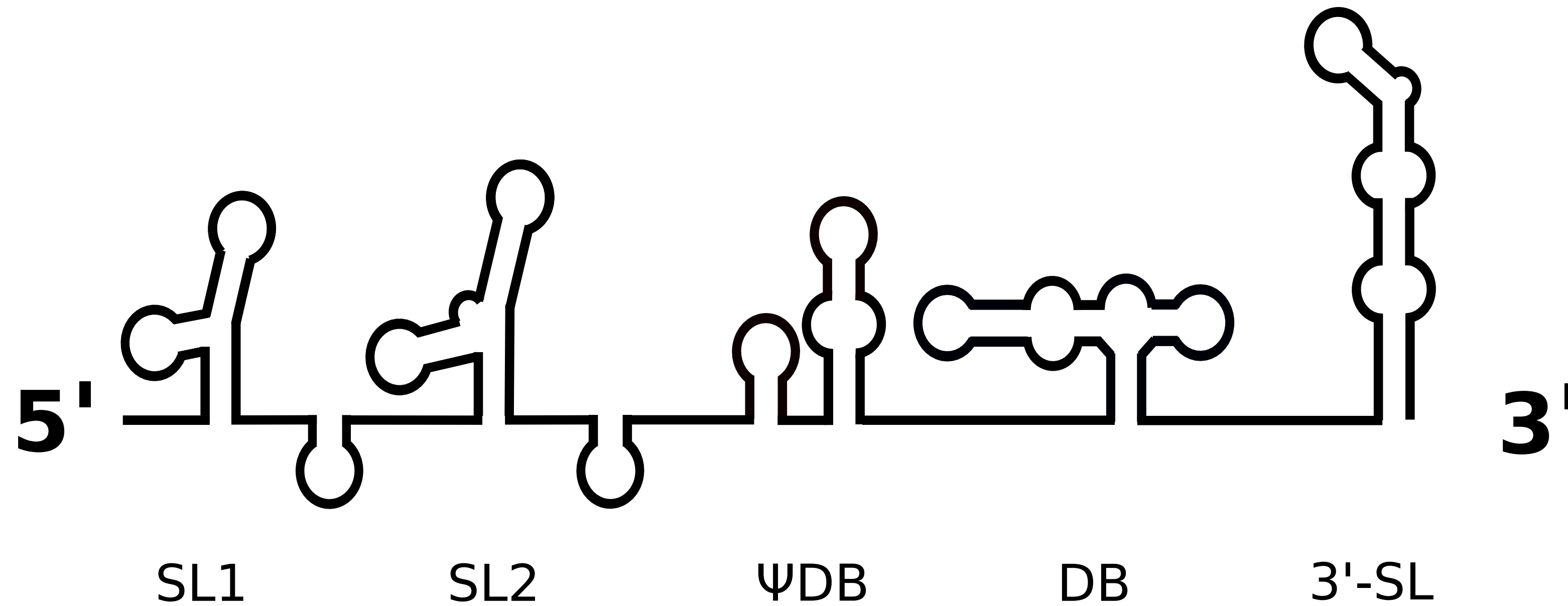
Flavivirus Genome Organization



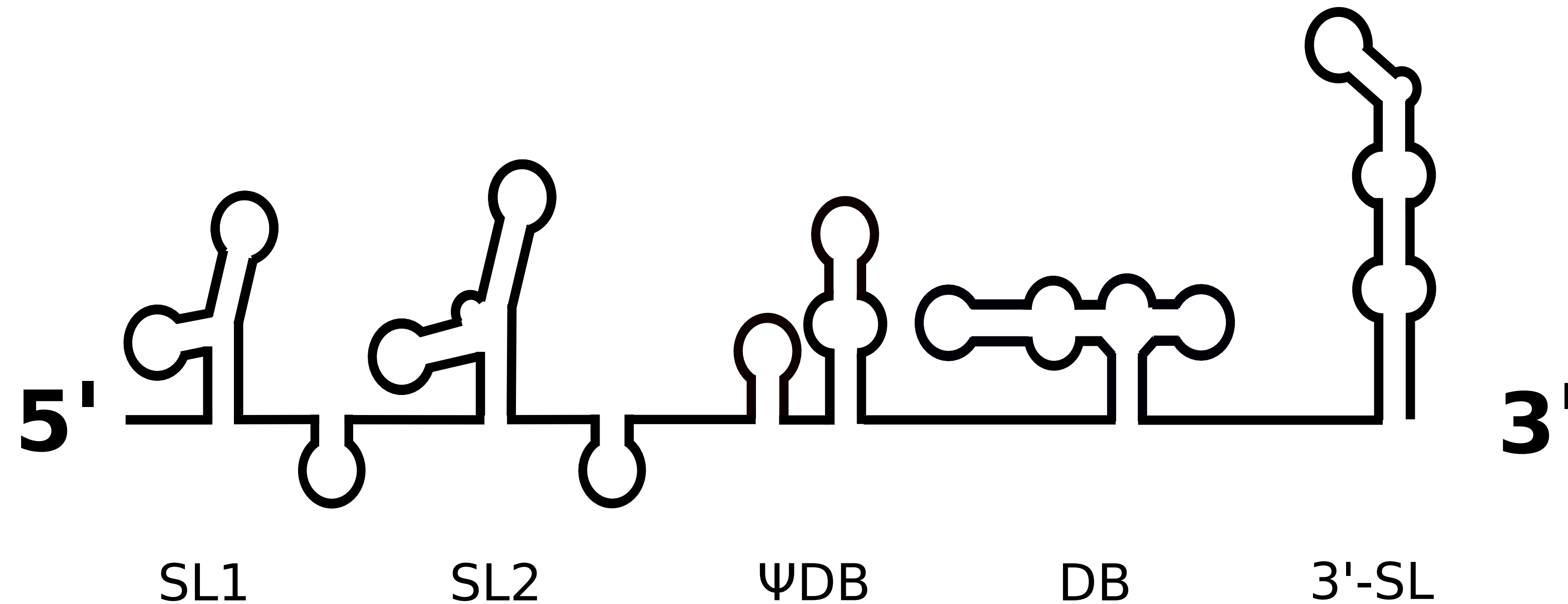
<https://viralzone.expasy.org/>

- Non-segmented, single-stranded, (+)-sense RNA genomes of 10-12kB length
- Capped, non-polyadenylated
- Encode a single ORF, flanked by highly structures 5'UTR and 3'UTR

Conserved RNAs in Viral 3'UTR: ZIKV

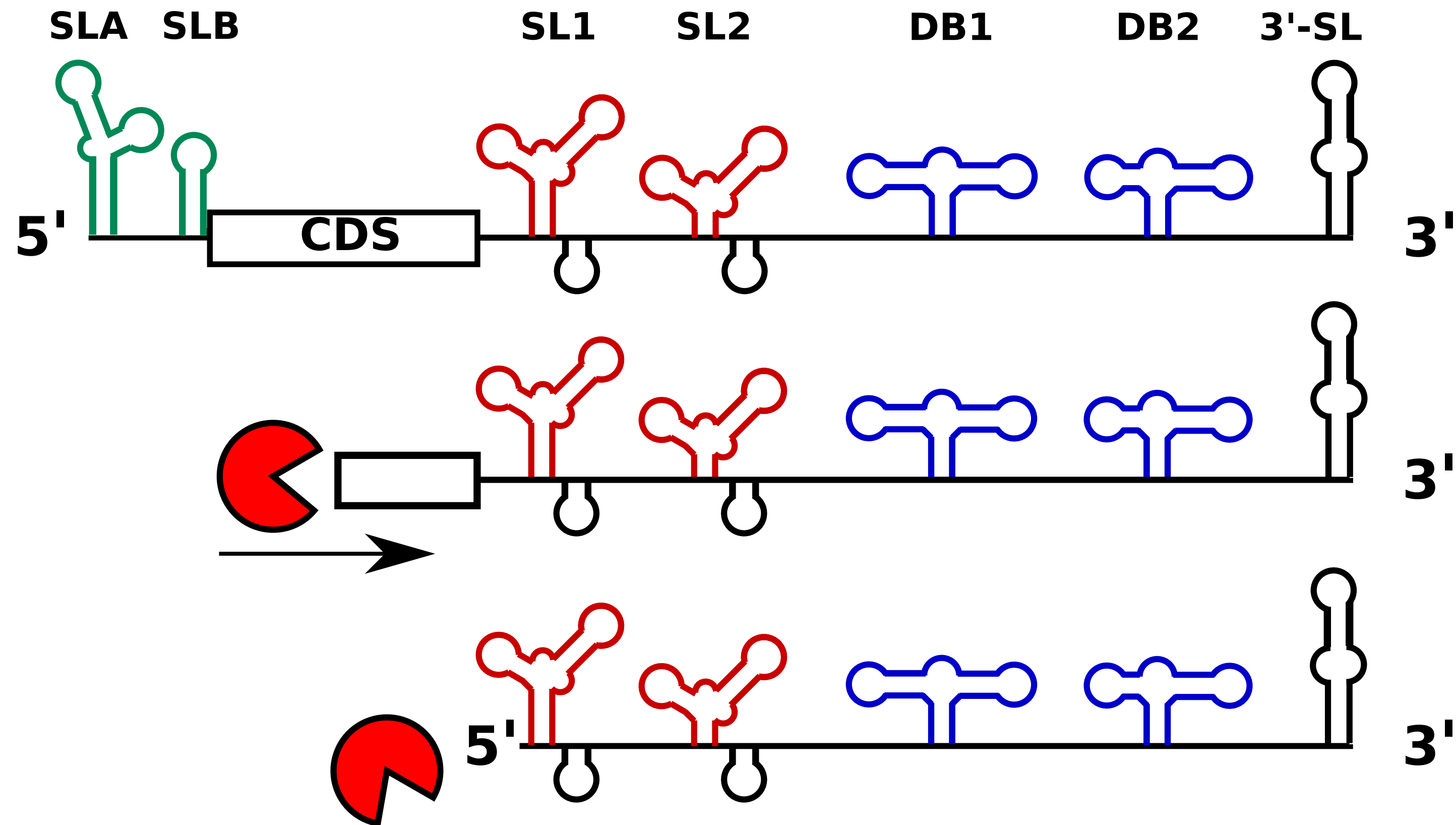


Conserved RNAs in Viral 3'UTR: ZIKV



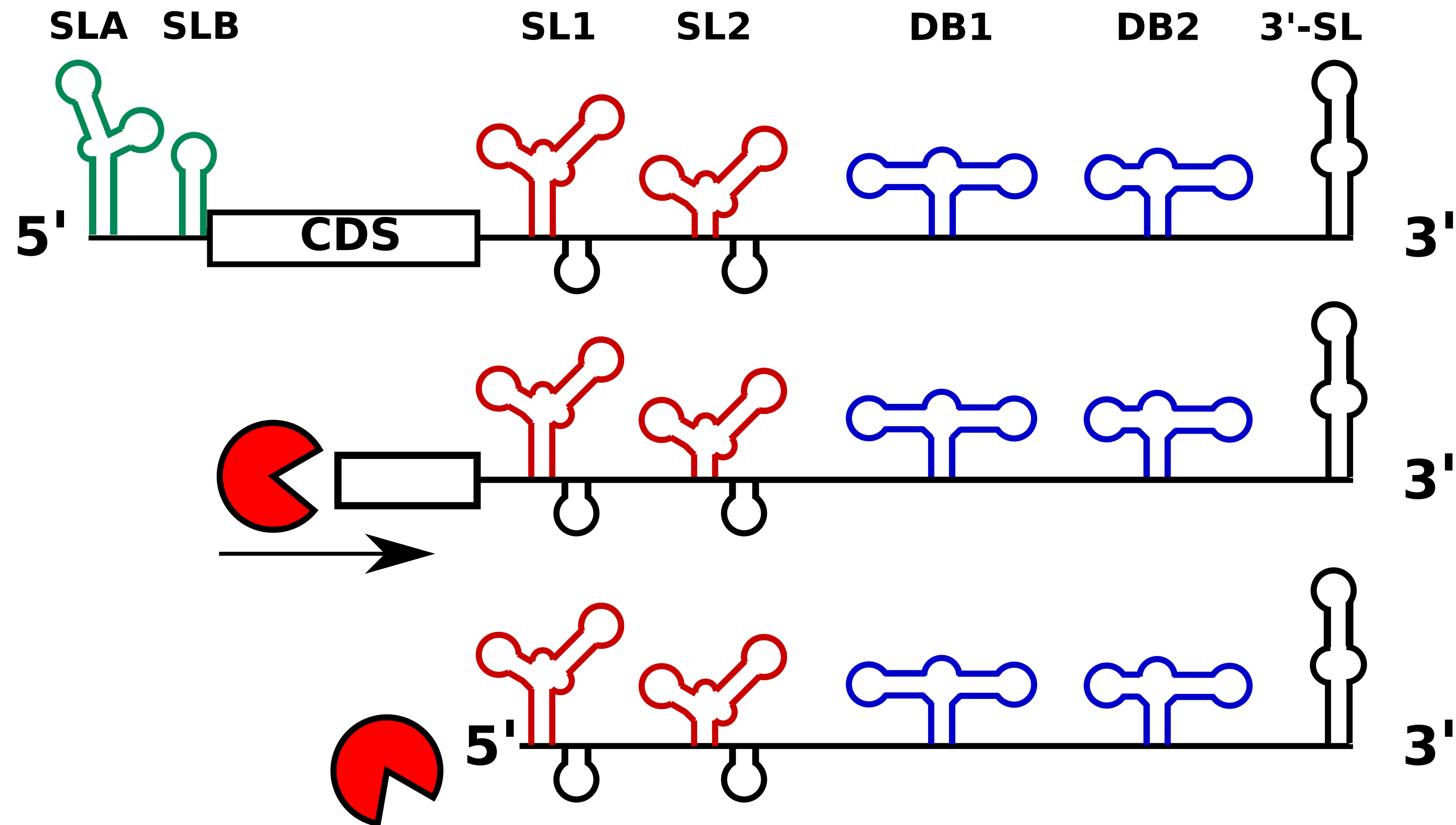
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- Stable decay intermediates produced by partial exoribonuclease degradation
- Xrn1 is efficiently stalled at conserved xrRNA structures

Conserved RNAs in Viral 3'UTR



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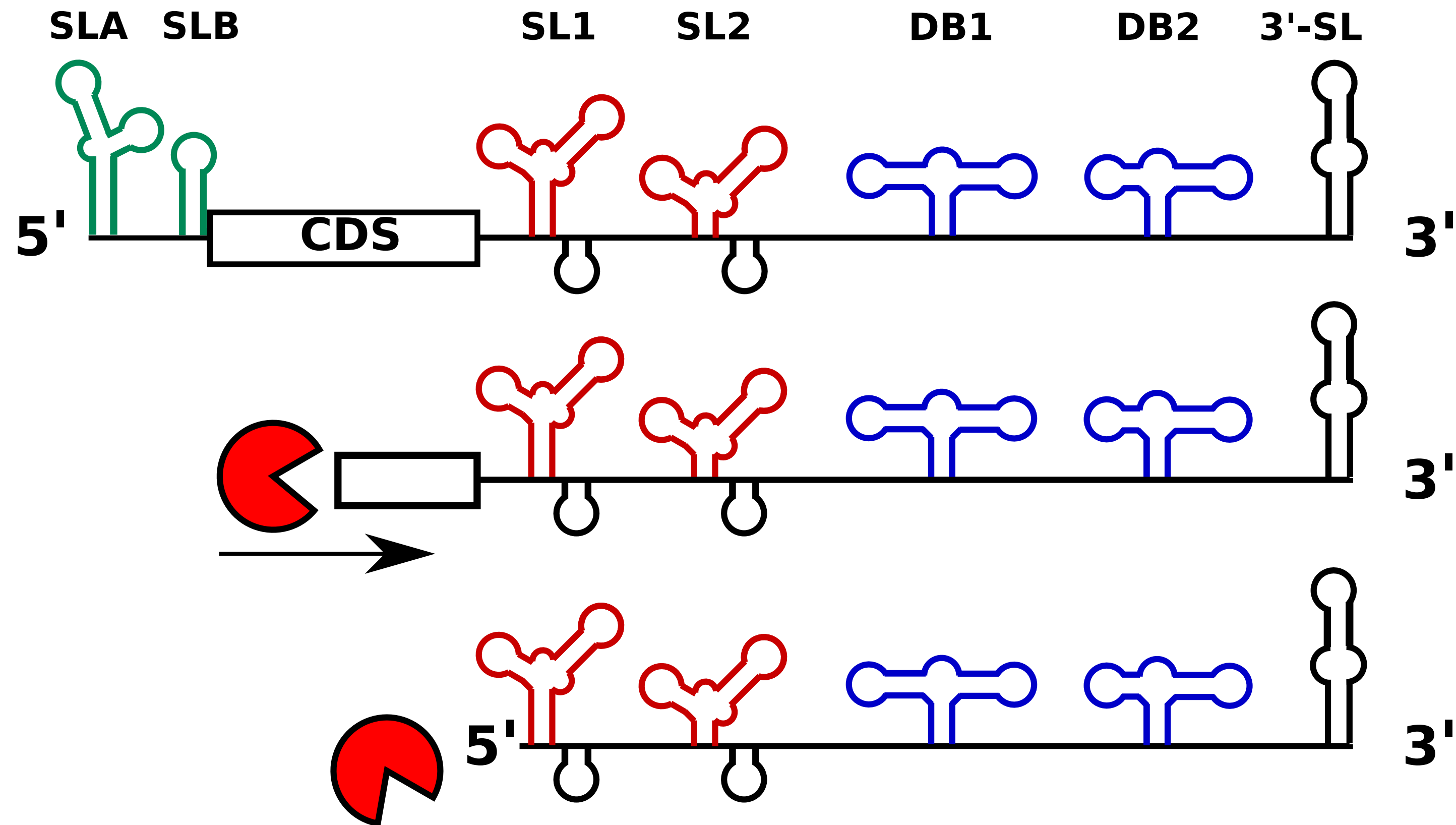
Conserved RNAs in Viral 3'UTR



**sfRNA mediates
pathogenicity !**

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Conserved RNAs in Viral 3'UTR

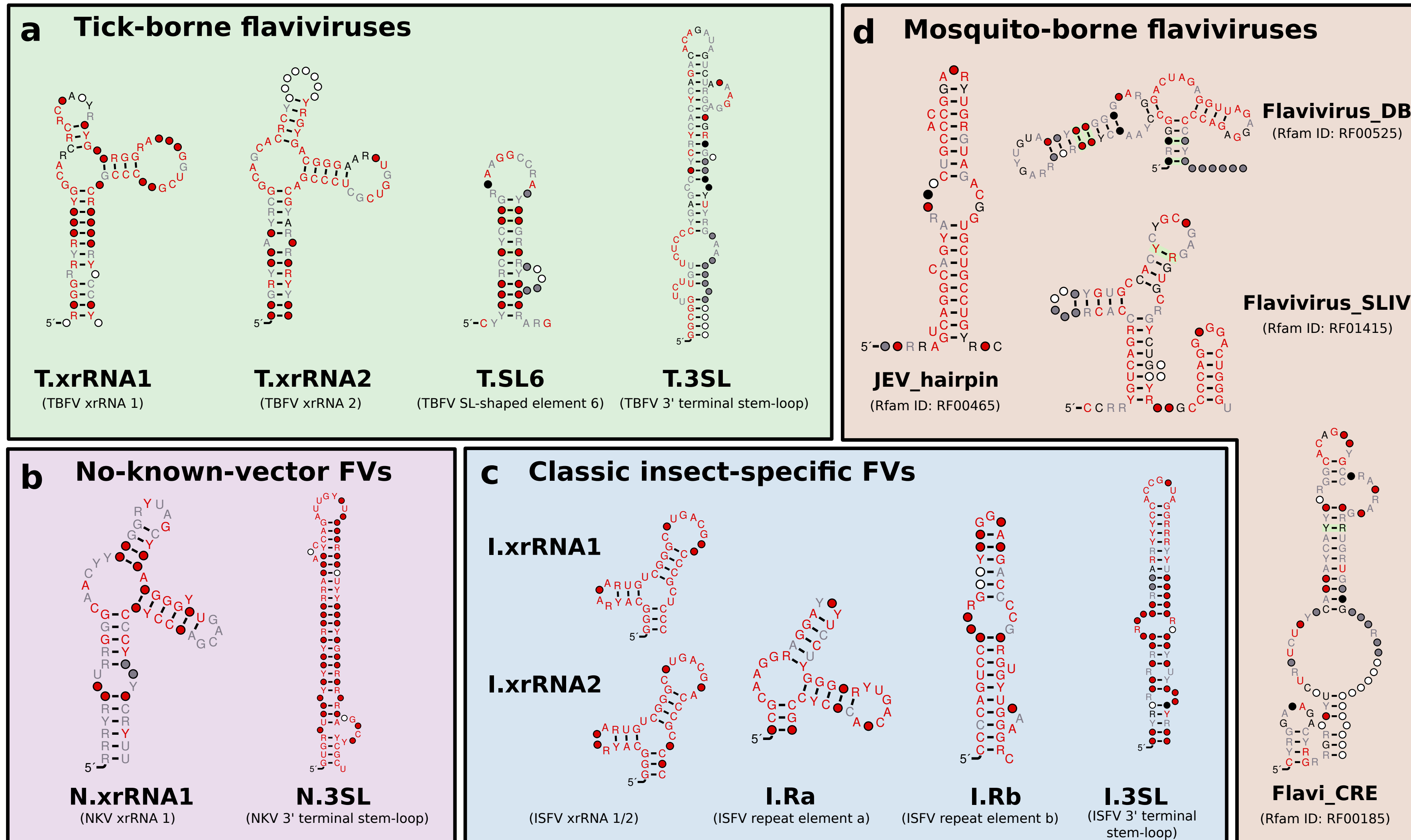


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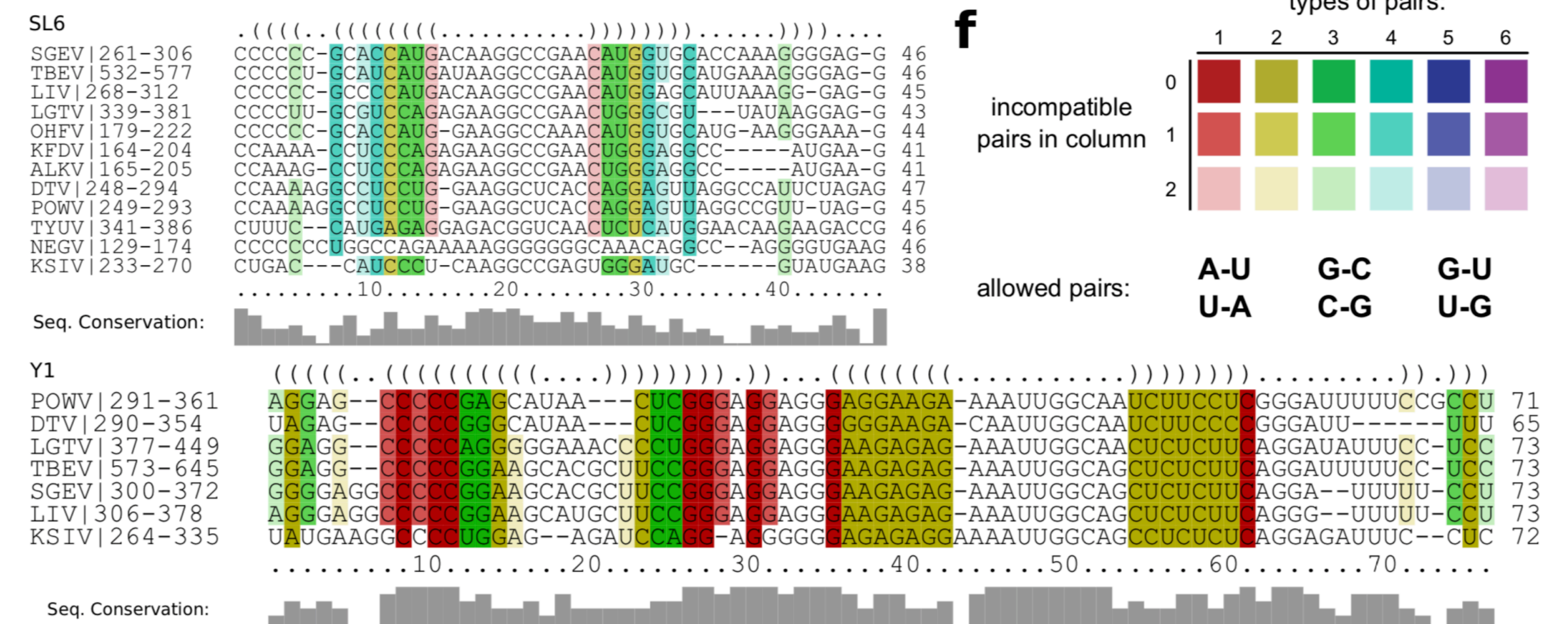
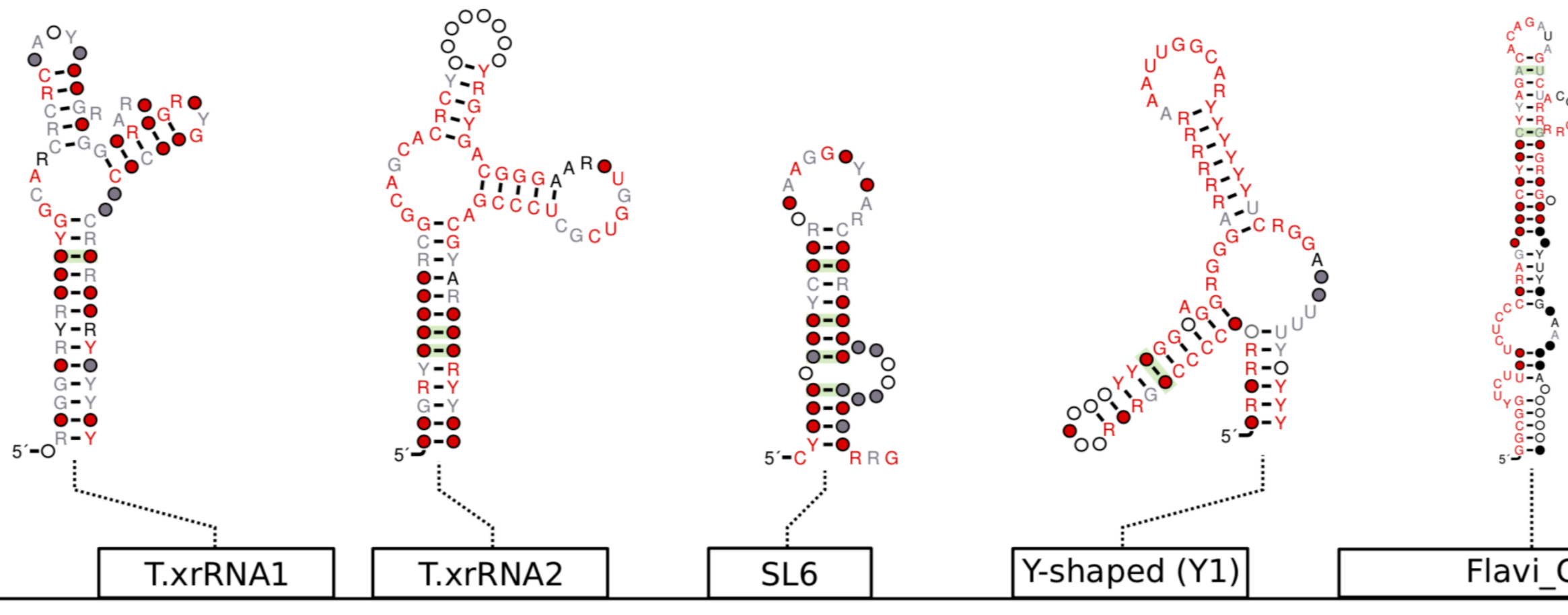
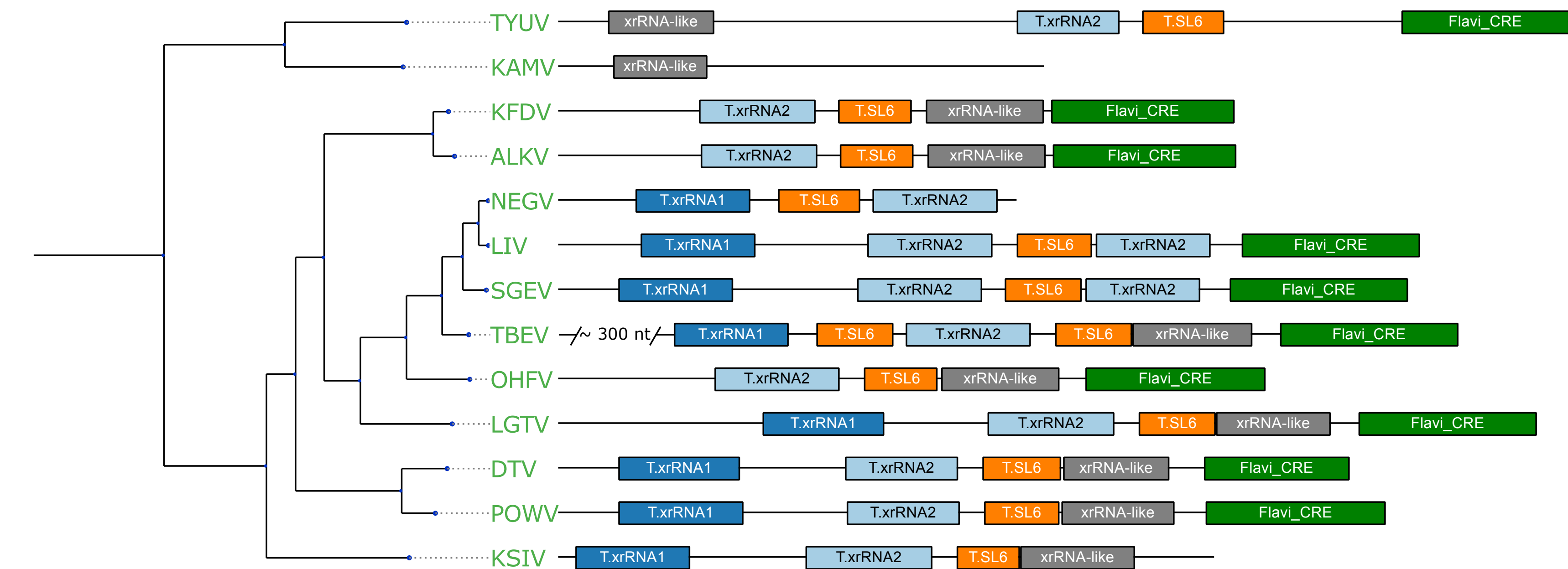
How to find them?

- Accumulation of short flavivirus RNA (sfRNA) upon infection
- Stable decay intermediates produced by partial exoribonuclease degradation
- Xrn1 is efficiently stalled at conserved xrRNA structures

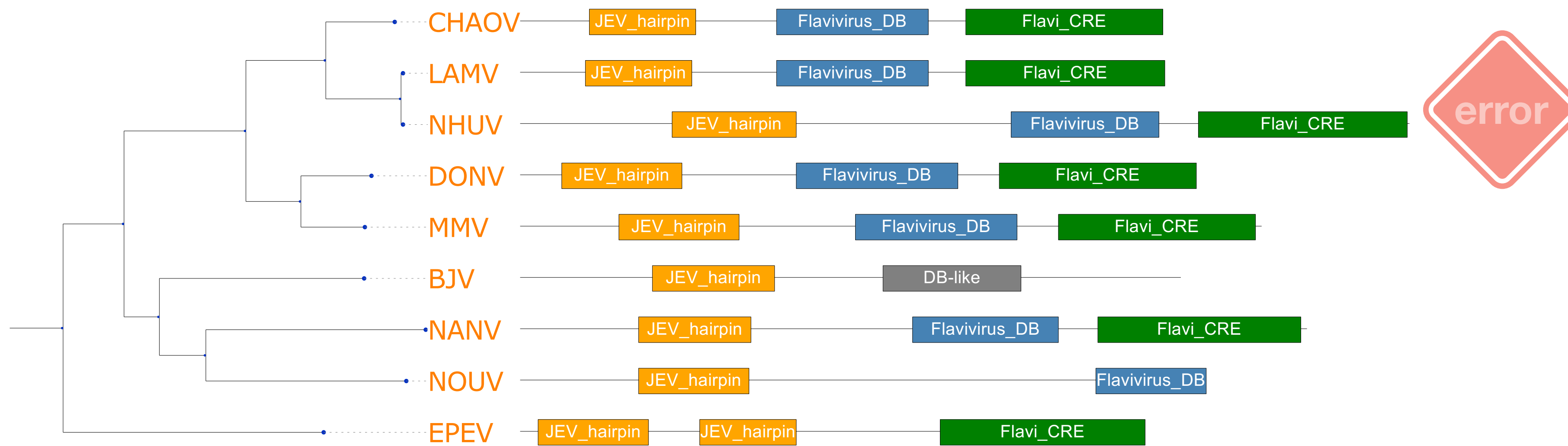
Consensus Structures & Covariance Models



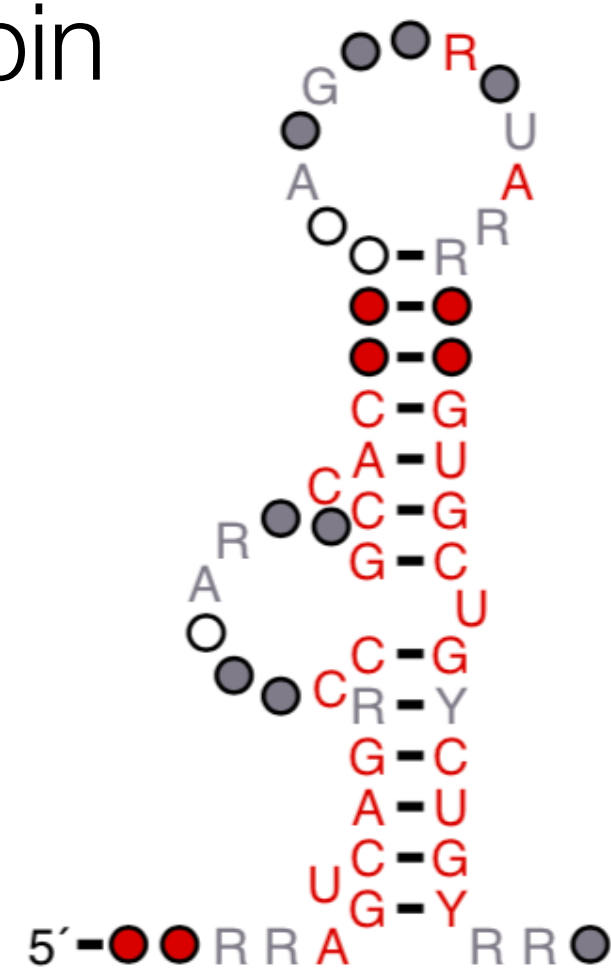
Tick-Borne Flaviviruses



Dual-Host Affiliated Insect-Specific FVs



JEV_hairpin



EPEV	10-64
EPEV	92-139
CHAOV	36-88
LAMV	34-86
NHUV	78-139
DONV	22-81
MMV	51-110
BJV	68-128
NANV	61-116
NOUV	61-115

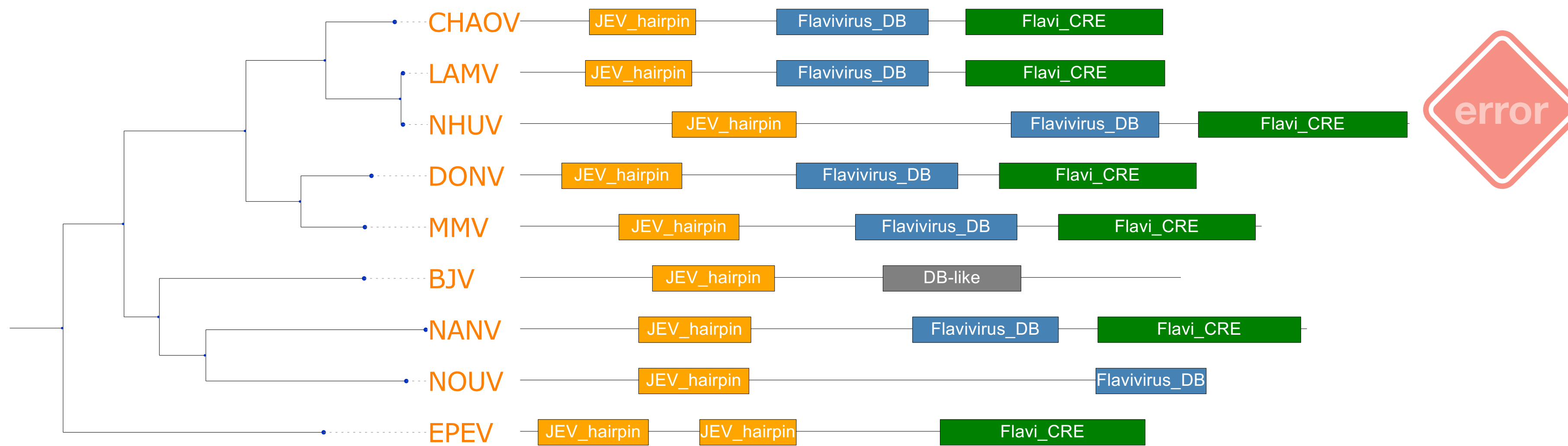
.....(.((((.....(((.....)))..)))..)).....

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AAAGA	U	CAG	G	CGAGAA	-----		GC	CAC	AC	-----	AGCAA	AAGU	GUGC	U	CCUG	U	AA--	48	
UGAGA	U	CAG	G	CU--AAA	----	U	GC	CAC	CGG	-AUGAUA	-GUAGAC	G	GUGC	U	CCUG	C	CAG--	53	
UGGGA	U	CAG	G	CU--AAA	----	U	GC	CAC	CGG	-AUGAUA	-GUAGAC	G	GUGC	U	CCUG	C	CAG--	53	
UUUGA	U	CAG	A	CAGGAUGAA	ACC	U	GC	CAC	C	UCUAUGGA	AGCUAAGU	G	GUGC	U	GU	CUG	U	AA--	62
UUGCAG	U	CAG	G	CUCACGAA	UGUGA		GC	CAC	CG	-GAUGGGA	-CUAGAC	G	GUGC	U	CCUG	C	GC--	60	
UUGCAG	U	CAG	G	CUCAAGACA	UUGA		GC	CAC	CG	-GAUGGGA	-CUAGAC	G	GUGC	U	CCUG	C	GC--	60	
AACGA	U	CAG	G	C--GGCUUA	UGC	U	GC	CAC	C	UCUAUGGA	AACUAGGG	G	GUGC	U	CCUG	C	GU--	61	
AUGAA	U	CAG	G	CUU--AAG	----	U	GC	CAC	C	CGUAUGAGA	-CUAUCG	G	GUGC	U	CCUG	U	CGUC	56	
ACGAA	U	CAG	G	CCU--AAC	----	G	GC	CAC	U	CUUAAGA	AUG--CUAAAG	A	GUGC	U	GU	CUG	U	AAG--	55

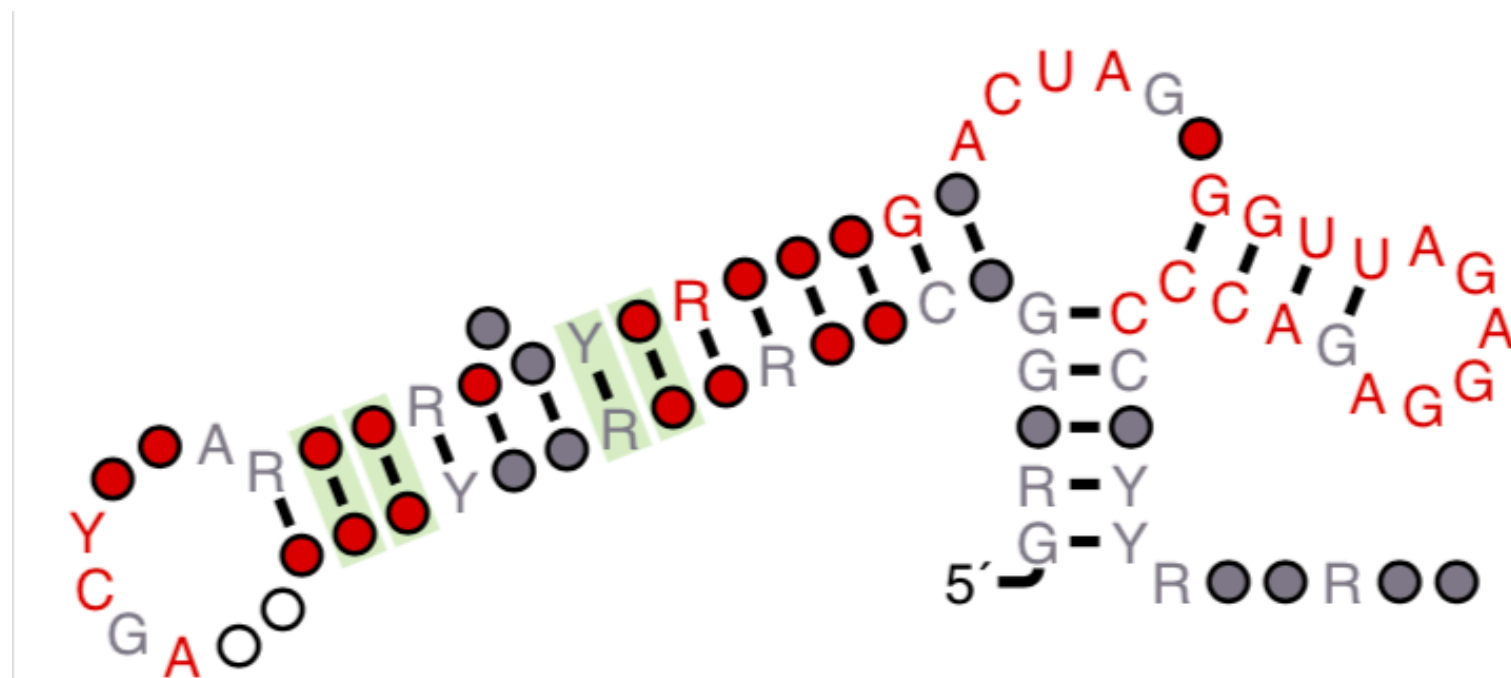
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Dual-Host Affiliated Insect-Specific FVs



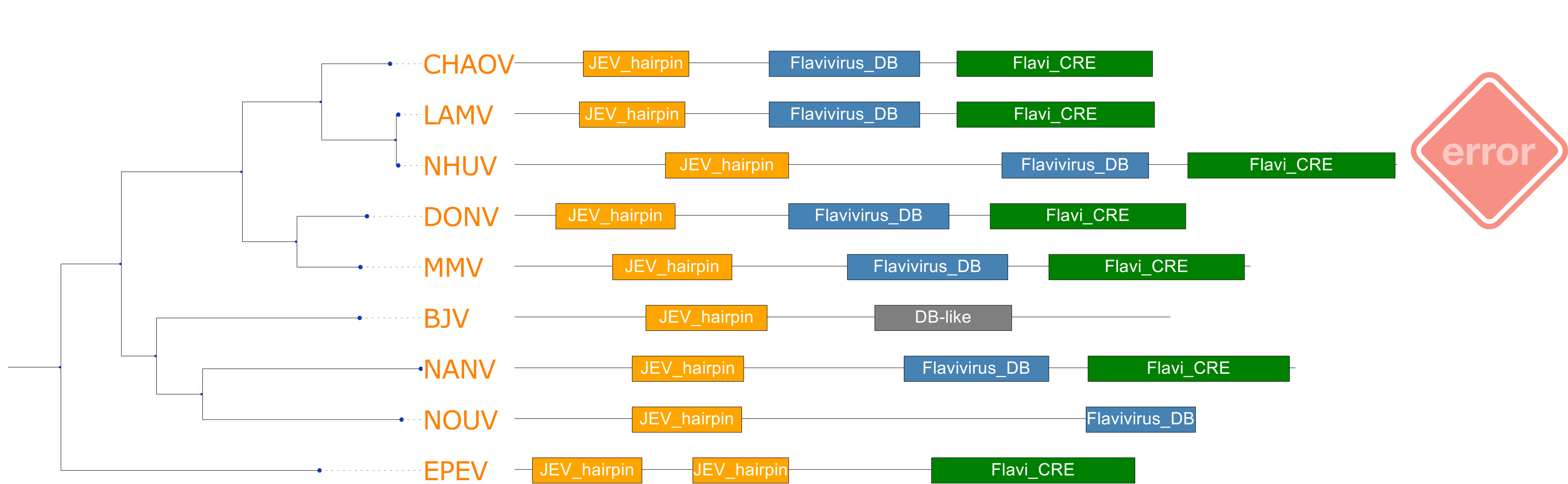
Flavivirus_DB



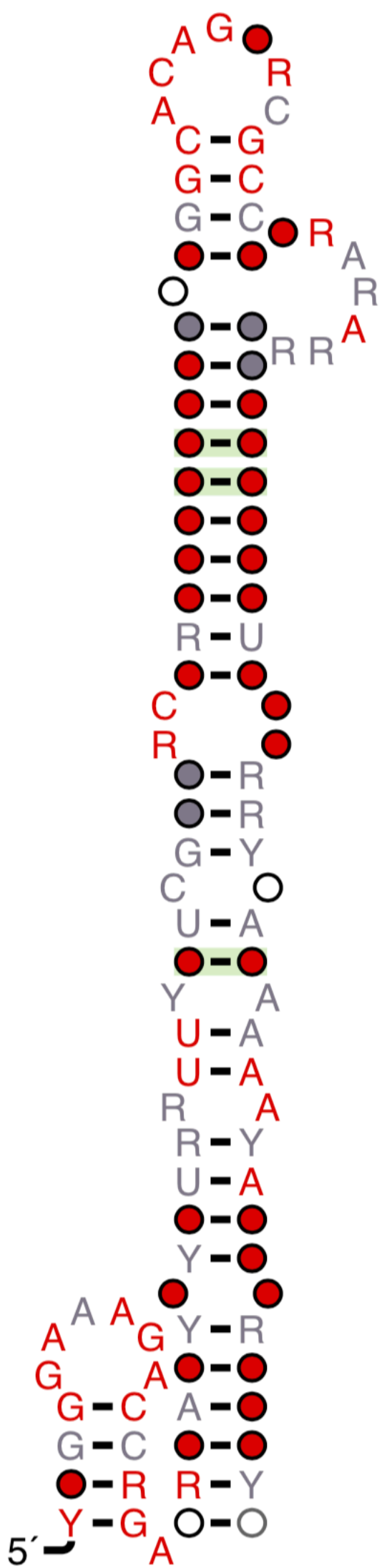
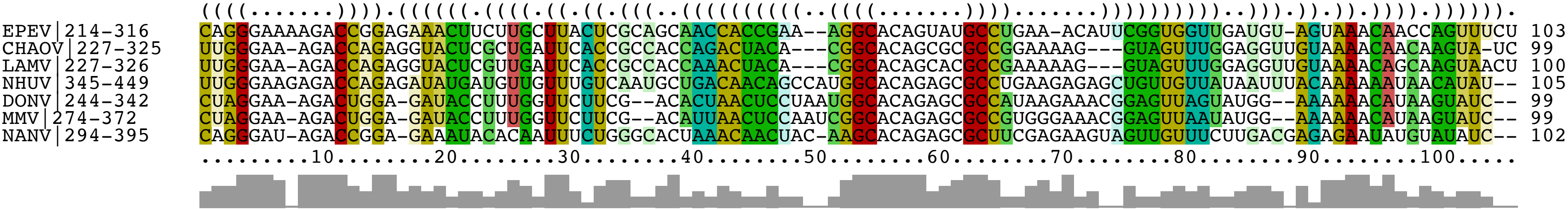
CHAOV	131-206
LAMV	131-206
NHUV	250-323
DONV	141-221
MMV	171-251
NANV	200-272
NOUV	293-347

[illegible]

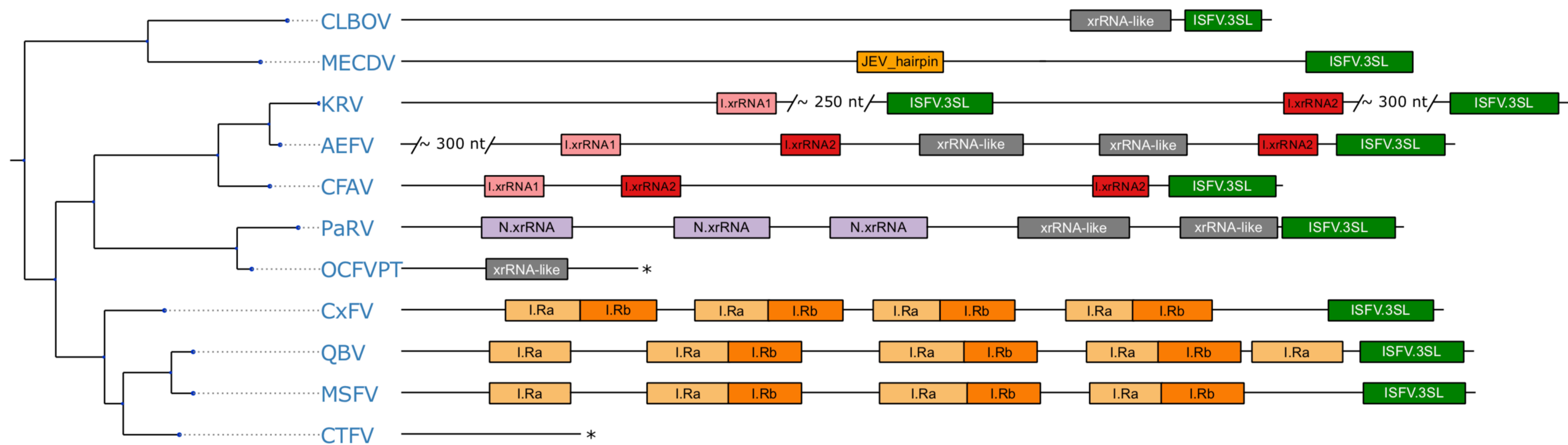
Dual-Host Affiliated Insect-Specific FVs



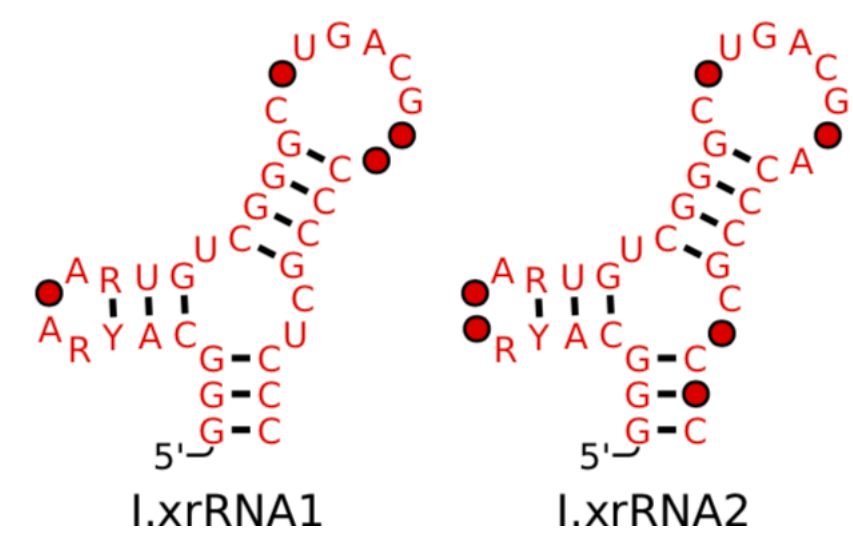
Flavi_CRE



Classic Insect-Specific FVs



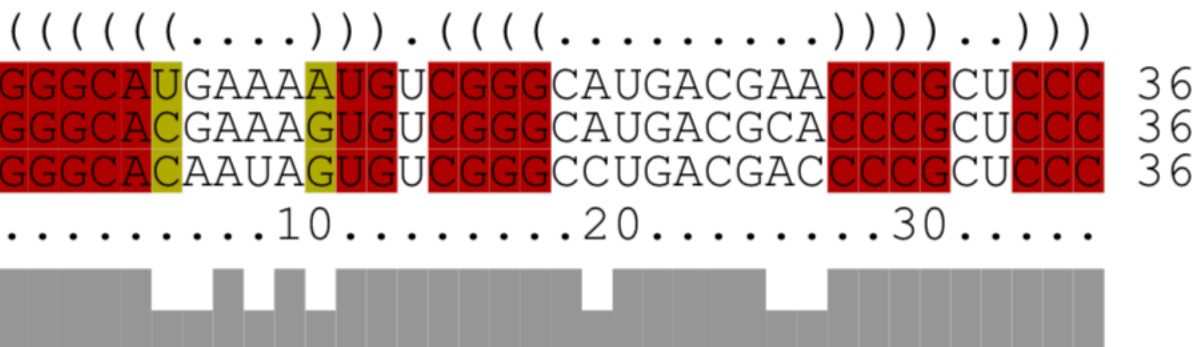
KRV / AEFV / CFAV



I.xrRNA1

CFAV| 53-88
KRV| 199-234
AEFV| 382-417

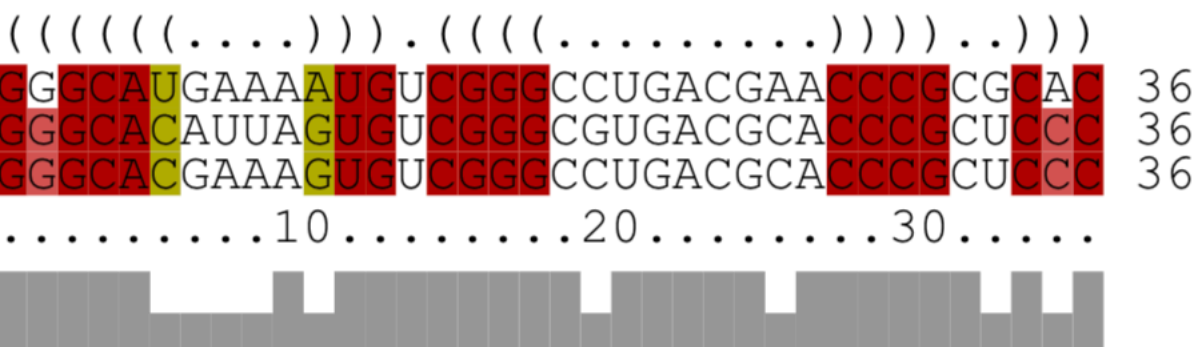
Seq. conservation



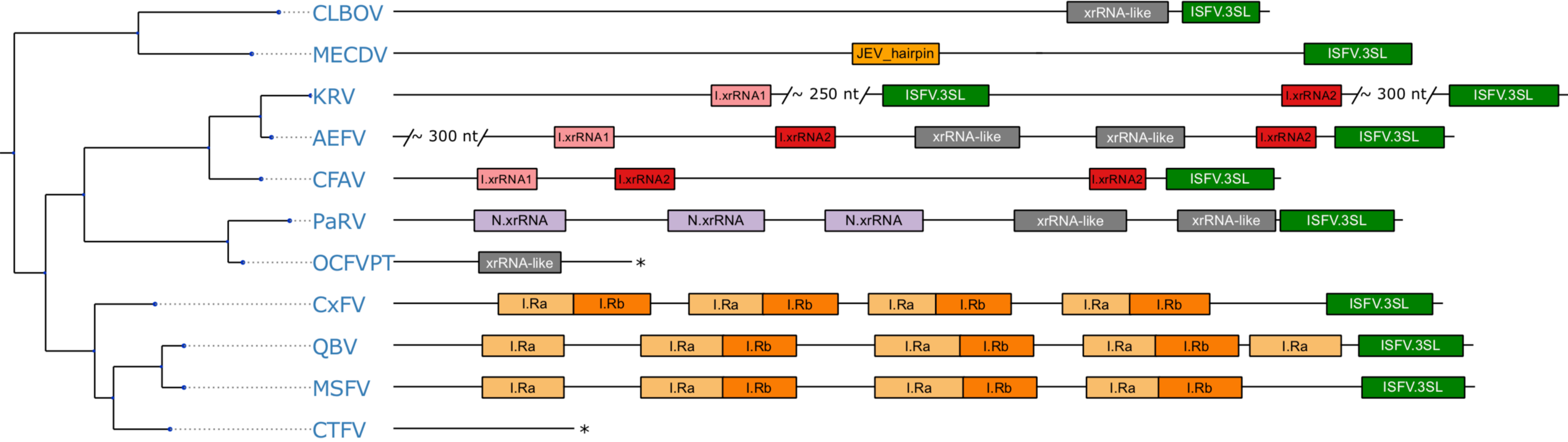
I.xrRNA2

AEFV| 520-555
CFAV| 139-174
KRV| 795-830

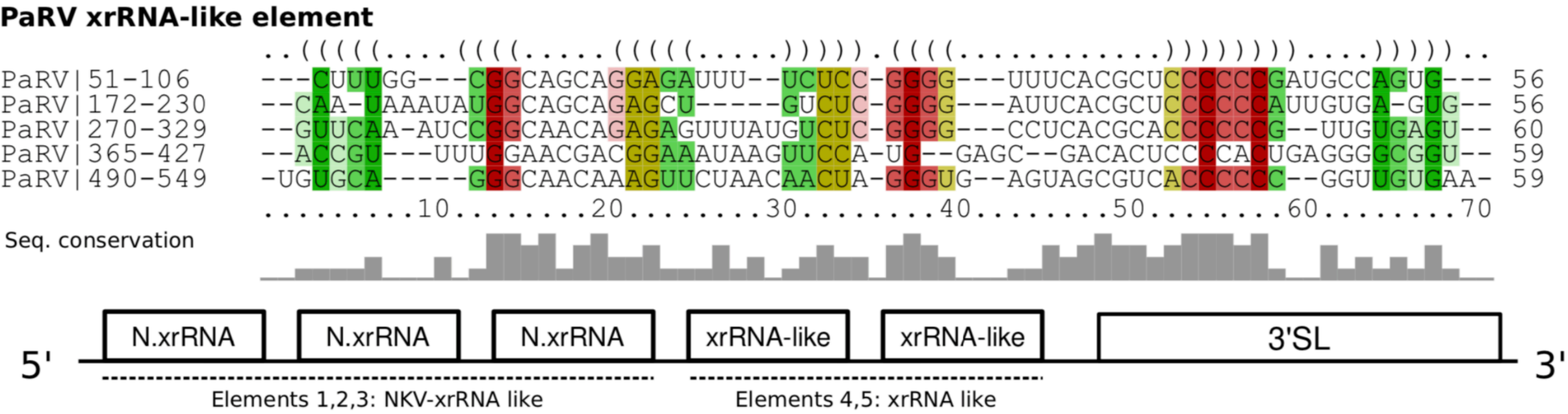
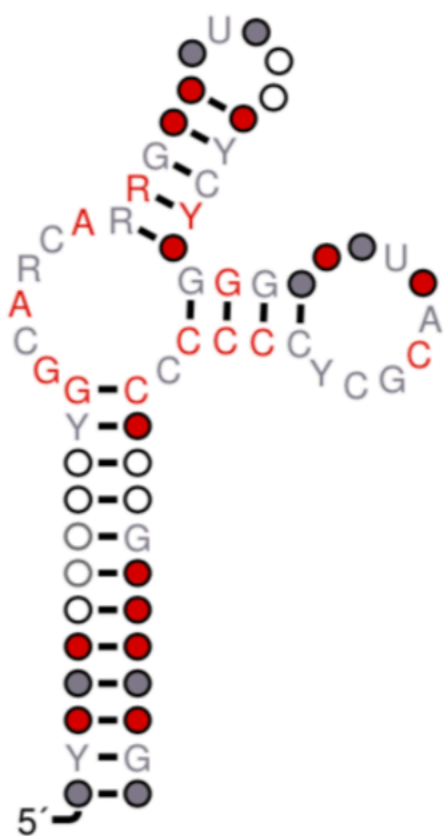
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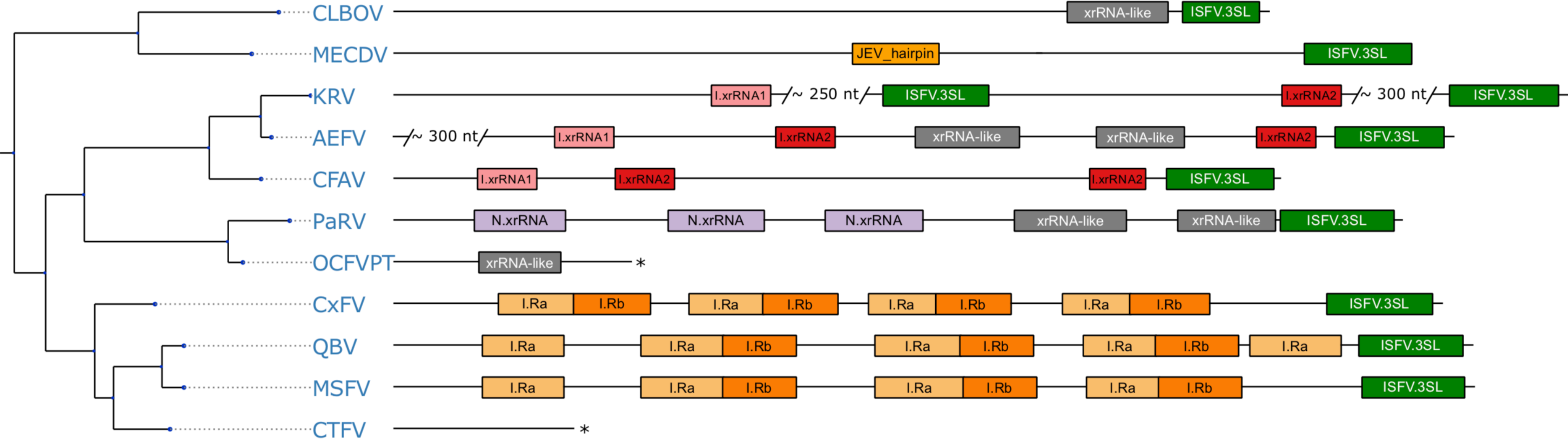
Classic Insect-Specific FVs



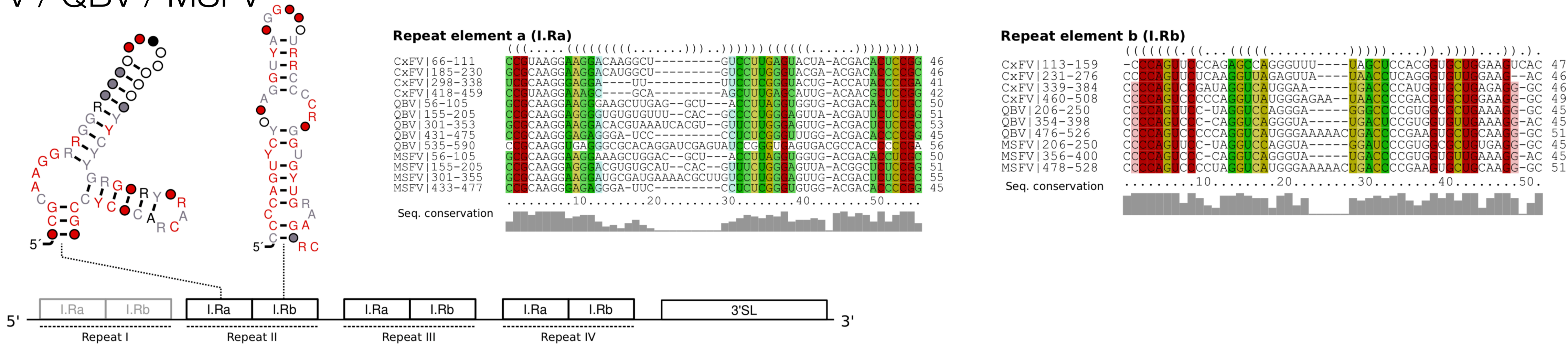
PaRV



Classic Insect-Specific FVs



CxFV / QBV / MSFV

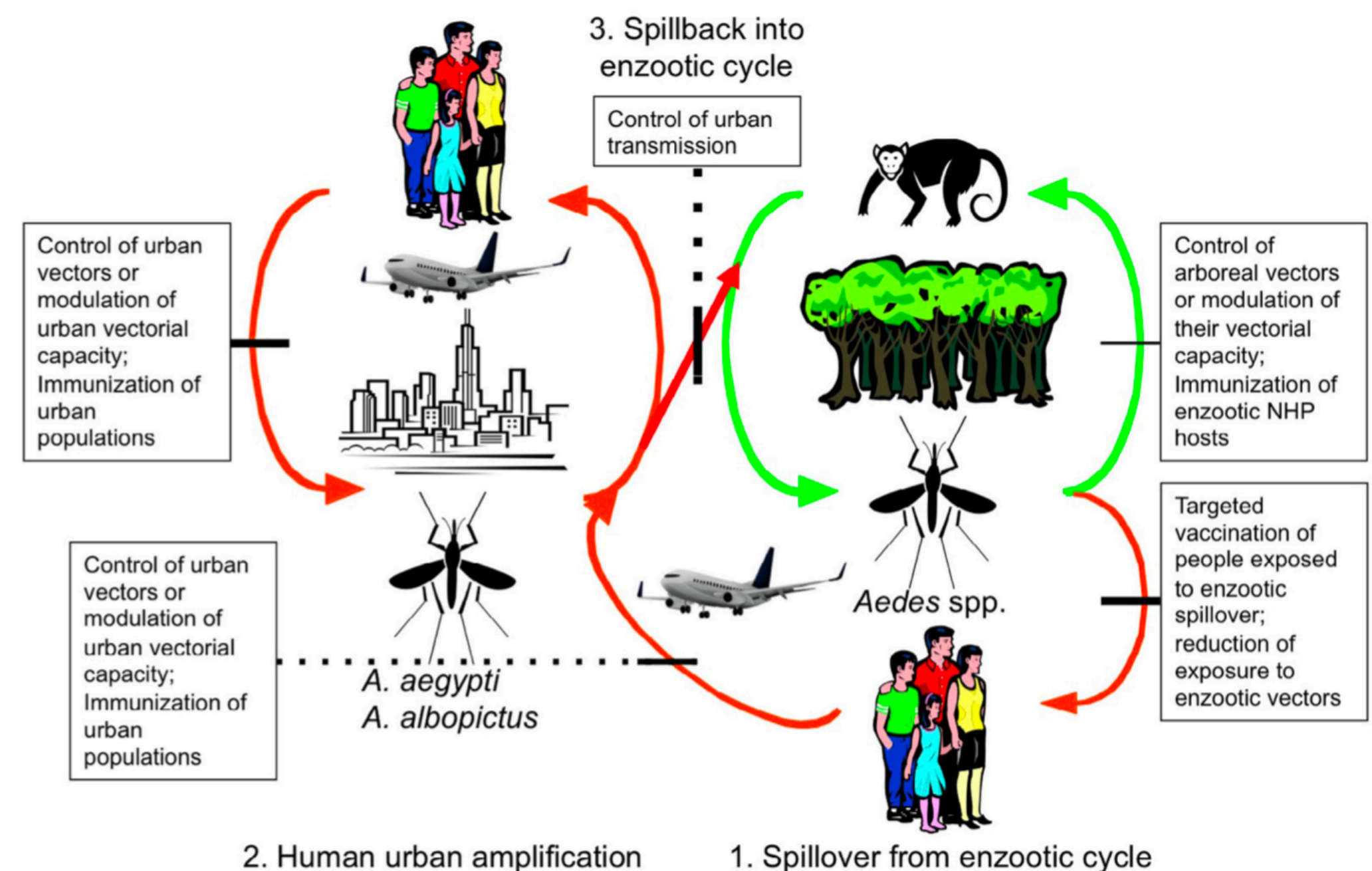


Part II:

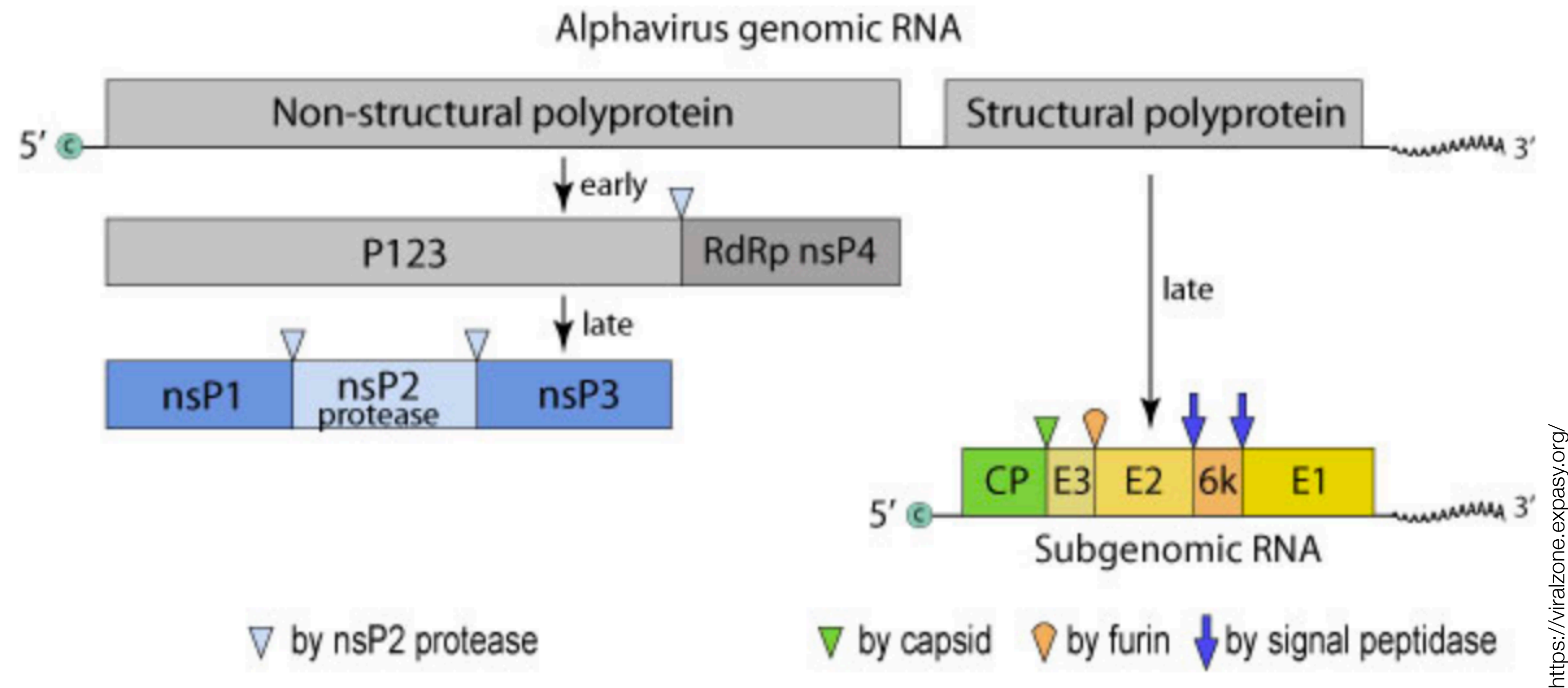
Conserved RNA structures in alphaviruses

Chikungunya Virus (CHIKV)

- Family *Togaviridae* / genus *Alphavirus*; mosquito-borne (*Aedes* spp.)
- Single-stranded (+) sense RNA virus
- Chikungunya fever: febrile illness, arthralgia, rash, rarely causes hemorrhagic complications
- Enzootic in tropical and subtropical regions of Africa
- First outbreak described 1952 in Tanzania
- No vaccine available

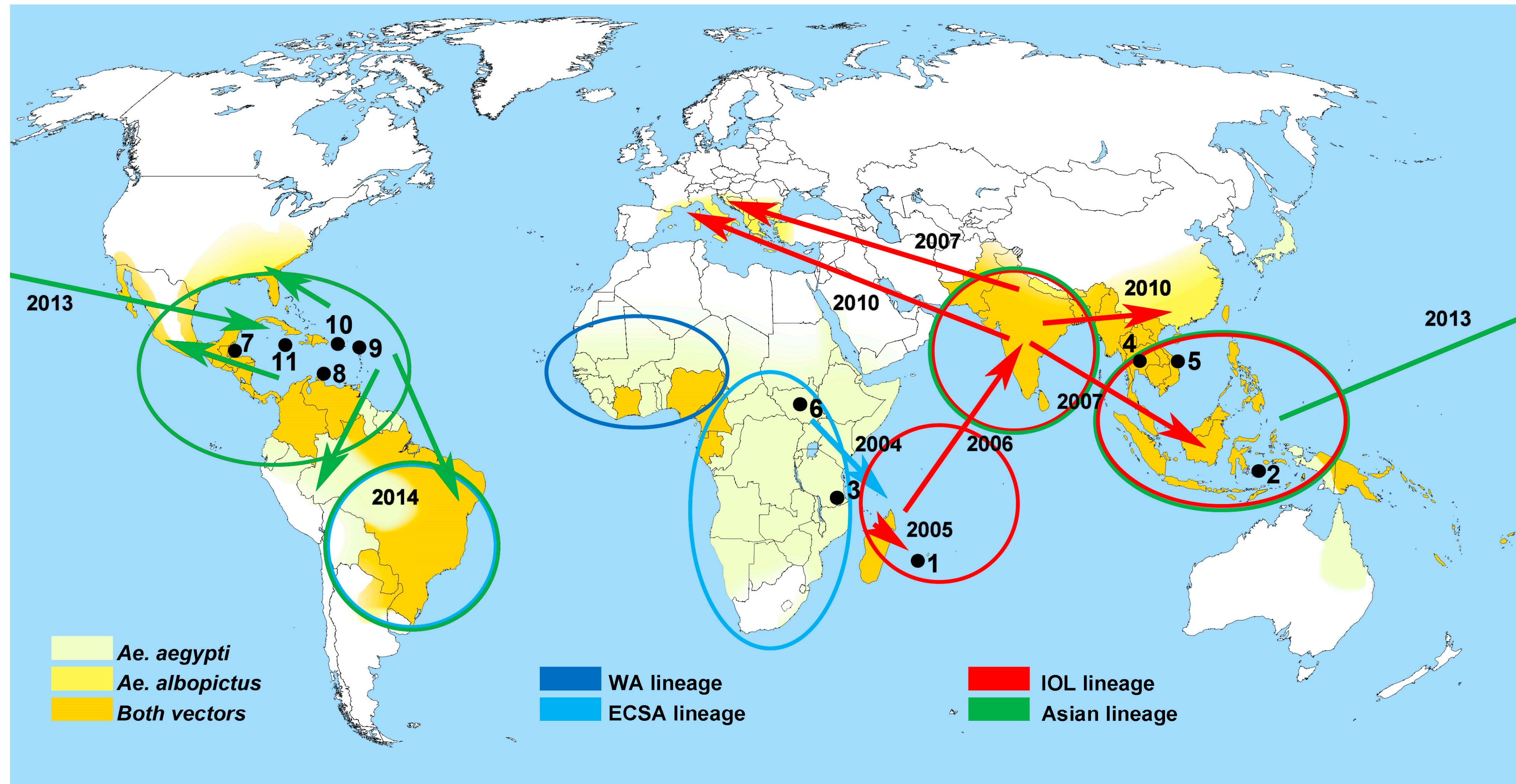


Alphavirus Genome Organization



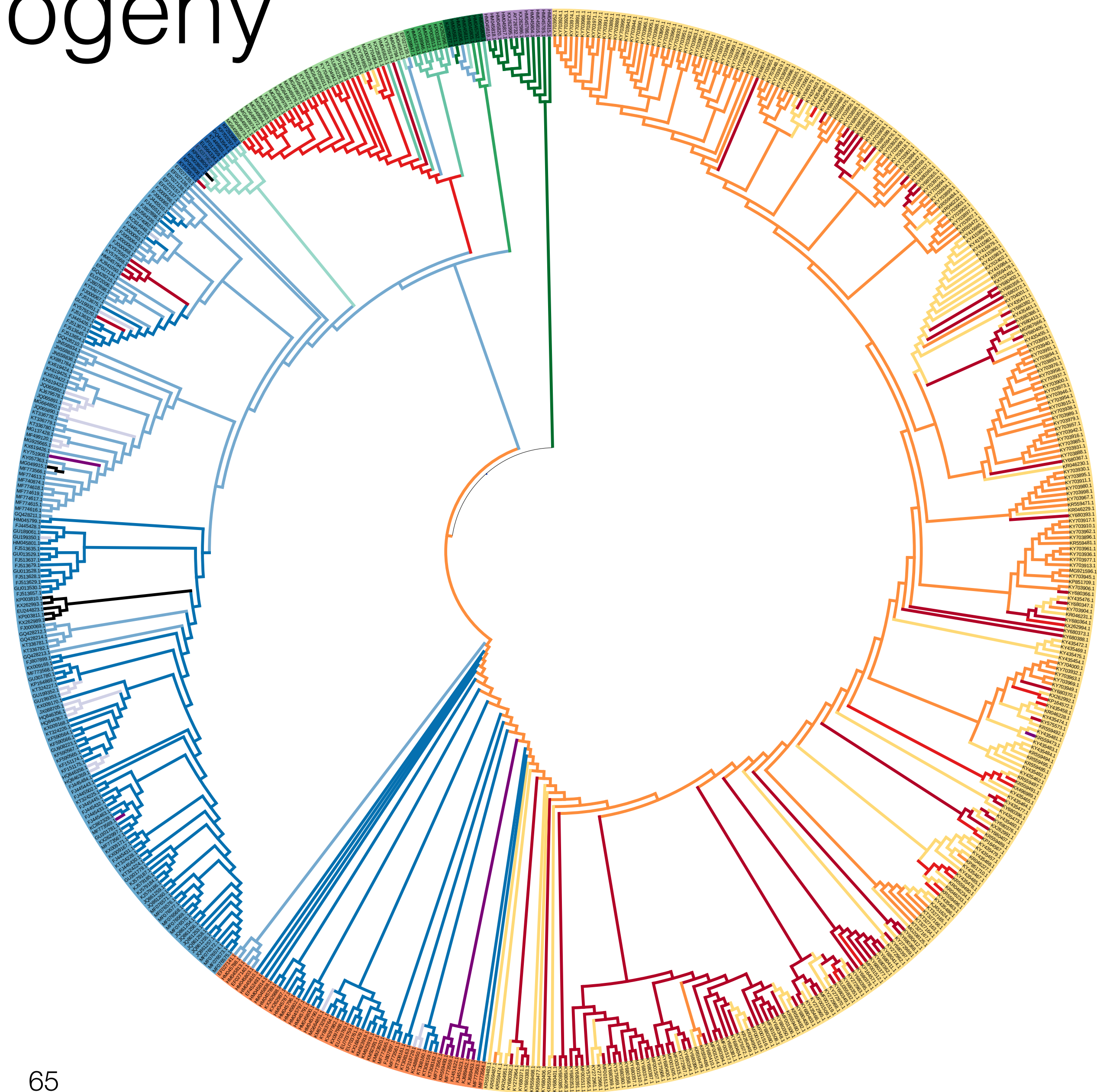
- Non-segmented, single-stranded, (+)-sense RNA genomes of 11-12kB length
- Capped and polyadenylated
- AV genomes appear to host cells as mRNA for immediate translation upon entry into the cytoplasm

CHIKV Epidemic Spread



Frickmann et al., Viruses (2019)

Updated CHIKV Phylogeny



- 590 CHIKV genomes
- iq-tree / SH-aLRT

CHIKV 3'UTR



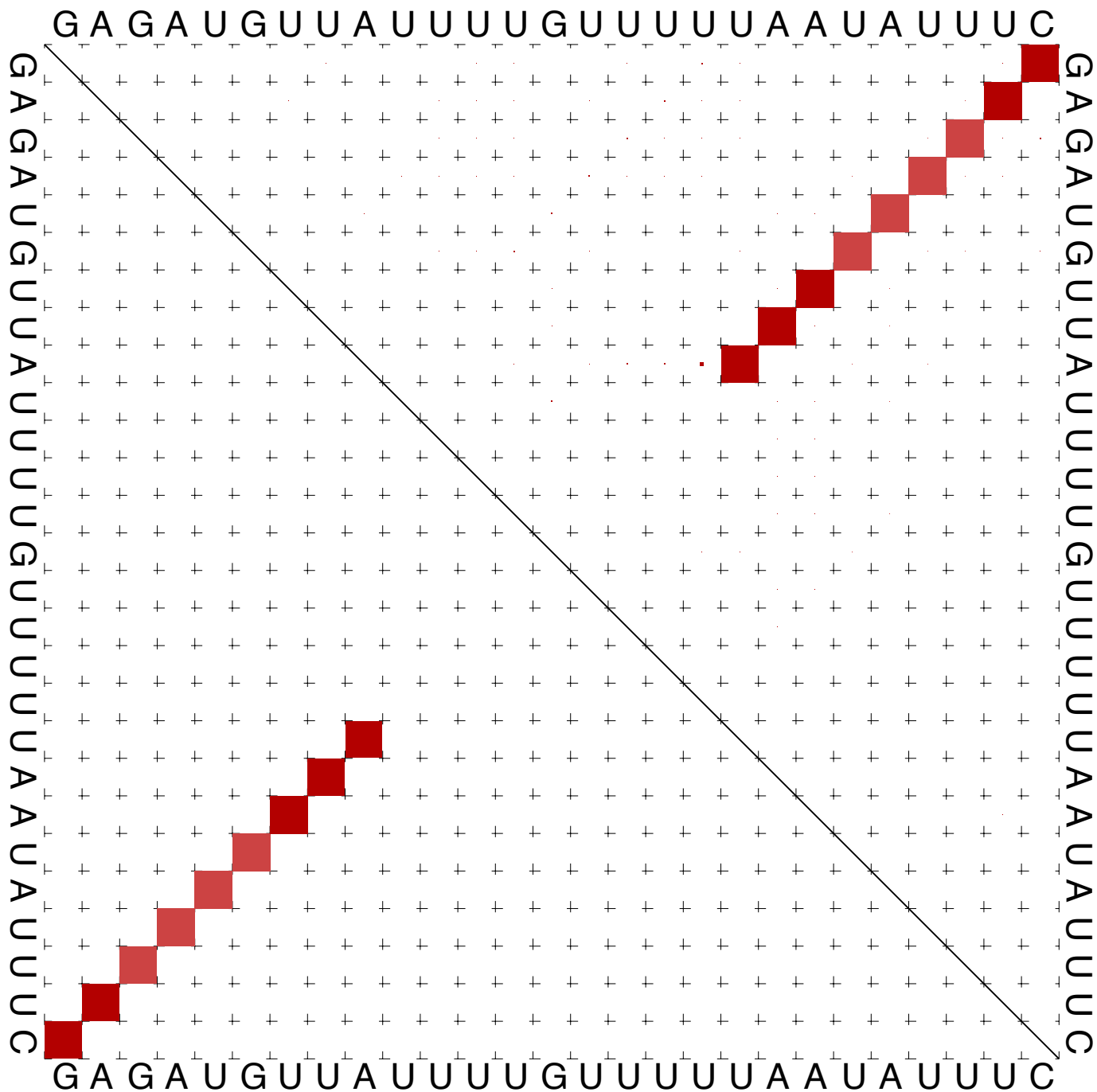
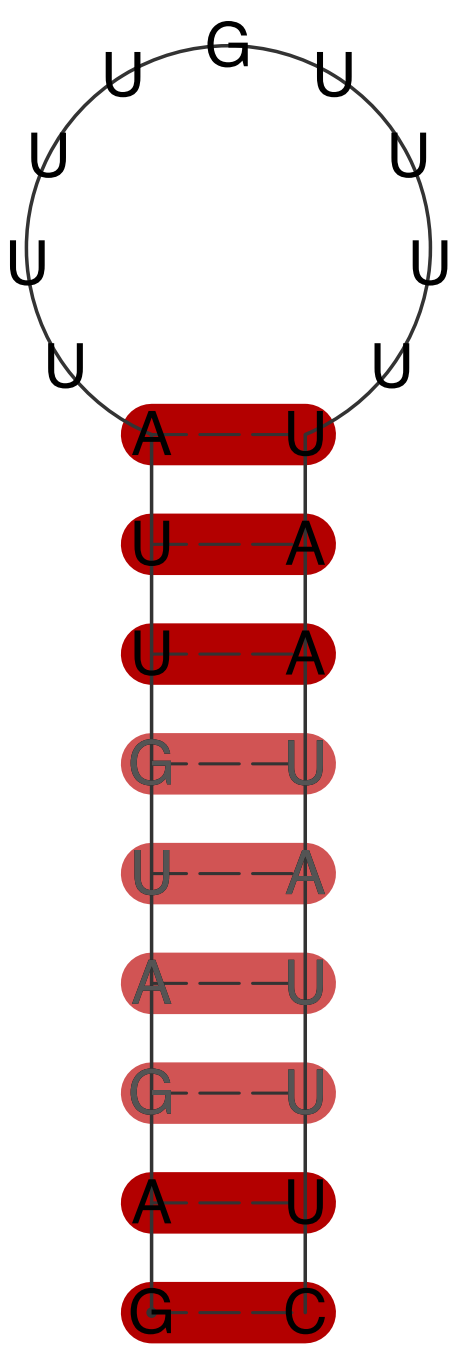
- 110 full length CHIKV 3'UTRs
- Variable length (510nt — 930nt)
- Structural alignments + Covariance models
- Thermodynamic modelling based on ViennaRNA Package

CHIKV 3'UTR: Conserved RNA Structures

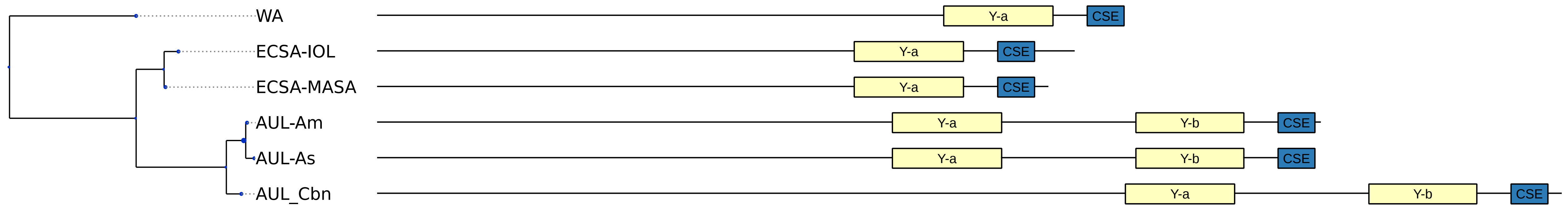


Conserved Sequence Element (CSE)

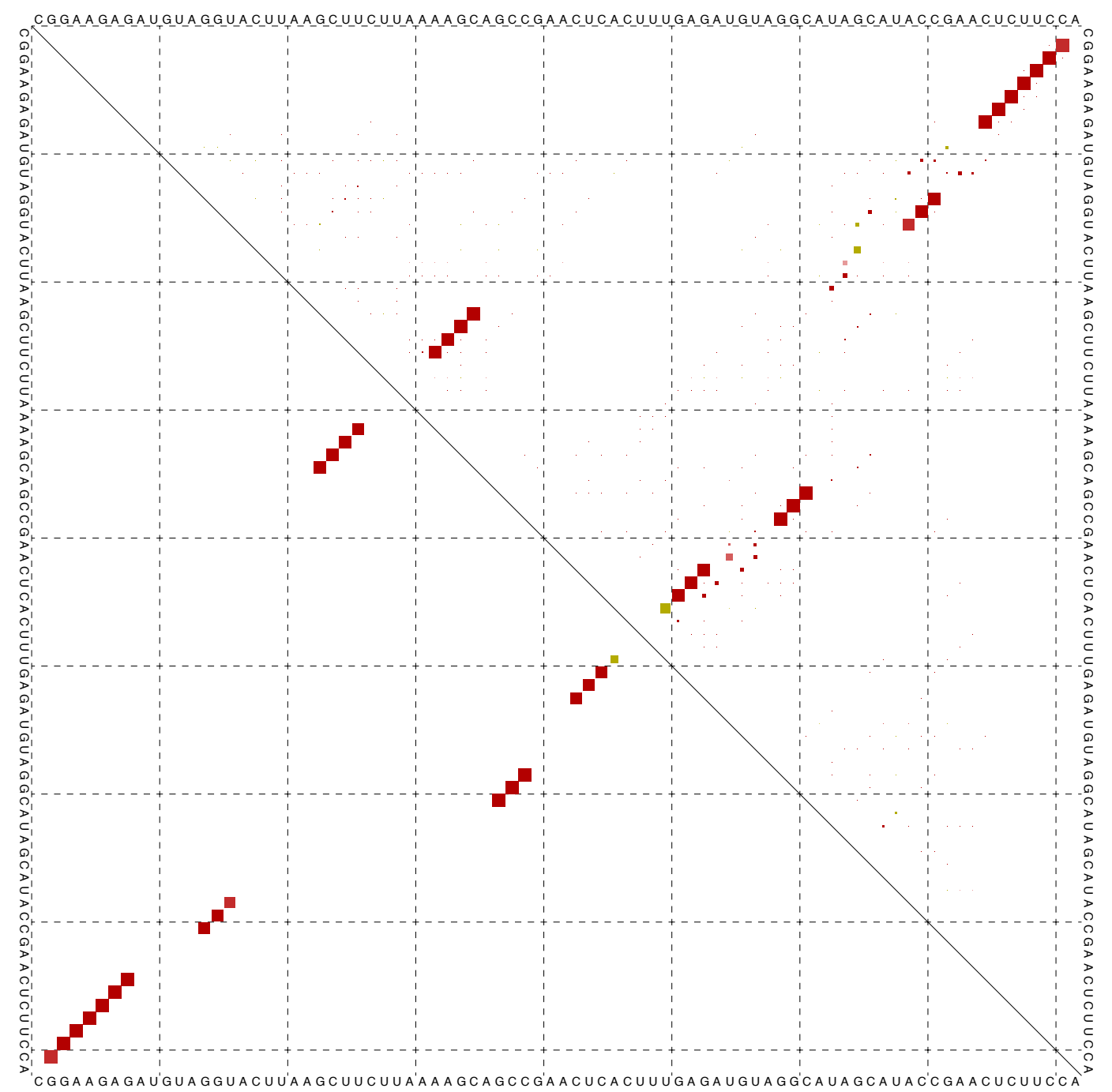
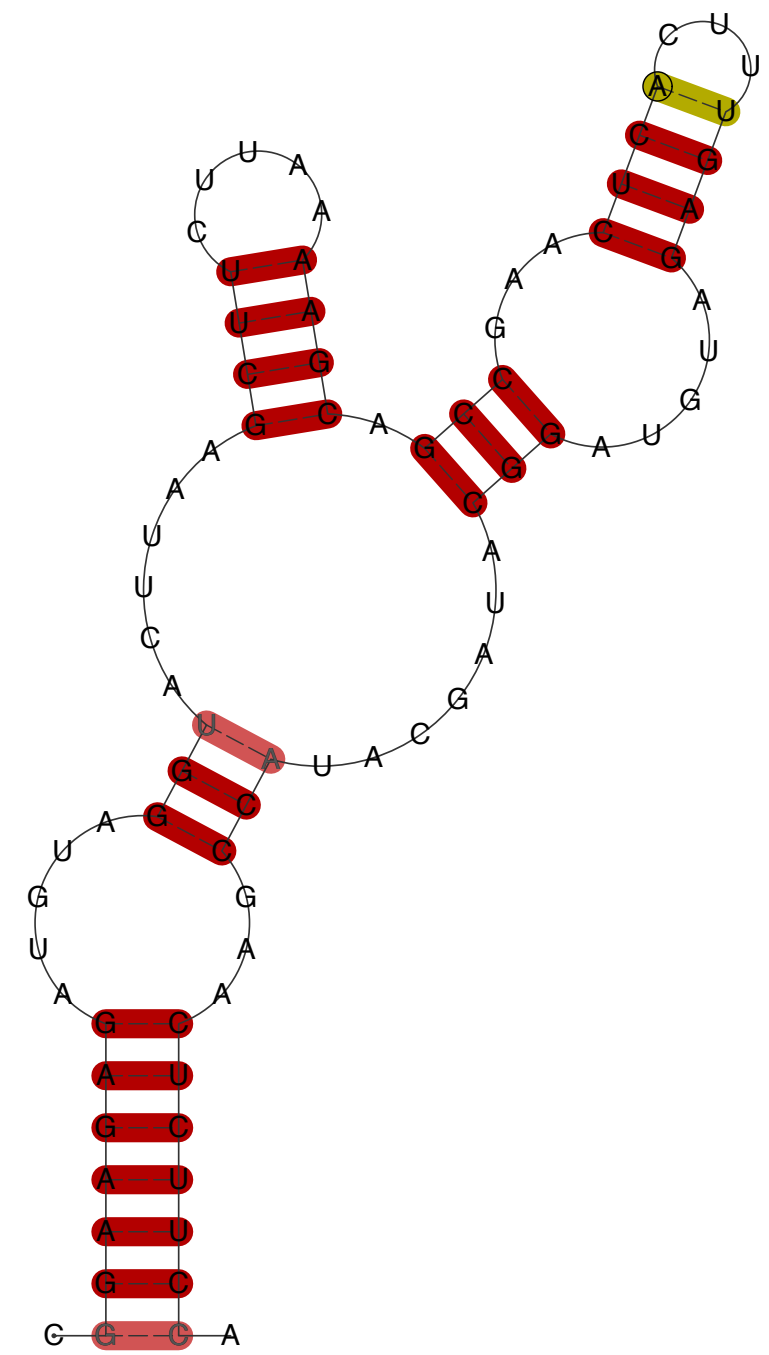
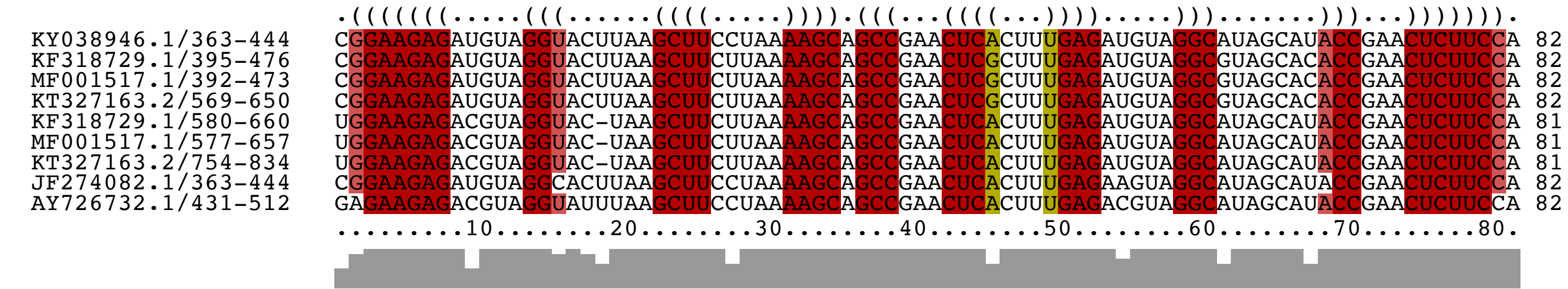
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KF318729.1/688-714	GAGAUGUUAUUUUGUUUUUAAUAUUUC	27
KT327163.2/862-888	GAGAUGUUAUUUUGUUUUUAAUAUUUC	27
JF274082.1/472-498	GAGAUGUUAUUUUGUUUUUAAUAUUUC	27
KY038946.1/472-498	GAGAUGUUAUUUUGUUUUUAAUAUUUC	27
AY726732.1/540-566	GAGAUGUUAUUUUGUUUUUAAUAUUUC	27
MF001517.1/685-707	GAGAUGUUAUUUUGUUUUUAA----UC	23
	10.....20.....	



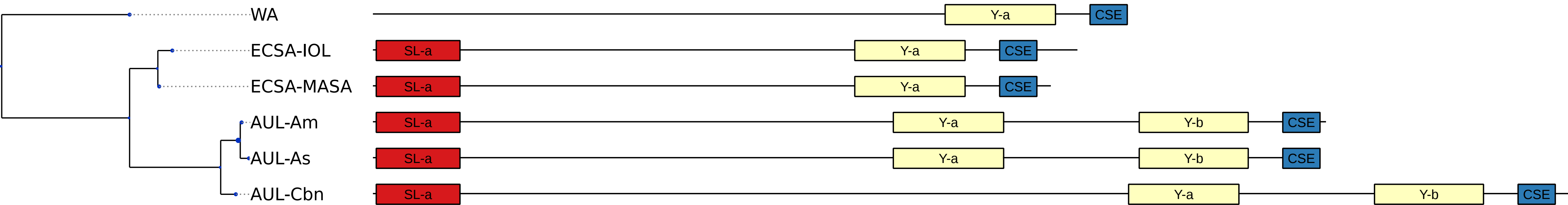
CHIKV 3'UTR: Conserved RNA Structures



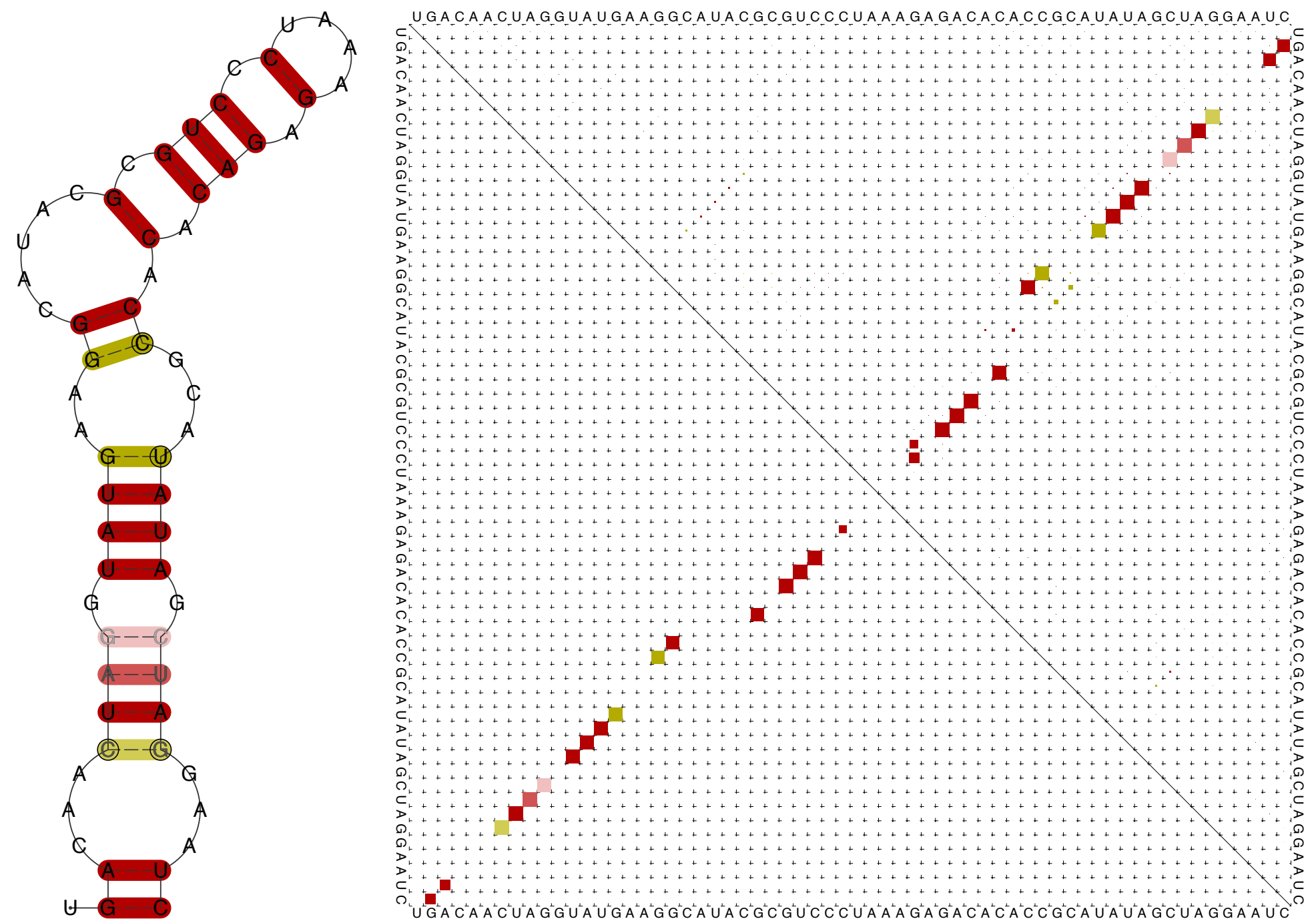
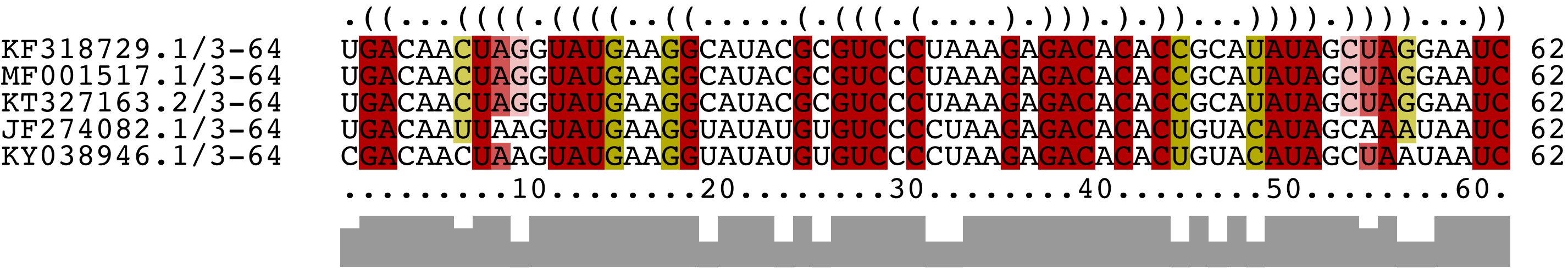
Y-shaped Element (SL-Y)



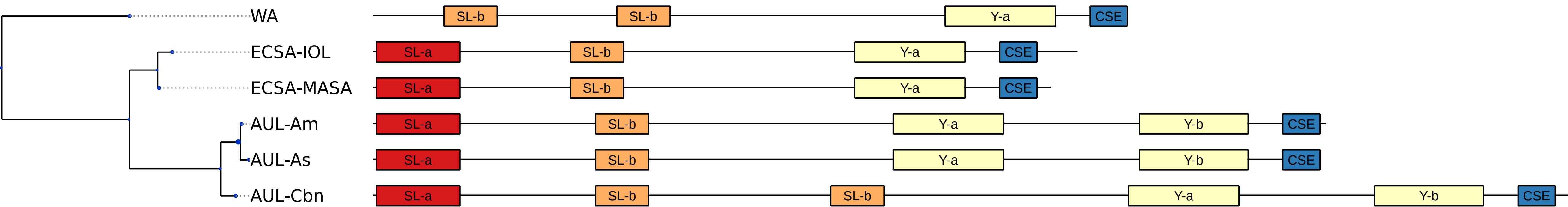
CHIKV 3'UTR: Conserved RNA Structures



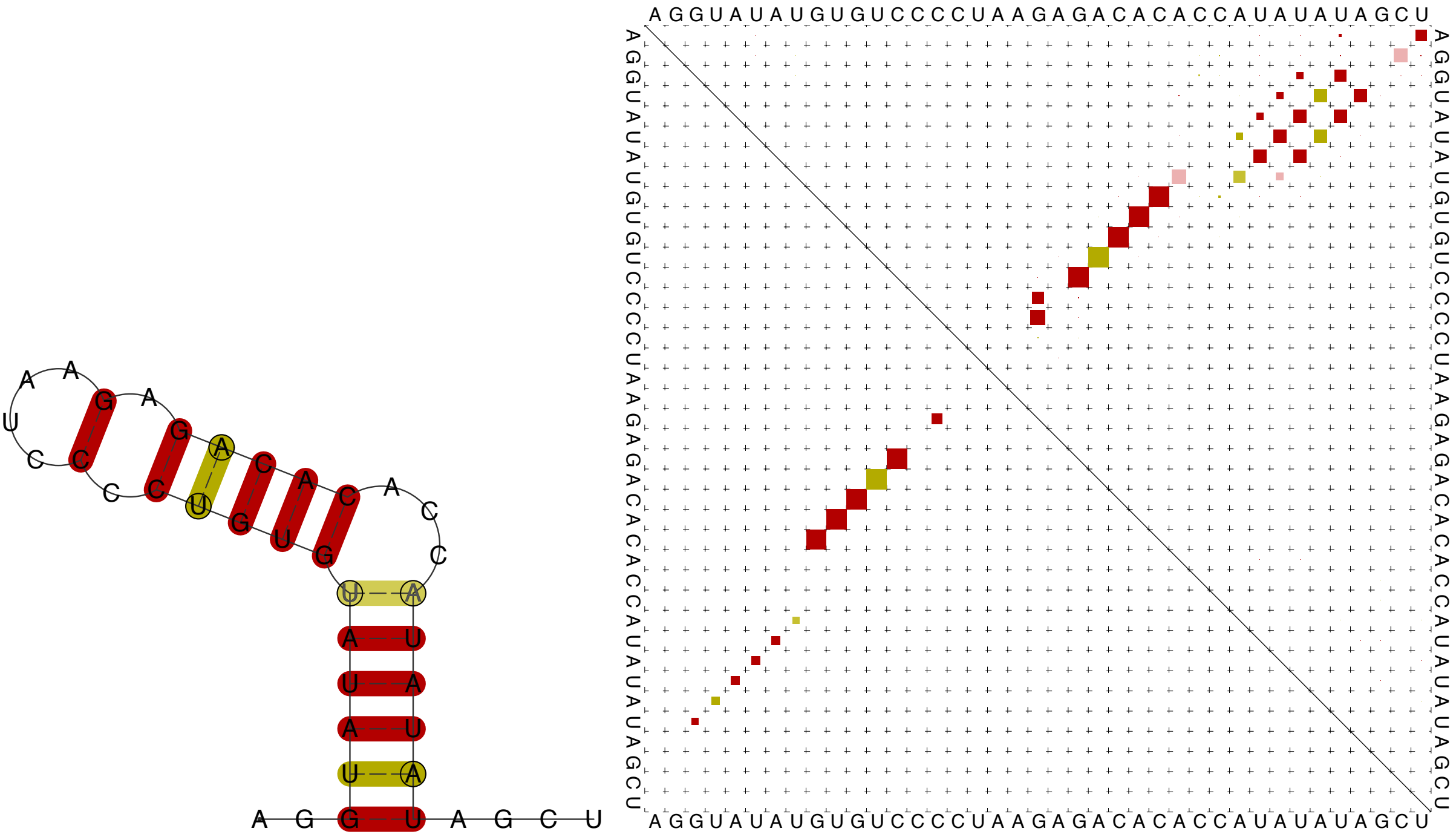
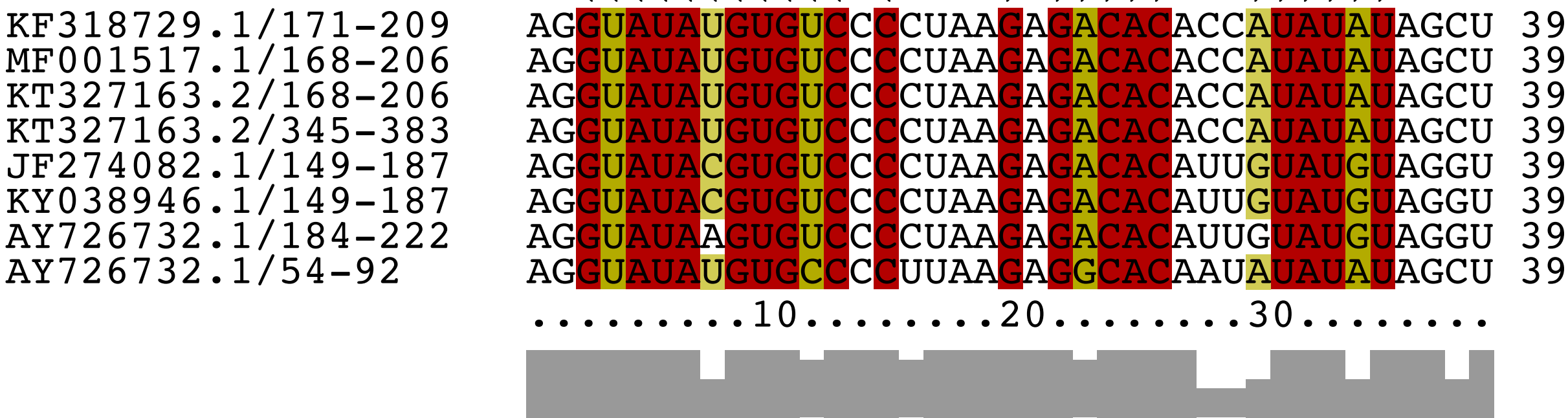
Stem-loop a (SL-a)



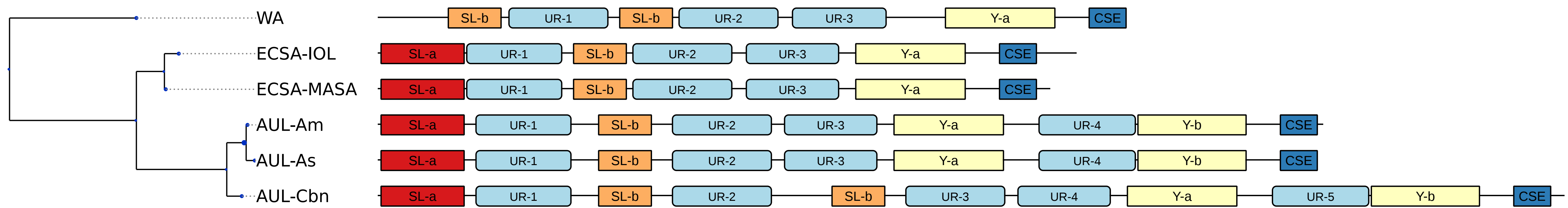
CHIKV 3'UTR: Conserved RNA Structures



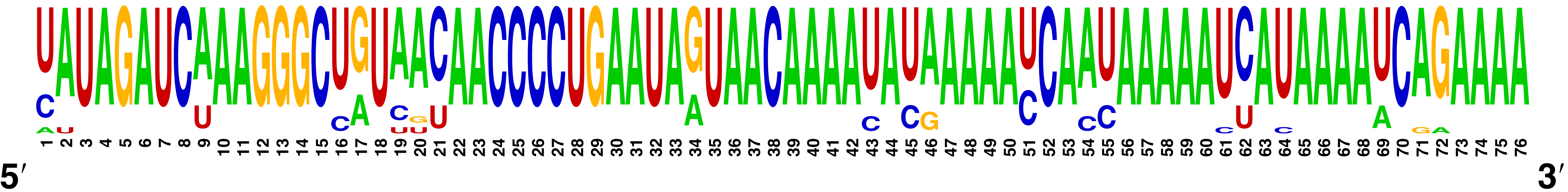
Stem-loop b (SL-b)



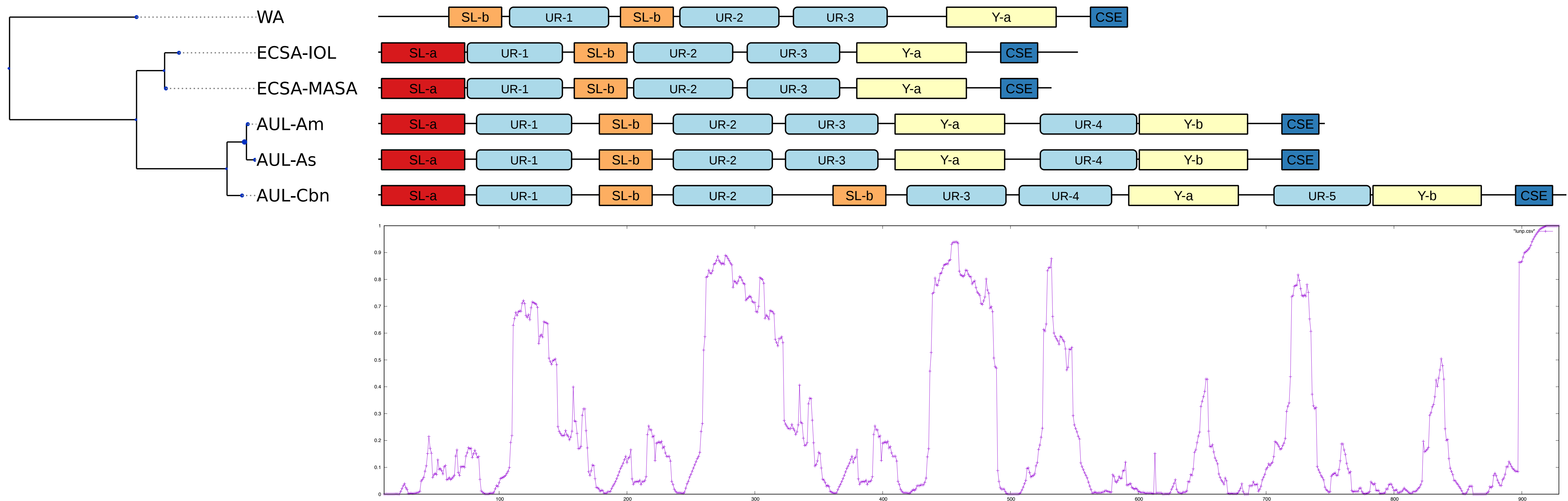
CHIKV 3'UTR: Conserved RNA Elements



Unstructured repeat (UR)



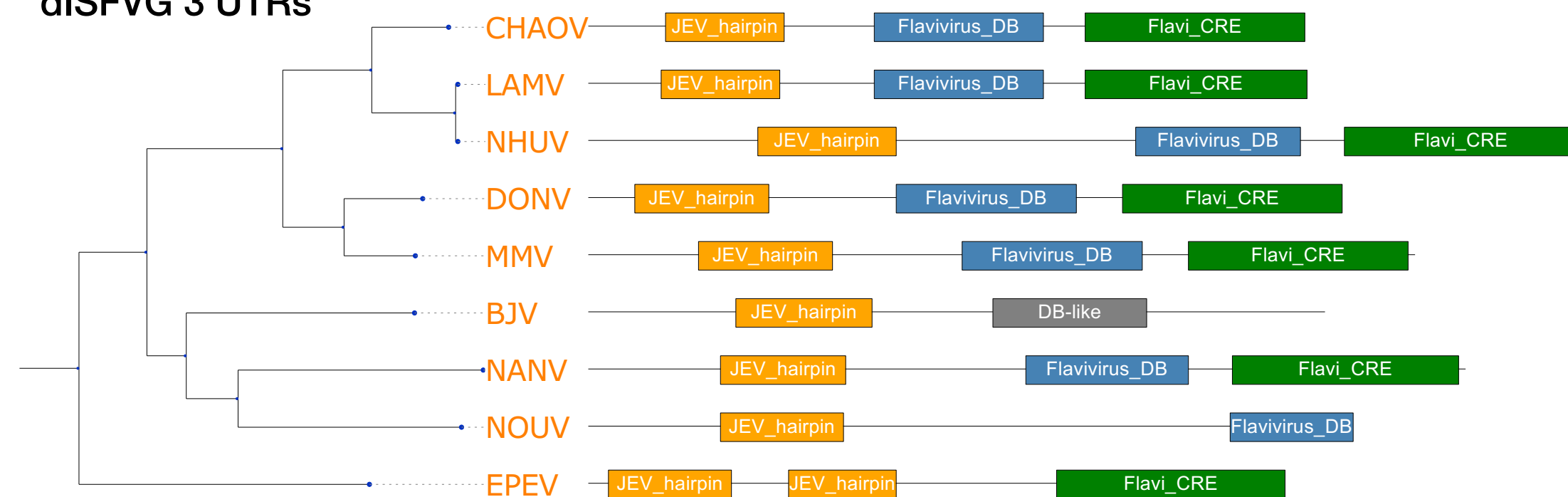
CHIKV-Cbn Accessibility



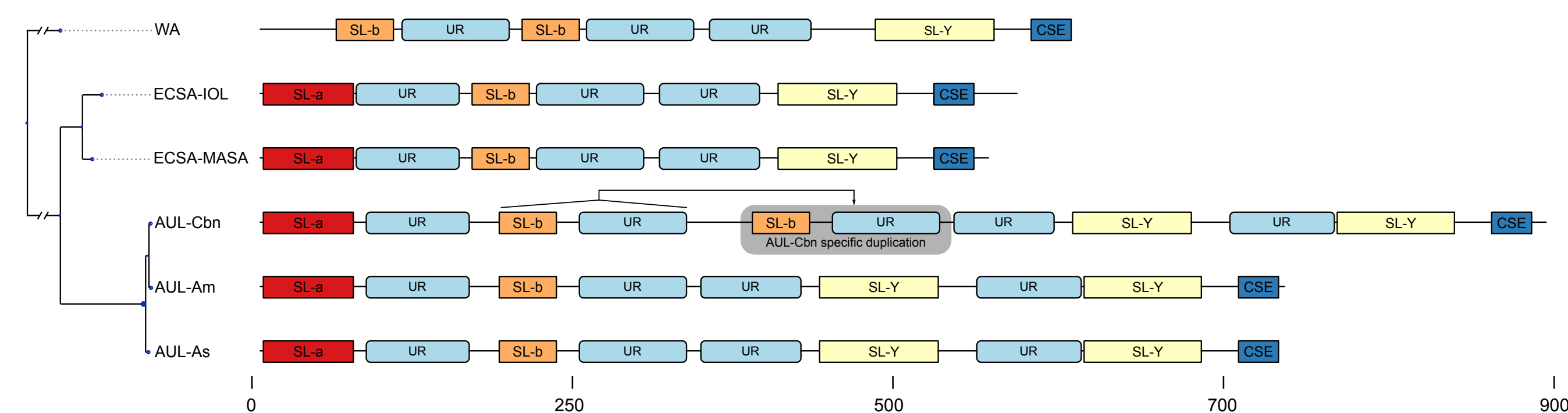
Summary

- Functional RNA structures are evolutionarily conserved
- Structural alignments and covariance models
- Varied architecture of conserved RNAs in flavivirus UTRs
- Lineage-specific non-coding RNA architecture in Alphaviruses
- Tools freely available through ViennaRNA Package

dISFVG 3'UTRs



CHIKV 3'UTRs



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